What aggregate gains conceal

Migration and agricultural transformation in Brazil, flood-resilient urban growth in Viet Nam, and a fiscal-incidence classic across developing countries · three rigorous economics papers, including two peer-reviewed development-journal articles. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

These papers share a suspicion of headline totals. In historical Brazil, immigrant knowledge appears in more intensive land use and higher farm values, not simply more workers. In Ho Chi Minh City, a spatial model separates genuinely new metropolitan output from activity shifted across neighborhoods when flood, transport, and innovation investments are bundled. Across fifteen developing countries, progressive tax-and-transfer systems can reduce poverty in the aggregate while still leaving many poor households net losers. **Literature trend:** rigorous development economics increasingly traces where aggregate gains come from, where they move, and who pays for them. Mechanisms, spatial displacement, and household-level incidence are not footnotes; they determine whether a policy creates development or merely improves an average.

01 CURRENT / MIGRATION AND AGRICULTURAL DEVELOPMENT · STARTS ON PAGE 2 · Journal of Development Economics 183 · 2026 · peer reviewed · CC BY-NC 4.0

The Effects of Immigration on Agricultural Development: Brazil in the Age of Mass Migration

Did European mass migration change Brazilian agriculture rather than merely add labor? The authors digitize the 1920 population and agricultural censuses and instrument municipal immigrant shares with national inflow waves interacted with the timing of railway access. A one-standard-deviation increase in the immigrant share raises farm value per hectare by roughly 0.7 standard deviations, alongside more intensive cultivation and greater use of tools, without slowing structural transformation. The design depends on railway timing affecting farm outcomes only through immigration, and the historical setting limits external validity. The paper reframes migration as a source of production knowledge and land-use change, not only a labor-supply shock.

02 CURRENT / CITIES, FLOODS, AND INFRASTRUCTURE · STARTS ON PAGE 17 · World Bank Policy Research Working Paper 11444 · 2026 · CC BY 3.0 IGO

Evaluating Spatial Underperformance: A Spatial Computable General Equilibrium Model Applied to Ho Chi Minh City

Why does well-located Thu Duc produce far less than its metropolitan position suggests, and how much can flood protection and transport recover? A spatial general-equilibrium model of 116 cells endogenizes firm and household location, commuting, land use, and agglomeration, calibrating productivity and amenity to observed distributions. Flood mitigation raises modeled productivity by 12–21%; a combined package lifts Thu Duc output about two-thirds above business as usual, with 56% of the gain new to the metropolis rather than displaced from elsewhere. These are calibrated counterfactuals, not causal estimates, and depend on functional-form and growth assumptions. The payoff is a disciplined way to separate local redevelopment from metropolitan growth.

03 CLASSIC / FISCAL INCIDENCE AND POVERTY · STARTS ON PAGE 32 · Journal of Development Economics 122 · 2016 · peer reviewed classic · CC BY 4.0

Can a Poverty-Reducing and Progressive Tax and Transfer System Hurt the Poor?

Can a fiscal system reduce poverty overall yet make many poor households poorer? Higgins and Lustig formalize “fiscal impoverishment” and apply incidence data from fifteen developing countries. In ten systems conventionally classified as both progressive and poverty reducing, at least one-quarter of the poor pay more in taxes than they receive in transfers. Their axiomatic measures and decompositions reveal losses hidden by net poverty statistics. Incidence accounting cannot fully capture behavioral responses, general-equilibrium effects, or differences in service quality, and cross-country harmonization is imperfect. The enduring lesson is that a favorable average can coexist with serious harm to identifiable poor households.

Editorial note: references and appendices were omitted where possible; no included article contributes more than about 15 text pages, excluding table-heavy pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Journal of Development Economics 183 (2026) 103861

Contents lists available at ScienceDirect

Journal of Development Economics

journal homepage: www.elsevier.com/locate/devec

Regular article

The effects of immigration on agricultural development Brazil in the Age of Mass Migration

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ARTICLE INFO

Dataset link: https://data.mendeley.com/datasets/pzgkvp3mnk/1

JEL classification:
F22
J61
N36
N56
O13
O15
Q15

Keywords:
Immigration
Agricultural effects of immigration
Age of Mass Migration
Brazil
Agricultural development
Land use

ABSTRACT

We study the effects of immigration during the Age of Mass Migration (1855–1920) on Brazil's agricultural development in 1920. Instrumenting for a municipality's immigrant share using the interaction of aggregate immigrant inflows and the expansion of Brazil's railway network, we find that a greater share of European immigrants in a municipality led to an increase in farm values. The bulk of the effect was the product of more intense cultivation of land. We also find that it is unlikely that immigration slowed Brazil's structural transformation. Our results imply that immigration into an emerging agricultural economy can contribute substantially to agricultural development.

1. Introduction

Immigration and agriculture are intimately connected. In modern developed economies, particularly in the United States, immigration is an important source of labor for the agricultural sector.[1] Moreover, the major immigrant countries of the Americas—the United States, Argentina, and Brazil—all experienced a formative period of agricultural frontier expansion during the *Age of Mass Migration*—a period, roughly 1840–1920, in which over 50 million people migrated from Europe to the Americas (Hatton and Williamson, 1998; Hatton and Ward, 2019; O'Rourke and Williamson, 1999). Within this historical context, this paper investigates the role of immigration in driving the development of the destination country's agricultural sector.

In particular, we study the effects of European immigration to Brazil in the period 1855–1920 on the development of the Brazilian agricultural sector in 1920. Brazil at this time was an overwhelmingly agrarian economy with a nascent industrial sector. Brazilian policymakers, responding to labor demand driven by the expanding agricultural frontier and growing coffee sector, as well as by the decline and eventual abolition of slavery in 1888, initially developed immigration subsidy programs encouraging immigration from Europe to provide labor for the agricultural sector. Even outside of these programs, immigrants entered the agricultural sector in large numbers (Klein, 1995). We investigate whether and through what mechanisms immigration supported the development of this sector, which we operationalize by

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[1] See, for instance, Kostandini et al. (2014), Richards (2018), Baldone et al. (2021), Ifft and Jodlowski (2022), Zahniser et al. (2012).

https://doi.org/10.1016/j.jdeveco.2026.103861
Received 29 January 2025; Received in revised form 28 April 2026; Accepted 11 June 2026
Available online 27 June 2026
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comparing the per-hectare value of farms across Brazilian municipalities with different European immigrant shares of population. Given the long-standing debate in development economics regarding the role of agricultural development in delaying or promoting economic progress in general,[2] we also investigate the contribution of immigration to the country's ongoing structural transformation.

The main empirical challenge that we face is the potential endogeneity of immigrants' location choice within Brazil. To address this challenge, we adapt to the Brazilian case an identification strategy developed by Sequeira et al. (2020) to study the effects of immigration in the United States during the Age of Mass Migration. This strategy creates an instrumental variable for a municipality's immigrant share of population based on the interaction of aggregate immigrant arrivals and the rollout of the rail network, which was the main means of transportation for immigrants to reach their destinations (Holloway, 1980; Lanza et al., 2023; Minale et al., 2024). Intuitively, we compare two municipalities, one of which was linked to the railroad just before a large national-level immigration inflow, and the other of which was linked just after. The former, by virtue of being more easily accessible to the large number of immigrant arrivals, is predicted to have had a greater population share of immigrants by 1920. This method enables us to control for rail linkage directly, addressing concerns that rail was built targeting specific areas or had direct effects on the local economy, and ensures that we do not simply compare linked and unlinked places. The main identification assumption is that a municipality's *timing* of rail construction (conditional on ever being linked) was independent of *aggregate* immigration. To support this assumption, we show that municipalities linked just before immigration boom years were similar to those linked just before immigration lull years.

We digitized a wide array of municipality-level data from the 1920 Brazilian census (Directoria Geral de Estatística, 1922), which was the first successful Brazilian census after 1872 and the first to collect agricultural data for the whole country.[3] Our main finding is that municipalities with a higher population share of European immigrants in 1920 had greater farm values per hectare, which we interpret as indicating greater development of the agricultural sector. Our preferred IV specification implies that a one-standard deviation increase in a municipality's population share of European immigrants generated a 0.7-standard deviation increase in its farm values per hectare. The effect on farm values is present for total farm value and separately for each of its three constituent components—land, infrastructure, and tools and machines. The finding that immigration affected components of farm value besides land is crucial for two reasons. First, it alleviates the concern that our outcome variable captures local characteristics other than agricultural development, which would be capitalized in land values.[4] Second, the value of infrastructure, tools, and machines is linked directly to the development and productivity of the agricultural sector, which is our ultimate object of interest.

We find that changing land use patterns were the primary mechanism for immigration's effect on agricultural development. We show that a greater share of European immigrants in population led to an increase in the share of farmland cultivated (as opposed to being left fallow or as forest). This increase in cultivation intensity was responsible for about one quarter of the effect of immigration on farm values.[5] We interpret these findings as exemplifying the important role that immigrant incentives, either unofficial or official, can play in shaping the effect of immigration in general and land-use transformation in particular. Specifically, we argue that our results can be explained, at least in part, by a combination of the unique incentives offered to immigrants in the government-subsidized immigration program—in which expanding cultivation was necessary to continue to provide immigrants with their main form of compensation—as well as by the differences in incentives faced by temporary immigrants relative to natives and permanent immigrants.[6]

Mirroring the broader debate in development economics, Brazilian historiography (Suzigan, 2000, pp. 23–47) raises the possibility that immigration, by facilitating agricultural development, slowed Brazil's structural transformation. To shed light on this issue, we study the total effect of immigration on a variety of outcomes related to Brazil's structural transformation.[7] We find that immigration did not slow, and in fact may have accelerated, Brazil's structural transformation. In particular, we find that immigration contributed to a decline in the agricultural share of the labor force. Together with our results for farm values, this implies that immigration fostered an agricultural sector that was more productive while demanding less labor—a factor that has been identified as an essential driver of structural transformation (e.g., Herrendorf et al., 2014; Caselli, 2005; Montero and Yang, 2022). We also find a positive effect of immigration on the literacy of both native- and foreign-born individuals, suggestive evidence that immigration was associated with increased employment in industry and services, and no evidence that immigration reduced the participation of women in the labor force relative to men. We find no evidence of an adverse impact of immigration on institutions in the form of an increased numbers of rentiers, whose presence may have held back development, or of an inflated public sector arising in response to immigration.

1.1. Literature review and contribution

The main contribution of this paper is to advance the literature on the effects of immigration on the destination economy, and in particular that focusing on the effects of immigration on the destination country's agricultural sector.[8] The bulk of this literature focuses on the effects of immigration restriction on outcomes such as agricultural productivity, mechanization, crop mix, and innovation, typically over a short time horizon and in settings where agriculture was already well established.[9] In modern settings, much of this work is based on increases in the intensity of immigration enforcement in the United States, which is found to reduce immigrant labor supply and lead to

[2] See, for instance, Asher et al. (2023), Dinkelman et al. (2024), Prebisch (1962), Singer (1950).

[3] We also use data from the São Paulo Census of 1904–1905 (Secreteria da Agricultura, Commercio e Obras Publicas do Estado de São Paulo, 1906–1910), which includes detailed agricultural information, but only for the state of São Paulo.

[4] This is further supported by the fact that a higher share of European immigrants in population also generated an increase in the value of agricultural output per hectare. These findings indicate that, even if expectations of future productivity were priced into the value of land, immediate effects were also important.

[5] We also find that there was no effect of immigration on the value of agricultural output per *cultivated* hectare; that is, the increase in output per hectare coming from immigration was driven by the cultivation of additional land.

[6] We do not claim that these incentives were the sole drivers of the increase in cultivation intensity, nor do we quantify the extent of their influence. Our goal is to show that the mechanism that we document is plausible in our context by discussing aspects of Brazilian immigration that may have contributed to increased cultivation intensity.

[7] Our goal here is to determine the *total* effect of immigration on structural transformation, not to separate the direct effect of immigration on structural transformation and the indirect effect of immigration-induced agricultural development.

[8] Other papers on this topic include Baldone et al. (2021), Richards (2018), San (2023), Abramitzky et al. (2023), Sequeira et al. (2020), Lafortune et al. (2015), Bazzi et al. (2016), Ifft and Jodlowski (2022), Kostandini et al. (2014), Lanza et al. (2023), Droller et al. (2023), Lew and Cater (2018), Clemens et al. (2018), Cance (1925). Escamilla-Guerrero et al. (2026) provide a more detailed review of this literature and the others discussed below.

[9] For instance, studies of the effects of immigration in the United States do not focus on the period of frontier expansion.

declines in area cultivated, increases in innovation, shifts to a less labor-intensive crop mix, and rising agricultural wages (Ifft and Jodlowski, 2022; Kostandini et al., 2014; Richards, 2018). In historical settings, attention is concentrated on the effects in the United States of the 1920s immigration quotas and of the termination of the Bracero program in 1964, which are found to have had minimal labor market effects (Clemens et al., 2018; Abramitzky et al., 2023), but to have driven mechanization (Lew and Cater, 2018; Abramitzky et al., 2023) and innovation (San, 2023).[10]

The effects of immigration restriction and enforcement, however, need not parallel the effects of immigration itself. A smaller set of historical studies focuses on the effects of immigration *per se* on agriculture. Using a network-based shift-share instrument, Lafortune et al. (2015) study the effects of US immigration in the period 1910–1940 on the US agricultural sector. They find that immigration-induced increases in labor supply generated a shift towards more labor-intensive crops and a reduction in the capital-labor ratio, with little or no impact on land prices or farm size. Sequeira et al. (2020) also analyze the effects of immigration on agricultural outcomes in the United States, though these are a secondary focus of their analysis. They find little evidence of an impact of immigration on farm values from 1860–1920, but a substantial positive effect in 1930, which they attribute to superior agricultural expertise among immigrants.

This paper—benefiting from unique features of the Brazilian economy, agricultural sector, and immigration experience—directly studies the effects of immigration on agriculture over a longer period in an emerging frontier setting in which immigration was, in large part, targeted towards areas in which agriculture was nascent. That is, our focus is on the seldom-studied interplay between agricultural development and immigration.[11] We show that immigration into an emerging agricultural economy can have substantial effects on the course of agricultural development, and in particular on land-use transformation.

Ultimately, these results shed light on the effects of immigration in a less-studied context—developing countries. The effects of immigration in developing countries may be quite different from those in developed countries because of substantial differences in economic structure (including a larger agricultural sector) and institutions, and because these countries may attract different types of migrants or migrants from different places of origin than do more developed countries. But the literature on the effects of immigration focuses largely on the modern United States (e.g., Abramitzky and Boustan, 2017; Card, 1990; Hanson, 2009; Kerr and Kerr, 2011; Borjas, 2003; Card, 2005) and other modern developed countries (e.g., Dustmann et al., 2016; Manacorda et al., 2012). Even historical studies of the United States during the Age of Mass Migration shed light on migration from poor to rich countries, as most US immigrants in this period came from countries with substantially lower wages (Williamson, 1995). Although there is a growing literature on the effects of immigration in developing countries,[12] more such work is needed. We enrich this literature by shedding new light on the effects of immigration in an agrarian economy at an early stage of development. Our results show that migration can have positive effects in a developing country despite weak institutions, inequality, and an economic structure geared towards primary production.

This paper also contributes to the literature on the Age of Mass Migration (e.g., Abramitzky and Boustan, 2017), which has focused largely on the United States, especially in the latter years of this phenomenon (e.g., Tabellini, 2020; Abramitzky et al., 2023). Latin America remains understudied, with the existing literature focusing primarily on Argentina.[13] Brazil, despite being the third-most popular overall destination and the second-most popular destination after the United States in the 1890s (Ferenczi and Willcox, 1929; Sánchez-Alonso, 2019), has received relatively little attention. The value of studying Brazil is enhanced by the fact that its economy, institutions, and immigration experience were quite different from those of the United States and Argentina.

Finally, we add to the economics literature on immigration to Brazil during the Age of Mass Migration. Within this literature, this paper is most closely related to Lanza et al. (2023). They use the same farm-level data for the state of São Paulo that we use in part of our analysis (though aggregated to the municipality level in their case) to study the effect of immigration as part of the subsidized immigration program on coffee production and capital adoption in agriculture. Arguing that the assignment of immigrants to municipalities within this program was quasi-random and comparing municipalities with a greater immigrant share to those with a smaller share, they find that a greater share of immigrants was associated with greater coffee output per farm and the adoption of more agricultural tools in 1920. Beyond this, most research focuses on the long-run effects of European immigration, showing that nineteenth-century settlement patterns had persistent effects on development and human capital accumulation, measured in the later twentieth and twenty-first centuries.[14]

This paper provides the first causal study of the contemporaneous effects of immigration on agricultural development and structural change that covers the entirety of Brazil, all crops, and the whole of the immigration flow, whether subsidized or unsubsidized. It thus contributes to the long-standing debate about the role of immigration in the development of the largest Latin American economy (e.g., Alston et al., 2016; Holloway, 1980; Papadia, 2019). Our findings help to clarify the long-debated role of immigration in the development of the Brazilian economy. Confirming the assertions of the historical literature (Holloway, 1980; Milliet, 1941; Monbeig, 1984; Vidal Luna et al., 2014; Colistete, 2015; Buciferro, 2021), we find that immigrant labor was an important force in driving the growth of Brazil's agricultural sector. But contrary to this prior scholarship, we find that this was not primarily because immigrants uniquely enabled the production of coffee or solved agricultural labor shortages.[15] Instead, immigrant labor was unique in that it drove land-use transformation, causing land to be cultivated more intensely. Importantly, we do not find that the gains that immigration brought to the agricultural sector were accompanied by effects that limited structural transformation. On the whole, then, our results imply that immigration enabled Brazil to exploit its natural resource endowments, and may have set it on the course of broad-based economic development.

2. Brazil in the Age of Mass Migration

Over 3.5 million European immigrants entered Brazil between 1850 and 1920, with nearly 1.5 million of these arriving between 1888—when the government implemented a subsidized immigration

[10] Lee et al. (2022) study Mexican repatriations from the United States in the 1930s, finding negative labor market effects on natives.

[11] Bazzi et al. (2016) study the effects of internal migration to Indonesia's agricultural frontier.

[12] See Böhme and Kups (2017), Verme and Schuettler (2021) for a review.

[13] The overall picture regarding the effects of immigration emerging from this literature is that immigration during this period induced industrialization, innovation, and overall development in the United States without harming native workers (e.g., Sequeira et al., 2020; Tabellini, 2020). Similarly, immigration enhanced industrialization in Argentina by providing capital and high-skilled labor to the industrial sector (Droller, 2018; Droller et al., 2023).

[14] See de Carvalho Filho and Monasterio (2012), Rocha et al. (2017), Witzel de Souza (2018), de Carvalho Filho and Colistete (2010), Stolz et al. (2013), Feler et al. (2016), Tang and Monteiro (2023), Minale et al. (2024), Nakaguma et al. (2023), Klein (1995).

[15] This is not to say that immigration on aggregate was not important in these regards from a macroeconomic perspective. We discuss this distinction in further detail below.

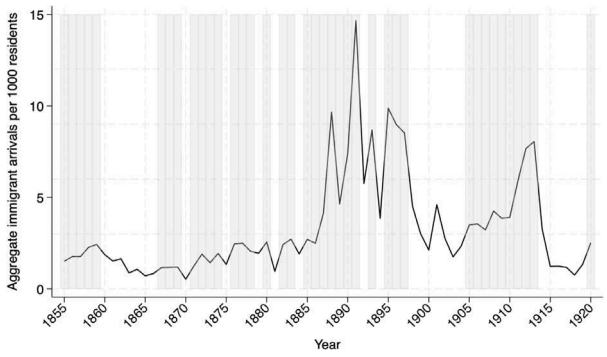

Fig. 1. Immigrant arrivals to Brazil by year, and immigration lulls and booms. *Note:* The graph shows immigrant arrivals in Brazil (1850–1920). Shaded areas are immigration booms, defined as years with an immigrant flow above the previous five years' moving average. The remaining periods are lulls. Gaps in the population data were linearly interpolated to calculate arrival rates. *Source:* Directoria Geral de Estatística (1908), Instituto Brasileiro de Geografia e Estatística (1954) for aggregate immigration numbers. Instituto Brasileiro de Geografia e Estatística (1954), Bolt and van Zanden (2020) for population numbers.

program—and 1900 (Fig. 1 plots immigration rates relative to population over this period). This made Brazil the second most popular immigrant destination in the New World in this period, trailing only the United States (Ferenczi and Willcox, 1929, p. 550). The largest single group of immigrants in this period were from Italy, where immigrants from the north of the country were recruited for subsidized labor contracts. Over 80 percent of immigrants to Brazil were subsidized in the pre-1900 period, possibly reflecting the limited attractiveness of Brazil as a destination due to its low wages and living standards relative to other main immigrant receiving countries, the fresh memory of slavery, and the perceptions of an adverse disease environment (Papadia, 2019).

São Paulo state was the destination of most immigrants.[16] But all southeastern states received large numbers of migrants, as did the south of Brazil, the latter despite being unsuitable for coffee production and being characterized by a climate and productive structure closer to that of Europe than to that of the rest of the country. Even the sparsely populated north and center-west featured municipalities where immigration had a large impact on the composition of the local population. The only region to have almost entirely missed out on mass immigration was the northeast, which was in sharp economic decline following the decreased importance of its two main cash crops—sugar and cotton (Leff, 1997).

Despite this decline, cotton and sugar combined still made up around a quarter of the total value of agricultural production in 1920. Coffee on its own amounted to around the same value, making it Brazil's most valuable crop. Among food crops, maize stands out, making up about 24 percent of agricultural production. Overall, however, the agricultural sector was geared towards exports, despite strong growth in the internal market for food crops starting from the mid 19th century, principally driven by the rapidly expanding southeast (Pereira, 2026). Indeed, around 55 percent of the value of agricultural production in 1920 was made up of export-oriented crops (Instituto Brasileiro de Geografia e Estatística, 1990).

The *colonato* contract was at the center of the subsidized immigration program, which was particularly important in São Paulo as the epicenter of coffee production. Under this scheme, immigrants received free passage and lodging, and were responsible for caring for and harvesting coffee trees for a three-year contract period. The typical *colono* had three main sources of compensation—the fixed wage paid for caring for coffee trees, the piece rate paid at harvest, and access to land on which to grow crops for his own consumption and for sale on the market.

The provision of land for *colonos* to cultivate food crops was seen as the most important part of their compensation. In practice, food crops were often cultivated on the same land on which coffee trees were planted—interrow cropping. The need to continue to provide this compensation to immigrants both led to more intensive cultivation of land in new coffee groves and required the opening of new land to coffee production in order to enable the continued provision of the interrow cropping privileges, which would have competed with more mature trees (Holloway 1980, pp. 74–78, 87–89; Sánchez-Alonso 2019, p. 13). The peculiar incentives faced by migrants did not escape foreign observers at the time. The French agronomist J. Picard (1903) observed that, because the *colonato* program gave immigrants ownership of the staple crops they planted, more care was exercised in their cultivation—care that also benefitted the cash crops for which the migrants were responsible.

In order to reduce the incentive for return migration, the *colonato* contract required adult male immigrants to be accompanied by their families (Klein, 1995). However, the evidence shows that this requirement was not strictly enforced (Lesser, 2013). Ultimately, it is estimated that about 46 percent of immigrants eventually returned to Europe (Levy 1974, p. 66, cited by Sánchez-Alonso 2019, p. 8),[17] with the highest rates among Italians, who were noted for aiming to accumulate savings before returning home (Sánchez-Alonso, 2019, p. 9) and for whom migration (and eventual return migration) was "part of a lifetime strategy for improving living standards at home" (Sánchez-Alonso, 2019, p. 19).[18]

The immigrant experience for *colonos* in Brazil began at reception centers, known as Immigrant Hostels (*Hospedarias dos Imigrantes*), which were present in each of the three main entry ports—Santos in the state of São Paulo, Rio de Janeiro, and Salvador in the state of Bahia. At these hostels, immigrants could sleep, eat, and receive medical attention or vaccinations after being registered. The São Paulo hostel, which processed most migrants, was the main venue for the matching of immigrants arriving under the subsidy program and the employers. In general, immigrants had one week after arrival to find a farm on which to work, and upon signing a contract would be provided with a train ticket to their destination (Holloway, 1980; Lanza et al., 2023). The short time horizon in which to choose their final destination meant that immigrants had very little information about their final destination. This lack of choice and the limitations to free movement within Brazil for some time after arriving in the country led to widespread discontent

[16] Online Appendix Fig. A.1 presents a map of political divisions of Brazil, showing the country's regions and the states that each contains.

[17] Only 46 percent of immigrants remained permanently in the state of São Paulo (Sánchez-Albornoz, 1986) and about 35 percent of all immigrant arrivals processed at São Paulo from 1911 to 1920 were returnees (Holloway, 1980, p. 56; see also Klein, 1995, p. 210). Note that it was not the case that 46 percent of immigrants returned to their home countries. Instead, 46 percent of arrivals resulted in a return, with many immigrants arriving and returning several times. For instance, if an immigrant entered four times and returned three times, this is a 75-percent gross return rate, but a zero percent net return rate because the immigrant remains in Brazil.

[18] Italian data on return migration, which are available from 1905 onwards, indicate that about 50 percent of immigrants returned to Italy at least once (Cinel, 1991). Beyond the size of the return flow, Sánchez-Alonso (2019, p. 9) points out that "with the available evidence, we cannot assume either positive or negative selection in return migrations because no relevant data on the characteristics of returned migrants exist". She does, however, indicate that most Italians returning were likely not returning after failure in the Brazilian economy. Cinel (1991, p. 112) argues that return rates for Italians were higher for those working in industry or commerce than for those working in agriculture, that returns from the states of Rio Grande do Sul and Santa Catarina were uncommon, and that returns from the state of São Paulo were much more common in comparison.

among immigrants, especially in the earlier phases of mass immigration (da Costa, 2000; Fausto, 1999).

The rail network on which these immigrants traveled has been identified as a major factor in Brazil's economic development in the late nineteenth and early twentieth centuries (Summerhill, 2003, 2005). This role is remarkable given the relatively slow and fragmented development of the network (Herranz-Loncán, 2014; Summerhill, 2003) and the fact that decisions regarding the placement of railways did not always follow an economic logic. Although existing settlements of population and economic activity were targeted, the placement of rail was heavily influenced by the interests of local landowners and the government's efforts to appease them (Summerhill, 1997; Bignon et al., 2015). The railways were financed by both private local and foreign (especially British) capital. The central and local governments were not as active as they were, for example, in United States (Leff, 1997; Summerhill, 1997). Nonetheless, government intervention was essential in kick-starting railway construction and in sustaining unprofitable lines, which were politically important due to the influence of local landowners.

Brazilian immigration changed substantially at the beginning of the twentieth century (Klein, 1995, p. 208). Concerns about the conditions of migrants in Brazil led to bans on subsidized migration by several European governments (Lesser, 2013). These concerns encompassed both the conditions of immigrants on coffee plantations and those in migrant colonies—rural settlements for migrants created by private planters in cooperation with the state and which were viewed as being located in economically unfavorable locations (Cameron, 1931). Most notably, the 1902 Italian *Decreto Prinetti*, which made subsidized emigration illegal, was targeted mainly at limiting migration to Brazil.[19] In part as a result of these bans, but also because of the opportunities offered by the growth of the frontier economy (Klein, 1995), subsidized migrants no longer made up the majority of entrants to the country, with their share falling to about 40 percent (Cameron, 1931). Although these bans did not halt mass migration to Brazil, as Fig. 1 shows, they did reduce its magnitude relative to population and led to a change in the distribution of migrants' countries of origin: Fig. 2 shows a decline in the share of immigration coming from Italy with a commensurate increase in the share from Portugal and Spain. There was also a change in the main immigrant sources within Italy, with migrants before 1902 coming primarily from the north and those after coming primarily from the south.[20] The Spanish immigrants resembled the Italians in many ways—they were largely brought in by immigration subsidies, settled primarily in rural areas, and had high rates of return migration (or migration to other locations in Latin America). The Portuguese, on the other hand, were quite different. They were less likely to be subsidized, relying instead on networks deriving from Brazil's colonial history. They were also most likely of the three major immigrant groups to settle in urban areas, where they often entered the service sector (Sánchez-Alonso, 2007; Klein, 1992, 1993; Sánchez-Alonso, 2019).

The shift in migrant origins was coupled with changes in settlement and occupations: whereas immigrants arriving before 1900 were almost entirely directed to agricultural areas, those entering after 1900 settled in large numbers in cities in the states of São Paulo and Rio de Janeiro (Foerster, 1919, p. 289), where they worked in small workshops or as unskilled industrial labor. As a result, the share of immigrants employed in manufacturing increased from about 0.8 percent in 1900 to 13 percent in 1920.

Although it is well established that immigrants were more numerate and literate than the Brazilian population (Rocha et al., 2017; Stolz et al., 2013), direct evidence on the selection of immigrants is limited. It is possible that subsidized immigrants were negatively selected, as subsidies significantly relaxed liquidity constraints, allowing the relatively poor and unskilled to migrate. The few studies addressing selection of Italian and Portuguese migrants, the two major immigrant groups in Brazil, provide mixed evidence.[21]

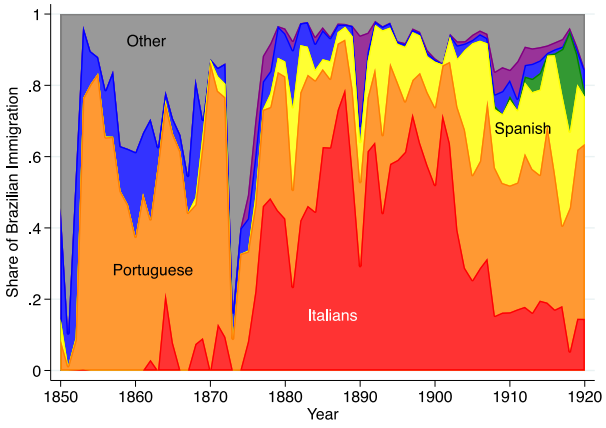

Fig. 2. Distribution of origin countries for Brazilian immigration.
Note: The graph shows the share of immigration to Brazil by source country (1850–1920). The color key is as follows: Italians (red), Portuguese (orange), Spanish (yellow), Japanese (green), Germans (blue), Russians (purple), and other (gray). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Source: Directoria Geral de Estatística (1908), Instituto Brasileiro de Geografia e Estatística (1954).

Buciferro (2021) argues that immigrant laborers were intensively exploited by native landowners—a condition that immigrants were willing to endure only because working in agriculture offered opportunities to transition into farm ownership. Indeed, one of the incentives driving immigrants in exerting labor effort was the possibility of eventually owning land—the prospect of which was a strong pull factor for migrants, especially for Italians—and immigrants were reputed for their hard work and desire to save (Cinel, 1991; Sánchez-Alonso, 2007; Florea, 2023; Holloway, 1980; Lesser, 2013). Holloway (1980, pp. 140–142) points out that the relatively poor conditions faced by plantation workers provided an incentive to save in order to move up the agricultural ladder, and the cultivation of food crops was one of the main mechanisms by which this was done. Recent scholarship, however, has debated to what extent immigrants were able to achieve upward mobility. The consensus in the literature is that immigrants were able to transition to land ownership after a few years on the *fazenda* (Klein, 1995; Lanza, 2021; Sánchez-Alonso, 2019). More specifically, Holloway (1980, p. xvi) documents that many first-generation

[19] The decline in immigration was also driven in part by shocks to coffee demand due to business cycles in Europe and difficulties in adjusting coffee supply to market conditions as a result of the long life of coffee trees and the lag time between their planting and entrance into production (Vidal Luna and Klein, 2014).

[20] See Online Appendix Fig. A.2 for the division of immigrants by origin in the 1920 census.

[21] Fernández-Sánchez and Tortorici (2024) show that Portuguese migrants, who mostly moved to Brazil, were, on average, positively selected on the basis of literacy. Determining the selection of Italians is more difficult given the highly segmented destination choice patterns of this group (Hatton and Williamson, 1998; Spitzer and Zimran, 2024). But given the early dominance of northern Italy in the migratory flow to Brazil, due in part to the fact that migration subsidies were offered exclusively to migrants from the north of Italy (Hatton and Williamson, 1998, p. 102), and that migrants from this region to the United States were negatively selected (Spitzer and Zimran, 2018), it is likely that the early migrants were negatively selected. Hatton and Williamson (1998, p. 121) also find that farmers from Italy, who tended to be relatively poor, were more likely to travel to Brazil than they were to the United States. However, the shift of Italian emigration to a largely southern-Italian phenomenon at the turn of the twentieth century may have been associated with a more positive selection, as migrants from this region who emigrated to the United States were positively selected (Spitzer and Zimran, 2018, 2024).

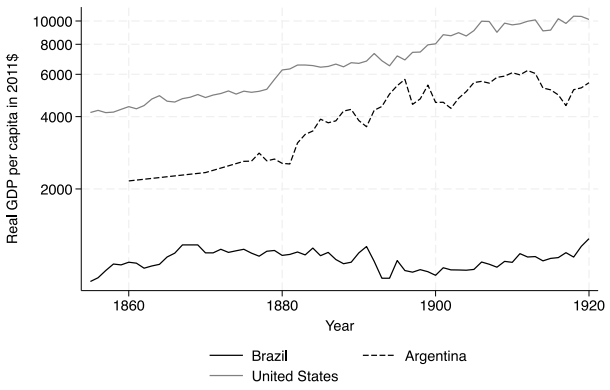

Fig. 3. Real GDP per capita in major immigrant destinations.
Note: The graph shows real GDP per capita (ca. 1850 to 1920) for Brazil and the other two major immigrant destinations in the Americas: Argentina and the United States.
Source: Brazil: Prados de la Escosura (2009), Barro and Ursúa (2008), Bolt and van Zanden (2020); US: Sutch (2006), Prados de la Escosura (2009), Bolt and van Zanden (2020); Argentina: Prados de la Escosura (2009), Bértola and Ocampo (2012), Bolt and van Zanden (2020).

immigrants were likely to become owners of small and medium-sized farms.

The effects of immigration on other aspects of Brazilian economic development remain the subject of debate. One body of literature argues that industry only started to develop rapidly once the agricultural export sector, led by coffee, was disrupted by events such as the First World War and the Great Depression. Another, however, has argued that the development of the agricultural export sector was a crucial precursor for the development of other sectors of the economy, particularly industry (Suzigan, 2000). Insofar as immigration drove the development of this sector, it may have thus delayed or accelerated industrialization, compounding or counteracting any direct effects of immigration on industry.

Altogether, Brazil's immigration and economy bear some important similarities to those of the other major migrant destinations of the Age of Mass Migration—the prevalence, at least from the later nineteenth century, of migrants from the poorer European periphery, the high rate of return migration, and, in Argentina's case, the agricultural sector's share of the economy (Lesser, 2013; Droller and Fiszbein, 2021). But many features set it apart. Most dramatically, there were substantial differences in development between Brazil and the other two major migrant-receiving countries in the Americas, as shown in Fig. 3: Brazil was, by a substantial margin, the poorest of the three countries (Bolt and van Zanden, 2020); indeed, Argentina was one of the 10 richest countries in the world at the eve of World War I and had living standards similar to those of the United States (Spruk, 2019). Commodity production was also uniquely important in Brazil relative to these other countries. In addition, the concentration of Brazilian immigrants in agriculture was unique in comparison to Argentina and the United States, where most immigrants provided labor outside the agricultural sector (Pérez, 2017): in Brazil, approximately 40 percent of immigrant men worked in agriculture in 1920 (Online Appendix Fig. A.3) as opposed to about 15 percent in the United States and 17 percent in Argentina in 1895. Subsidized migration was also uniquely important in Brazil. The United States explicitly banned subsidized migration beginning in 1885, and although there is evidence of subsidized immigration in Argentina, it did not have the relevance it did in Brazil (Lesser, 2013). Brazil also had weaker institutions and worse governance, reflected in low access to justice and an inefficient public sector, as well as a high concentration of economic power and a highly unequal land distribution (Naritomi et al., 2012).

3. Conceptual framework

The simplest way in which immigration could influence agricultural development is by adding to the agricultural labor force. In an environment in which the agricultural frontier is rapidly expanding, a shortage of labor could prevent land from being cultivated; therefore, additional labor provided by immigration can increase the value of previously unused or underused land. Such an effect need not prevail, however, if, for instance, native and immigrant labor were perfectly substitutable and natives relocated within Brazil in response to the arrival of immigrant labor.[22] Similar arguments could be made for immigrants increasing the local demand for land or agricultural products and investing in farm capital in the form of tools and machines where land would otherwise be uncultivated or in lower demand.

The predicted effect of immigration is also complicated by the likelihood that immigrants and natives were fundamentally different from one another, and thus not perfectly substitutable. In the Brazilian context, immigrants may have possessed certain characteristics that made them particularly productive agricultural workers, such as specialized agricultural knowledge or greater human capital. In this sense, a greater concentration of immigrants would spur local agricultural development. Similarly, the complementarity of immigrant labor with capital may have differed from that of native labor. Such differences could, for instance, imply that immigrant labor was uniquely suited (or not) to enabling the cultivation of coffee.

Immigrants also faced different incentives than natives. For instance, temporary migrants, who made up a substantial share of migrants to Brazil, may have been incentivized to supply more labor than natives or permanent immigrants while in Brazil, which would in turn contribute to greater land values where they settled. This mechanism is consistent with evidence that temporary immigrants substitute inter-temporally, supplying more labor in the destination and enjoying greater leisure at home.[23] Similarly, as discussed above, the *colonato* contract, under which many immigrants worked, also created a unique set of incentives that may have driven land to be cultivated more intensively than it otherwise would have been.

Immigration's effect on agriculture would likely have also spilled over into other sectors of the economy (Lewis, 1954; Schultz, 1964). It has been posited that greater agricultural productivity could encourage structural transformation (Herrendorf et al., 2014; Caselli, 2005; Dinkelman et al., 2024; Johnston and Mellor, 1961; Timmer, 1988; Montero and Yang, 2022). On the other hand, it has also been suggested that it can lead to specialization in primary production, discourage human and physical capital accumulation, hinder the development of other sectors of the economy, and ultimately delay structural change (Matsuyama, 1982; Singer, 1950; Prebisch, 1962). The existing empirical literature is equally ambiguous (Foster and Rosenzweig, 1996; Hornbeck and Keskin, 2015; Bustos et al., 2016, 2020; Asher et al., 2023), suggesting that the effect of agricultural development may be strongly context-dependent.

[22] This relocation could also come in the form of immigration to a municipality deterring in-migration by natives. Note that immigrant arrivals could lead to the increase of land values elsewhere in Brazil, if, for instance, immigrant arrivals led natives to relocate to the frontier where labor was previously scarce. But no identification strategy based on the comparison of different municipalities within Brazil can identify such an effect absent detailed data on internal migration, which to our knowledge do not exist. Similarly, such an identification strategy will not be able to capture effects of immigration on the national-level labor force, which may have been important in helping to drive frontier development.

[23] See Dustmann (1994, 2000), Dustmann et al. (1996), Dustmann and Görlach (2016), Epstein and Venturini (2011), Galor and Stark (1991), Hill (1987), Holloway (1980), Klinthäll (2006), Kyarko and Chartouni (2017), Vijverberg and Zeager (1994), Wahba (2022).

4. Data

Our analysis is based primarily on data that we digitized for this project from the 1920 Population and Agricultural Census of Brazil (Directoria Geral de Estatística, 1922).[24] The 1920 census was the first complete population census successfully carried out in Brazil since 1872, and the first ever agricultural census covering the whole country. While this constrains our analysis to a cross-sectional design, the census provides a very rich set of variables at the municipality level that capture economic, population, human capital, and labor force characteristics. We also use complementary data from a variety of other sources, mainly as control variables throughout the analysis, as well as farm-level data from the state of São Paulo from its 1904–1905 census. Altogether, to our knowledge, this is the most comprehensive municipality-level dataset that has been assembled to date for early-twentieth-century Brazil.

4.1. Main outcome variables

The 1920 census provides data on the average monetary value of farms (agricultural establishments) by municipality in *milréis*. These establishments were often made up of a single plot of land, but could also refer to multiple plots in the same municipality managed by the same person, group of people, or organization (e.g., the government). Combined with information on the size of farms, these data enable us to compute our main outcome variable—the average value of farms per hectare of land. The census reports farm value in total and broken down into three components—land, infrastructure, and tools and machinery—which we also analyze separately.

Farm values are intended to measure agricultural development, which includes any factor making the local agricultural sector more productive and valuable. Two components of farm value—infrastructure and tools and machines—are directly related to the development and productivity of the agricultural sector. The other component, land, as an immobile factor of production, will capitalize in its value any local characteristic that improves agricultural productivity. Land values will also capture both backward- (resulting from past investments capitalized in farm values) and forward-looking (resulting from the expected trajectory of the local economy) aspects of development, thus capturing the long-term productive potential of the local agricultural sector.

While agricultural development will, for these reasons, lead to increases in farm values, this measure may also capture some other features of development and of local economies. In addition to agricultural development, land values will also capitalize local amenities and other characteristics, including expectations about other aspects of a location's future development, that are unrelated to agricultural development.

For two main reasons, we are confident that the results reported below are not confounded by these potential issues of interpretation. That is, that even if local non-agricultural amenities and expectations (including expectations about agricultural productivity, which we are interested in capturing) were priced into land values, effects on agriculture, both immediate and long-term, were also important. First, we will show that our results hold not only for farm values in total or for land values, but for each of the several components of farm value.

Second, we also find supportive evidence from an alternative measure of agricultural productivity. In particular, we use data from the 1920 census on total agricultural production broken down into 14 crops. We combine this information with current prices for each of these crops, also from the Census, to compute the value of agricultural production per hectare and per cultivated hectare—variables that capture agricultural productivity explicitly. We use these variables as alternative measures of agricultural development, and our results for these outcomes further support our findings. For our main analysis, we prefer to focus on farm values rather than on output. As argued, farm values reflect the long-run productivity of the local agricultural sector, and are therefore a more reliable measure of the permanent effects of immigration on agricultural development than output observed at a single point in time. Output in 1920, by contrast, would have been determined in large part by idiosyncratic shocks and short-run cropping decisions, making it a noisier proxy for the underlying productivity gains we seek to capture.

4.2. Other outcome variables

We also collect data on a series of factors that may have affected the value of farms through their relationship with immigration. The first set of variables concerns the labor force. We construct measures of population density (inhabitants per square kilometer) and agricultural employment density (workers employed in agriculture per square kilometer). To measure land use, we use data on the share of farm land cultivated and the share of cultivated farmland by planted crop. We also measure the use of tools and machines in agriculture (beyond the value measures). We use data on the share of farms with tools—plows, harrows, seeders, cultivators, harvesters, and tractors—and machines in the form of devices employed for processing crops.

We also examine four sets of outcome variables related to structural change. To measure human capital formation, we use data on literacy—the share of individuals who could read and write—which the census reports separately for immigrants and natives. As indicators of economic structure, we use data on the share of workers employed in agriculture, in industry, in services, and in the public sector. We also use the population share of individuals living on income from property or investments (rentiers) as a measure of the presence of local landed elites. Finally, we collect data on the female to male ratio among the employed population by economic sector (agriculture and industry).

4.3. Explanatory variables

We use data on the total population and number of European immigrants by municipality to construct our main explanatory variable—the population share of European immigrants by municipality. We also calculate the share of each individual immigrant group (e.g., Italians, Portuguese, etc.) in population, as the immigration data are reported by nationality. We focus on European immigration for two reasons. First, the historical literature has documented that, during the Age of Mass Migration, nearly 90 percent of the immigrant flows to Brazil came from Europe (Lesser, 2013; Sánchez-Alonso, 2019). Second, the exogenous variation that our identification strategy relies on is partially the product of shocks to European immigration. Because European immigration to Brazil began as early as the 1820s, this 1920 measure of the immigrant share of population captures enduring settlement patterns that account for return migration and life cycle changes over several decades, fitting with our goal of studying the cumulative effects of immigration.[25] Because we focus on the share of Europeans in population, rather than solely on the number of immigrants, we are implicitly controlling for population in our analysis. The census also reports, and we collect, data on the share of landowners in a municipality who were foreign born.

Our identification strategy also relies on annual variation in aggregate immigrant arrivals in Brazil and variation in railway access

[24] Online Appendix B lists the specific sources within this census that we use.

[25] We discuss this potential issue of interpretation in more detail below.

at the municipality level. We obtain data on immigrant arrivals from Directoria Geral de Estatística (1908), Instituto Brasileiro de Geografia e Estatística (1954). Information provided by Giesbrecht (2023) enables us to identify the first station built in each municipality as well as the year when its construction was completed—that is, the year when each municipality was linked to the railway network. We then compute two further variables with this information—the number of years that a given municipality had been connected to the railway network by 1920 and an indicator variable for municipalities that were not connected by 1920.

4.4. Control variables

We collect data on a number of municipality characteristics that may have affected farm value and immigrant settlement to include as control variables in our analysis. To capture proximity to international and domestic markets, we create three variables—distance to the nearest port or frontier custom house, distance to the nearest principal city, and distance to the nearest principal town.[26] We also collect data on the location of immigrant colonies from the same source, complemented by data from Gagliardi (1958) for the state of São Paulo. We also create a battery of variables to control for differences in geographic conditions across municipalities, including surface area, ruggedness, altitude, latitude, and longitude. Next, we use data from the Global Agro-Ecological Zones project (Food and Agriculture Organization, 2021) to construct three variables that capture the suitability of land for agriculture and the adaptability of land for the production of different crops. These variables consist of the first two principal components of suitability for all major crops reported in the census except rubber, and the Herfindahl–Hirschman Index (HHI) of suitability, capturing how concentrated land suitability is in each municipality. Finally, we collect data from the 1872 census (Núcleo de Pesquisa em História Econômica e Demográfica, 2012), including the share of Europeans in population, agricultural employment, employment in the justice sector, the number of slaves, the share of the population that was white, literacy, and school attendance. These are useful in enabling a test of instrument validity, as they are predetermined relative to mass immigration. They also enable us to control for a variety of factors that might be concerning, such as the legacy of enslaved labor.[27]

4.5. Farm-level data

We complement our municipal-level analysis with data from the 1904–1905 Agricultural Census of the state of São Paulo (Secreteria da Agricultura, Commercio e Obras Publicas do Estado de São Paulo, 1906–1910). The data from this source are similar to those coming from the 1920 census, reporting the share of cultivated farm land, the share of cultivated farm land by planted crop, employment density (workers per hectare), and the breakdown of employment by foreign or native birth. But these data are reported at the level of the farm rather than the municipality. In total, we have information for over 40,000 farms across 163 municipalities. This source provides information on land values, as well as farm ownership—that is, whether the farm was owned by a foreign- or native-born person—and on the share of native and foreign workers.

[26] These features are identified using a map of Brazil created by the International Bureau of the American Republics (IBAR) in 1905, shown in Online Appendix Fig. A.5.

[27] Because they require us to change our unit of observation from the 1920 municipality to minimum comparable areas between 1872 and 1920, we do not include these controls in our main results, but we show that our results are robust to their inclusion.

4.6. Summary statistics

Online Appendix Tables A.1 and A.2 present summary statistics for all of the variables in our dataset. Fig. 4 presents maps displaying the geographic distribution of our main outcome variable (panel a) and our main explanatory variable of interest (panel b). The south and southeast are shown at increased magnification, as these regions received the majority of migrants. A number of features of our data are readily apparent. The first is the presence of substantial immigration in regions other than states of São Paulo and Rio Grande do Sul—the traditionally-examined migrant destinations. In fact, immigration was as intense as in these states as in many municipalities of the center-west, southeast, and north regions. The second is that there is a concentration of high farm values and high immigrant shares in the vicinity of São Paulo, in the main coffee-growing area of the southeast. The third is that there was considerable variation in farm values and immigrant shares throughout the country. The fourth is that immigrant origins varied significantly across municipalities, with immigrants drawn from a wide geographic range including European countries (Italy, Portugal, Germany, Spain), neighboring South American nations (Argentina, Paraguay, Uruguay), the Middle East, and Asia (see Online Appendix Fig. A.4).

Panel (a) of Fig. 5 presents the geographic evolution of the Brazilian railway network over time—part of the variation that contributes to our instrumental variables strategy. The expansion of the network inland from major ports over time is clear, with the share of municipalities connected to the rail network increasing from virtually 0 percent in 1860 to about 40 percent by 1920.[28] There appears to have been an important regional component to the rail network. Rail was virtually absent from the north, and evolved into several geographically distinct networks rather than one unified one.

5. Empirical strategy

5.1. Estimating equation

Our main estimating equation for the effect of immigration on farm value is given by

$$f_i = \psi s_i + \mathbf{x}_i' \Pi + \lambda_j + \epsilon_i,$$

where f_i is average farm value per hectare in municipality i, s_i is the population share of European immigrants in municipality i, λ_j are region or state fixed effects, and $\mathbf{x}_i$ is a vector of municipality-level covariates.[29] Our main results are reported with robust standard errors, but all results are robust to various corrections for spatial correlation (Colella et al., 2020; Conley, 1999), as we show in Online Appendix C.

Regardless of the richness of our controls, estimating the causal impact of immigration on agricultural development is complicated

[28] The first railway station was completed in 1854. Online Appendix Fig. A.6 illustrates the share of municipalities connected by decade. While Brazil's railway network expanded significantly, the share of connected municipalities was smaller than that in the United States, where the share of connected counties increased from 20 percent in 1850 to about 90 percent in 1920. Similarly, Brazil's railway mileage in 1914 was equivalent to that of the United States in the 1850s (Herranz-Loncán, 2014; Summerhill, 2003; Sequeira et al., 2020).

[29] These include the number of years with railway connection, an indicator variable for municipalities with no railway connection, geographic characteristics (surface area, ruggedness, altitude, latitude, and longitude), land characteristics (the Herfindahl–Hirschman Index of crop suitability and the first two principal components of suitability for all major crops reported in the census), and market access characteristics (linear and quadratic distance to the nearest city, principal town, port, and custom house). As we explain below in discussing our instrumentation strategy, we also control for the years a municipality had been connected to the rail network (or some function of this variable) and whether it had not yet been linked by 1920.

5.2. Instrumental variables strategy

To overcome the identification challenge, we implement an instrumental variables strategy based on the interaction of immigrant inflows and the expansion of the Brazilian rail network, following Sequeira et al.'s (2020) study of the effects of immigration during the Age of Mass Migration in the United States.[32] The intuition of this instrument is to compare two otherwise identical municipalities, one of which was linked just before a year of large immigrant inflows and the other of which was linked just after. Because of the importance of rail in transporting immigrants to their destinations, they would be more likely to settle in a municipality that was linked to rail at the time of their arrival. The municipality linked to rail just before the large immigrant inflow would thus receive immigrants from that wave, while that linked in the following year would not. These initial settlement patterns would then affect subsequent settlement patterns by creating migrant networks that subsequent immigrants might follow into these destinations. For this reason, even temporary immigration—important in light of the high rate of return migration—and immigration after the initial wave into a municipality would be affected by this variation.

We construct an instrument for the immigrant share of population, s_i, which we refer to as ς_i, of the form

$$\varsigma_i = \frac{1}{\theta_i} \sum_{t=1855}^{1920} m_t \times r_{it-1},$$

where m_t is the immigration flow to Brazil in year t, normalized by Brazil's population, r_{it-1} is an indicator variable for the presence of a train station in municipality i in year $t-1$,[33] and θ_i is the number of years that municipality i had been connected to the railway network by 1920.[34]

Panel (b) of Fig. 5 presents a map of the value of the instrument for each municipality. While the geographic extent of the non-zero instrument value matches that of the railroad (by construction), as shown in panel (a), this map also makes clear that the variation across space is not precisely the same as that in panel (a)—a fact arising from incorporating variation in immigrant inflows across years into the construction of the instrument. Given the potential for heterogeneity in the predictive power of the instrument in the first stage, we interact the instrument with region indicators.[35] The first-stage regression equation for our IV strategy is thus

$$s_i = \gamma_j \varsigma_i + x_i' \Omega + \lambda_j + u_i,$$

where γ_j is a region-specific coefficient. In Online Appendix E, we also show that our results are robust to allowing γ to vary at the more disaggregated state level instead.

The controls most crucial to our identification are, following Sequeira et al. (2020), the number of years that a municipality was connected to the railway network by 1920 and an indicator variable for municipalities that were not connected by 1920. In our baseline specifications, the control for years of rail linkage in 1920 is linear, but our results are qualitatively insensitive to the use of alternative functional forms.[36] These controls address the obvious concern that rail linkage may have had direct effects on economic activity (Summerhill, 2005), and that the location of rail construction was not random, likely targeting areas where economic activity was or would be greater (e.g., Atack et al., 2010; Donaldson and Hornbeck, 2016; Zimran, 2020): among other things, these controls ensure that we do not compare municipalities linked to the rail network to municipalities not linked to the network.

As in Sequeira et al.'s (2020) and Minale et al.'s (2024) application of this identification strategy, controlling for (some function of) years of linkage and the linkage indicator implies that identification arises from the functional form restriction in which the control for years of rail linkage is linear (or otherwise constrained) whereas the instrument is a non-linear function of a municipality's years of rail linkage (since the year of linkage is the sole determinant of years of linkage).[37] But, again as in their applications, the non-linearity of the instrument is not arbitrary, but is instead the product of actual variations in Brazilian immigration rates. The combination of the control for years of linkage, the control for whether a place was ever linked, and an instrument in which the variation is determined by immigrant arrivals ensures that the identification derives not from how long a place was linked to the rail network but by the interaction of when a municipality was first linked to the railroad and the state of the national-level immigration

[32] Minale et al. (2024), Droller (2018), Panza and Zylberberg (2024) use similar identification strategies in Brazil and in other settings.

[33] That is, whether municipality i was linked in year $t-1$ or in any year previously. We focus on railway linkage in year $t-1$ rather than in year t to ensure that municipality is linked for the entire year rather than only some, potentially small, fraction. In Online Appendix D, we show that our results are robust to an alternative definition of rail connectedness in which a municipality is considered connected to the rail network if another municipality whose centroid is within 100 km of its own has a rail station.

[34] This equation captures a number of refinements that we make to Sequeira et al.'s (2020) identification strategy and some departures that we make from it. The first is that we use annual data on the state of the rail network rather than data by decade. We also make a slight departure by dividing by the number of years that a municipality was linked to the rail network (θ_i) rather than the number of years in the study period, though our results are robust to using Sequeira et al.'s (2020) normalization. Dividing by the number of years linked gives a sense of the number of immigrants that we might expect to observe at a given point in time, which is what we observe in the 1920 census. In the presence of high rates of return migration, actual immigrant shares are likely to be more reflective of annual immigrant flows; normalizing by number of years linked implies that our measure approximates an annual inflow. Another departure that we make from Sequeira et al.'s (2020) implementation is that we do not multiply the instrument by a scaling factor, which they refer to as β—the coefficient from a so-called "zero-stage" panel regression of a place's immigrant share in one period on the product of the preceding period's rail access and aggregate immigrant inflow. We must make this departure because we lack panel data on immigrant shares and thus cannot estimate the "zero-stage" equation. From a technical perspective, the factor β is immaterial to the estimation, as it is simply a constant scaling factor and thus does not contribute to the first-stage relationship between the actual 1920 immigrant share and the instrument; this departure thus has no practical impact upon our results. Although the "zero-stage" estimation has no bearing on the results in Sequeira et al.'s (2020) identification strategy, it is used in that paper to establish the relevance of the interaction between railway access and immigrant inflows in predicting immigrant shares. In the Brazilian context, this validity is established both by our first-stage estimation and in panel data for the state of São Paulo by Minale et al. (2024).

[35] Abadie et al. (2024) show that limiting attention to subsamples in which the instrument does have a strong first stage can result in bias. They also show, however, that the method that we implement, also used by Deryugina et al. (2019), Jackson et al. (2016), Dix-Carneiro and Kovak (2017), Pascali (2017), provides more reliable estimates. Abadie et al. (2024) caution that this method may be problematic where there are many different subsamples, but our case, with five regions in Brazil, is unlikely to face such issues. In Online Appendix Fig. A.8, we illustrate the relationship between the instrument and the instrumented variable at the regional level, also splitting the sample along the time dimension. The time dimension indicates that the instrument is especially effective in predicting the immigrant share of a municipality for rail linkages taking place after 1900 in all regions except for the Northeast, where there was little immigration. This helps to better understand the source of identification.

[36] Online Appendix Table A.4 presents analogs of our main results in which we control for quadratic or exponential (instead of a linear) functions of years linked, or in which we include fixed effects for each decade of linkage. The results are qualitatively unchanged relative to our main specifications, presented in Table 2.

[37] We illustrate this nonlinearity graphically in Online Appendix Fig. A.9. Our normalization by years linked to the network provides further non-linearity.

cycle—specifically, fluctuations in the immigration cycle that deviate from the functional form used to control for years of linkage—when it first became linked. More simply, identification comes from when *in the immigration cycle* a municipality first becomes linked to the rail network. For instance, two municipalities connected just one year apart may have very different predicted immigration shares if one was connected just before an immigration boom and the other just before a lull, due to the sharp discontinuity in the immigration cycle between these years.

In this framework, compliers with the instrument — which drive the local average treatment effect (LATE) that it identifies — are municipalities that received a larger immigrant share in population because they happened to be linked to the rail network just before a large immigrant inflow (and vice versa), but that otherwise would not have attracted as high a share of immigrants. Such a municipality was likely on the frontier, and may have been linked only because it happened to lie between two targeted endpoints, and also happened to become linked at just the right time in the immigration cycle. Thus, for instance, large cities or extremely productive agricultural areas that would likely have attracted immigrants regardless of the timing of rail linkage do not drive the LATE. However, as we will show below, the IV and OLS estimates are quite similar, meaning that there is reason to believe that the LATE that we identify and the average treatment effect (ATE) of immigration were not very different.

As in other applications of this instrumentation approach, the main identification assumption is that when (but not whether) a municipality was linked to the rail network was not dependent on aggregate immigration levels.[38] Thus, perhaps the main identification concern facing our instrument is that municipalities connected to the railway network just before immigration boom years may have been systematically different from those connected just before lull years. For instance, such differences might arise if, in periods where a large immigrant inflow was expected, a particular effort was made to connect particularly productive or unproductive areas to the rail network. This concern is mitigated by the fact that the historical literature has documented that these fluctuations were influenced by a number of global and national macroeconomic factors, including changes in the price of coffee, the increase in labor demand due to the abolition of slavery, the implementation of the Decreto Prinetti in 1902, the First World War, and global macroeconomic shocks such as the Panic of 1907 (Spitzer, 2015; Hatton and Williamson, 1998; Spitzer et al., 2025; Sequeira et al., 2020).

To address these concerns more formally, we follow Sequeira et al. (2020) by comparing the observable characteristics of locations that became linked just before immigration lull years to those linked just before boom years. More specifically, we use data from the 1872 Population Census of Brazil to capture early migrant settlement (population share of European immigrants), racial composition (population share of whites), human capital (literacy rate and share of children attending school), state capacity (justice workers per thousand inhabitants), economic structure (share of employment in agriculture), and the presence of slavery (population share of slaves). We operationalize this test by defining booms as years with an immigrant inflow above the previous five years' moving average and defining lulls conversely. The boom and lull years are identified in Fig. 1. Columns (1)–(3) of Table 1 present the means of the observable characteristics for the full sample and for each group of locations. Column (4) tests for the statistical significance of differences between the groups, finding that the two sets of municipalities were, in fact, systematically different. These differences, however, come with respect to geographic variables, likely because of the gradual rollout of the railway from the coast. Columns (5) and (6) repeat the analysis controlling for region and state fixed effects respectively (which we include in our analysis). In column (6) there are statistically significant differences between the two groups only in ruggedness and longitude. There is also a marginally significant difference in literacy. As we will show below, our results are robust to controlling for all of these covariates. Thus, the evidence does not support concerns that railway expansion may have responded to immigrant inflows or vice versa.

Given the prevalence of return migration in the Brazilian context, it is important to consider its potential impact on the validity and interpretation of our results. If migrants systematically remained in more productive areas and left less productive ones, this would create an artificial correlation between the 1920 immigrant share and agricultural outcomes.[39] The historical evidence provides little support for such an effect: return migration appears to have been planned rather than a response to local economic conditions (Sánchez-Alonso, 2019), and return rates were higher among immigrants in industry and commerce than in agriculture (Cinel, 1991). Moreover, our instrumental-variables strategy directly addresses this concern, since return migration does not affect the relationship of the timing of rail linkage and the immigration cycle.

Return migration does, however, affect the interpretation of our estimates. Our data enable us to study the effect of a higher share of immigrants in population in 1920. Return migration could lead this share to fail to represent total immigration to a municipality over the duration of the Age of Mass Migration, the effect of which is of interest as well. The difference between these two objects depends on where immigrants were more likely to return from and on the characteristics of those who returned. If, as the limited existing evidence suggests (Sánchez-Alonso, 2019), return migrants were positively selected, then municipalities from which they departed would have had a lower immigrant share in 1920 despite having benefited from the presence of these more productive immigrants while they resided there. But those same municipalities would also have lost the ongoing contributions of their most productive immigrants. The net effect on our estimates is therefore ambiguous. Ultimately, the lack of any systematic data on return migration does not allow us to do more than note this potential issue of interpretation.

6. Main results and robustness

Table 2 presents our main results, estimated both by OLS (panel A) and using the instrumental-variables strategy introduced above (panel B). All variables discussed in this table are standardized, meaning that the coefficients can be interpreted as standardized coefficients. Columns (1)–(4) use total farm values per hectare as the outcome, while columns (5)–(7) divide the farm value outcome into its constituent components—land, infrastructure, and tools and machines. Our OLS results reveal a statistically significant positive relationship between the share of European immigrants and farm value per hectare: a one-standard deviation increase in the immigrant share was associated with a 0.6-standard deviation increase in farm value. Adding controls for land quality (column 2) and distance to domestic and international markets (column 3) has only a negligible effect on the magnitude of the estimates and has little impact on their precision. Our results are also insensitive to the inclusion of state fixed effects (column 4). Dividing farm value into its constituent components shows that the effect of immigration was realized for all three—roughly equally for land and infrastructure, and with a somewhat smaller effect for tools and machines. Notably, that the effect goes beyond the value of land alone suggests that the results are not purely the product of greater demand for land or of local amenities capitalized into the value of immobile factors.

[38] This implies that, even if areas with greater potential for economic or agricultural development were linked earlier, our identification would not be jeopardized unless this timing was also influenced by the cyclicality of the immigration cycle.

[39] The same would be true of systematic internal migration, insofar as internal migration out of a location is, for our purposes, equivalent to a return to the country of origin in that it removes an immigrant from a municipality.

Table 1
Balance test for observables: rail connection just before immigration boom years and lull years.

	Full sample	Boom	Lull	Difference	Cond. Diff. Regions	Cond. Diff. States
	(1)	(2)	(3)	(4)	(5)	(6)
Euro share 1872	0.01	0.02	0.02	−0.00	−0.00	−0.00
Population 1872 (1,000s)	16.41	20.05	19.67	−0.38	0.23	0.21
Agricultural employment 1872	0.31	0.31	0.31	−0.00	−0.00	−0.00
Justice work. 1872	0.24	0.24	0.28	0.04	0.05	−0.03
Slave share 1872	0.14	0.17	0.15	−0.02	−0.00	−0.01
White share 1872	0.38	0.42	0.41	−0.01	−0.01	0.00
Literacy 1872	0.15	0.16	0.14	−0.02	−0.02**	−0.02*
School attendance 1872	0.13	0.13	0.14	0.01	0.01	−0.00
Distance (port/custom house)	249.70	224.13	176.08	−48.05**	−41.67**	−11.44
Distance (city)	50.15	43.83	31.16	−12.67**	−14.36***	−4.93
Ruggedness	100.95	108.42	130.68	22.27**	28.25***	32.27***
Altitude	401.77	482.36	439.08	−43.28	−10.11	11.51
Latitude	−14.25	−17.99	−17.62	0.37	0.05	−0.10
Longitude	−43.58	−44.39	−42.83	1.56**	1.22***	0.48**
Land quality	−0.01	−0.37	−0.23	0.14	0.22	0.02
Observations	605	171	71	242	242	242

Note: *** p<0.01, ** p<0.05, * p<0.1. Column (1) shows the average value of each variable for the whole sample (including municipalities never linked to the railroad). Columns (2) and (3) show the averages for the two groups of municipalities—those connected just before a boom year in aggregate immigration and those connected just before a lull year. Booms and lulls are defined and illustrated in Fig. 1. Column (4) illustrates the results of a equality of means test; columns (5) and (6) do the same conditioning on regional and state fixed effects respectively. For this exercise we use minimum comparable areas rather than municipalities. This is necessary to link data across census years, given the creation and suppression of municipalities over time. Online Appendix Fig. A.7 provides an illustration of minimum comparable areas and municipalities across the Brazilian territory created to link the 1872 and 1920 census data.

Table 2
European immigration and farm value.

	Farm Value per hectare				Land	Infrastructure	Tools & Machines
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
				Panel A: OLS			
Share Europeans	0.587***	0.599***	0.596***	0.606***	0.570***	0.599***	0.430***
	(0.057)	(0.058)	(0.057)	(0.067)	(0.066)	(0.066)	(0.055)
Observations	1,289	1,289	1,289	1,289	1,289	1,289	1,289
R^2	0.542	0.557	0.575	0.595	0.605	0.463	0.404
				Panel B: IV			
Share Europeans	0.691***	0.671***	0.733***	0.741***	0.732***	0.575***	0.749***
	(0.152)	(0.117)	(0.117)	(0.118)	(0.120)	(0.140)	(0.124)
Observations	1,289	1,289	1,289	1,289	1,289	1,289	1,289
1st stage F-stat	16.87	22.21	22.61	22.31	22.31	22.31	22.31
Railway years	✓	✓	✓	✓	✓	✓	✓
No rail	✓	✓	✓	✓	✓	✓	✓
Geo Controls	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓				
Land adaptability		✓	✓	✓	✓	✓	✓
Land quality		✓	✓	✓	✓	✓	✓
Dom market access			✓	✓	✓	✓	✓
Int market access			✓	✓	✓	✓	✓
State FE				✓	✓	✓	✓

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. The unit of observation is a municipality. All municipalities are included in the analysis. The IV regressions are estimated using the limited information maximum likelihood (LIML) estimator. The instruments are the predicted share of Europeans interacted with region indicators, which instrument for the actual share of Europeans. *Railway years* is the number of years with railway connection. *No rail* is an indicator variable for municipalities with no railway connection. *Geo controls* include surface area, ruggedness, altitude, latitude, and longitude. *Land adaptability* is the Herfindahl–Hirschman Index (HHI) of crop suitability. *Land quality* includes the first two principal components of suitability for all major crops reported in the census. *Dom market access* includes the linear and quadratic distance to the nearest city and principal town. *Int market access* includes the linear and quadratic distance to the nearest port or custom house. All continuous variables are standardized to have mean zero and standard deviation one.

Panel B of Table 2 presents our instrumental variables estimates. This panel shows, in support of the relevance of our instrument, that our first-stage F-statistics are nearly 5 times larger than the weak instrument critical value of the LIML estimator with one endogenous regressor and 5 instruments, 4.84 (Stock and Yogo, 2005; Bound et al., 1995).[40] Our IV estimates corroborate the OLS estimates of panel A, revealing a statistically significant positive effect of a greater share of European immigrants on farm values in general and on each component. In our preferred specification in column (4), which controls for state fixed effects, we find that a one-standard deviation increase in the share of immigrants yielded approximately a 0.7-standard deviation increase in farm value.

[40] We use LIML given that the critical values for this estimator decrease as the number of excluded instruments increases, a property that fits our setting well, but the results are very similar if we employ two stage least squares, as shown in Online Appendix F. Our F-statistics are also greater than thresholds for smaller numbers of instruments, which may be important given that not all of the 5 instruments may enter into the first stage significantly.

In columns (1)–(4) of Online Appendix Table A.5, we extend the analysis to the value of agricultural output per hectare of land. As above, our results reveal a positive and statistically significant relationship between the share of European immigrants in population and this measure of agricultural development; in particular, we find that a one-standard deviation increase in the share of Europeans in population was associated with a 0.3-standard deviation increase in the value of output per hectare.

Online Appendix G presents the results of a number of tests of robustness of our main results. First, Online Appendix Tables G.1 and G.2 verifiy that our results are robust to the exclusion of certain groups of municipalities from the sample. Second, Online Appendix Table G.3 shows that randomly reassigning rail linkage years within the set of ever-linked municipalities fails to replicate our results. Finally, Online Appendix Table G.4 demonstrates that our results are robust to controlling for the 1872 characteristics listed in Table 1. We also find that the 1872 control variables have little to no power to predict 1920 farm values, implying that our findings are unlikely to be driven by persistent effects of historical factors, including the replacement of enslaved labor.

In Online Appendix Table A.6, we divide the immigrant share into its constituent national origin components—Italians, Spanish, Portuguese, and others.[41] Interestingly, we find that, only Italian and Spanish immigrants appear to have had an impact on farm values. As discussed above, the Italian and Spanish immigrants were similar in terms of their propensity to enter agriculture, their reliance on immigration subsidies, and their propensity to return migration, whereas Portuguese immigrants were more likely to be unsubsidized and to enter the urban service sector. It is thus unsurprising that the Portuguese do not drive agricultural development.

In Table 3, we test whether the impact of immigration was the product of immigrant labor or of immigrant land owners in order to better understand how immigrants affected the agricultural sector.[42] Columns (1) and (2) use the same municipality-level data from the 1920 Census as in our analysis above. Column (1) shows that a greater share of European immigrants led to a greater share of farms that were foreign owned.[43] But in column (2), we find no evidence that foreign ownership was associated with greater farm values; indeed, the point estimate for the coefficient on the share of farms owned by the foreign born is negative.[44]

The 1920 census permits only the observation of the share of Europeans in a municipality's population. The 1904–1905 Agricultural Census of the State of São Paulo, however, enables us to explicitly separate native- and foreign-born agricultural workers. It also provides information on land values, the nationality of farm owners, and number of foreign and domestic workers for over 40,000 farms across 163 municipalities (Secreteria da Agricultura, Commercio e Obras Publicas do Estado de São Paulo, 1906–1910). Moreover, because the data are reported at the level of the farm, we can include municipality fixed effects to control for unobserved factors at the municipality level that could have influenced land value. Column (3) of Table 3 substantiates our main result using the farm-level data, showing that a greater share of foreign-born workers was associated with greater land values per hectare. Column (4) shows that foreign ownership had no such association. Column (5) includes both variables in the regression. While we continue to find that a greater share of foreign-born workers was associated with greater land values, the coefficient on foreign ownership is statistically significant and *negative*.[45]

This finding is in line with previous literature arguing that bourgeois immigrants in the agricultural sector—both those who started off as landowners and those who became landowners at a later stage—had no detectable generalized positive effect on Brazilian local development because of an insurmountable advantage of domestic planters, potentially arising from superior knowledge, better access to capital, or better managerial skills (Dean, 1969). It is also consistent with the notion of immigrants who had previously worked on coffee plantations accumulating savings and purchasing smaller plots (Carlson, 2022, p. 705; Holloway, 1980, p. 147; Klein, 1992, p. 510), which may have been on lower quality land because of advantages enjoyed by native planters, and which may have not been able to realize the economies of scale of a larger plantation.

7. Mechanism

We now examine the primary mechanism for the effect of European immigration on farm values per hectare. Our focus in this section is on what we identify as the most important mechanism—immigration leading to more intense cultivation of land. We also briefly discuss other potential mechanisms—developed more completely in Online Appendix H—which we find to either have not been present or to have been present but less important than the cultivation intensity mechanism. The tables discussed in this section present estimates both for the cultivation intensity mechanism and for the mechanisms discussed in the appendix.

7.1. Empirical evidence

We find that the most important mechanism for the effect of immigration concerns land use, specifically cultivation intensity, which we capture with the share of farmland cultivated (as opposed to being left fallow or forest). Column (1) of Table 4 presents the results of our instrumental variables regression of the share of farm land cultivated on the European immigrant share of population. It shows that a one-standard deviation increase in the population share of European immigrants increased the share of farm land cultivated by 0.8 standard deviations.[46] Column (2) of Table 5 repeats our original IV regression

[41] Each column is a single regression. Because our IV and OLS estimates are largely similar, and because additional instruments would be required to separately estimate the effects of each nationality's share of population, we focus here only on OLS estimates. A map of the largest immigrant group by municipality is presented in Online Appendix Fig. A.4. Online Appendix Table A.7 presents analogous results to our main analysis in Table 2, in which we use immigrants from all places of origin, rather than just Europe. The results are qualitatively the same as those of the main analysis.

[42] Online Appendix Fig. A.10 presents the spatial distribution of foreign farm ownership.

[43] About 12 percent of nationwide agricultural establishments (Directoria Geral de Estatística, 1922, Volume III, 1ª Parte: Agricultura, p. XXVIII) and 8.6 percent in the average municipality (Online Appendix Table A.1) were foreign owned.

[44] We perform this exercise while recognizing the potential endogeneity of the share of farms owed by the foreign born in addition to the European share of population.

[45] In Online Appendix Table A.8, we show that the pattern, that it is the share Europeans and not the share foreign-owned farms that has explanatory power, holds also for output and for individual aspects of farm value.

[46] A natural concern is that, rather than being the product of immigration, increased cultivation intensity was a factor that attracted immigration. A related concern is that immigrants disproportionately moved to areas already on a path of agricultural expansion. While such sorting may have occurred, our identification strategy ensures that such issues do not drive our conclusions. In particular, the results of Table 4 are based on the use of our instrumental variables approach. If, as we have argued and demonstrated above, there was not a correlation between timing of rail linkage and the aggregate immigration cycle, then the results of this table indicate that there truly was an effect of immigration on cultivation intensity. Moreover, we show in Online Appendix Table A.9 that this finding is robust to adding 1872 controls. That said, we cannot rule out that there was also a degree of a relationship in the opposite direction, though the mere presence of such a relationship would not invalidate our results. However, the lack of an effect of immigration on population density or agricultural employment density, discussed below, suggests that a mechanism in which rising local demand spurred increased cultivation is unlikely to have been present.

Table 3
Farm values and foreign ownership.

	Municipality-level data		*Farm-level data*			
	Foreign-owned farms (1)	Farm value per hectare (2)		Land value per hectare (3)	(4)	(5)
Share Europeans	0.814***	0.890**	Share for. work.	0.159***		0.199***
	(0.069)	(0.355)		(0.024)		(0.025)
For.-owned farms		−0.292	For. owner		0.0656	−0.177***
		(0.268)			(0.053)	(0.061)
Observations	1,289	1,289		40,693	40,693	40,693
1st stage F-stat	22.31	9.71	R^2	0.395	0.380	0.398
All controls	✓	✓				
Municipality FE				✓	✓	✓

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. Regressions in columns (1) and (2) are estimated by instrumental variables using the limited information maximum likelihood (LIML) estimator. The instruments are the predicted share of Europeans interacted with region indicators, which instrument for the actual share of Europeans. All municipalities are included in the analysis. Regressions in columns (3) to (5) are estimated by ordinary least squares (OLS) using data for the state of São Paulo. *All controls* include the number of years with railway connection, an indicator variable for municipalities with no railway connection, geographic characteristics (surface area, ruggedness, altitude, latitude, and longitude), land characteristics (the Herfindahl–Hirschman Index of crop suitability and the first two principal components of suitability for all major crops reported in the census), market access characteristics (linear and quadratic distance to the nearest city, principal town, port, and custom house), and state fixed effects. All continuous variables are standardized to have mean zero and standard deviation one.

Table 4
Potential mechanisms and European immigration.

	Cultivated share of farms (1)	Coffee share (2)	Cash crops share (no coffee) (3)	Population density (4)	Agricultural emp. density (5)	Share farms with tools (6)	Share farms with machines (7)
Share Europeans 1920	0.824***	1.027***	0.024	−0.232	−0.094	0.701***	0.046
	(0.183)	(0.205)	(0.119)	(0.182)	(0.172)	(0.111)	(0.144)
Observations	1,287	1,285	1,285	1,296	1,296	1,289	1,289
1st stage F-stat	22.29	22.26	22.26	22.35	22.35	22.31	22.31
All controls	✓	✓	✓	✓	✓	✓	✓

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. The unit of observation is a municipality. All municipalities are included in the analysis. All regressions are estimated by instrumental variables using the limited information maximum likelihood (LIML) estimator. The instruments are the predicted share of Europeans interacted with region indicators, which instrument for the actual share of Europeans. *All controls* include the number of years with railway connection, an indicator variable for municipalities with no railway connection, geographic characteristics (surface area, ruggedness, altitude, latitude, and longitude), land characteristics (the Herfindahl–Hirschman Index of crop suitability and the first two principal components of suitability for all major crops reported in the census), market access characteristics (linear and quadratic distance to the nearest city, principal town, port, and custom house), and state fixed effects. All continuous variables are standardized to have mean zero and standard deviation one.

of farm value on the European share of population, but controls for the share of farmland cultivated.[47] This factor was associated with greater farm values—a one-standard deviation increase in the share cultivated was associated with a 0.4-standard deviation increase in farm value per hectare—and its inclusion leads to a substantial decline in the magnitude of the European share coefficient from a coefficient of over 0.7 to about 0.5, which constitutes evidence that this is a mechanism through which the effect of the immigration share passed. This is confirmed by the Gelbach (2016) decomposition in Table 6, which shows that it was the cultivated share of farms that was responsible for the bulk of the reduction in the effect of the population share of European immigrants on farm values when controls are added.[48] Based on these results, we conclude that European immigrants led a greater share of farm land to be cultivated, raising its value.

The results of columns (5)–(8) of Online Appendix Table A.5 provide further support for this mechanism. Whereas columns (1)–(4) find a strong positive effect of the population share of European immigrants on the value of agricultural production per hectare, columns (5)–(8) reveal no effect (or perhaps even a slight negative effect) on the value of agricultural production per *cultivated* hectare. This result is consistent with an increase in cultivation driving the effects that we document—when we condition on the extent of cultivation, there is no effect on output.

7.2. Discussion

The presence of a cultivation intensity mechanism is consistent with a number of aspects of Brazilian immigration, as discussed in Section 2, and with some of the potential mechanisms discussed in Section 3. In particular, the high rates of return migration and goal of accumulating savings to take back to the home country may have driven this increase. Efforts to accumulate savings in order to transition out of plantation work and into independent land ownership would have operated similarly.[49] Moreover, the unique features of the *colonato* contract also incentivized more intensive cultivation of the land, as discussed above.[50] The impacts of these factors would have been realized in the short term, before immigrants' returns to their home countries

[47] We do this while recognizing that potential endogeneity of the share of farmland cultivated is not addressed in this regression. However, having established a causal relationship between immigration and cultivation intensity in Table 4, we are not concerned that we are capturing a spurious relationship.

[48] Because of technical difficulties arising from our use of multiple instruments the decomposition is performed for the OLS analog of the regressions, not for the IV estimates, so the decomposition is not precise. But given the similarity of the OLS and IV results, this is not a major concern.

[49] This interpretation can explain both the results from the cultivation intensity mechanism and are consistent with our findings in Table 3 that land managed by immigrants was less valuable than that managed by natives—many of the incentives enumerated above would have faded when immigrants became landowners.

Table 5
Farm value and potential mechanisms.

	Farm Value per hectare					
	(1)	(2)	(3)	(4)	(5)	(6)
Share Europeans	0.726***	0.508***	0.521***	0.745***	0.577***	0.268*
	(0.118)	(0.153)	(0.143)	(0.102)	(0.142)	(0.143)
Panel A: Land use						
Cultivated share		0.396***				0.411***
		(0.065)				(0.063)
Coffee share			0.236***			0.064**
			(0.046)			(0.033)
Other cash crops share			0.132***			0.090***
			(0.025)			(0.021)
Panel B: Labor force						
Pop density				0.138**		0.255***
				(0.060)		(0.069)
Agr. emp. density				0.150***		−0.047
				(0.046)		(0.040)
Panel C: Tools & Machines						
Share farms with tools					0.022	0.004
					(0.019)	(0.016)
Share farms with machines					0.189***	0.189***
					(0.050)	(0.042)
Observations	1,232	1,232	1,232	1,232	1,232	1,232
1st stage F-stat	22.18	15.64	15.64	24.63	16.50	11.40
All controls	✓	✓	✓	✓	✓	✓

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. The unit of observation is a municipality. All municipalities are included in the analysis. All regressions are estimated by instrumental variables using the limited information maximum likelihood (LIML) estimator. The instruments are the predicted share of Europeans interacted with region indicators, which instrument for the actual share of Europeans. *All controls* include the number of years with railway connection, an indicator variable for municipalities with no railway connection, geographic characteristics (surface area, ruggedness, altitude, latitude, and longitude), land characteristics (the Herfindahl–Hirschman Index of crop suitability and the first two principal components of suitability for all major crops reported in the census), market access characteristics (linear and quadratic distance to the nearest city, principal town, port, and custom house), and state fixed effects. All continuous variables are standardized to have mean zero and standard deviation one.

or transitions into land ownership. But the impacts would also have been persistent. For instance, land cleared to be brought into cultivation would have then been available for future production; indeed the new coffee trees planted there would continue to be productive for many years thereafter. We do not claim that these are the only ways in which immigration drove increases in cultivation intensity; other aspects of immigration may also have been responsible. Instead, these interpretations demonstrate that the mechanism that we propose is plausible in our context.[51]

7.3. Other mechanisms

In Online Appendix H, we consider three additional mechanisms that we find to have not been present or to have been unimportant relative to the cultivation intensity mechanism. First, we find no evidence that a greater share of European immigrants was associated with greater population density or agricultural employment density (Table 4, columns 4 and 5), ruling out a mechanism in which immigrants were uniquely responsible for the provision of labor on the agricultural frontier or caused an increase in the local demand for land or agricultural products. The null effects of the share of immigrants in population on population density or the density of agricultural labor are surprising in light of the general view (e.g., Klein, 1995) that immigration was a crucial source of labor for the expanding agricultural frontier. Our interpretation of our results is that, while immigrants may have increased the size of Brazil's aggregate labor force, they "crowded out" natives, either by causing them to move elsewhere or in an indirect sense by deterring them from settling in areas on the expanding frontier. The results may also be the product of immigrants moving to the expanding frontier themselves to acquire land after accumulating sufficient savings (Carlson, 2022, p. 706).

Next, we test for a mechanism in which immigration led to a shift in the crop mix, in particular towards greater coffee production. While we do find that a greater immigrant share led to more coffee cultivation (Table 4, column 2), and that this increase in coffee cultivation was in part responsible for growing farm values, the magnitude of this mechanism is small relative to the cultivation intensity mechanism, as shown by the decomposition in Table 6.

Finally, we test whether immigration led to an increase in adoption of tools or machines in agriculture. We find evidence of an effect of immigration on tool adoption but not on machine adoption (Table 4, columns 6 and 7), of machine adoption on farm values (Table 5 columns 5 and 6), and of tool adoption on the coefficient on the population share of European immigrants (Table 6). But we also find that the magnitude of this mechanism was small relative to the cultivation intensity mechanism. In other contexts, such as the United States, there is evidence that immigrant labor was a substitute for mechanization

[50] Online Appendix Table A.10 studies the relationship between the share of Europeans and the share of foreign-owned farms on the one hand and measures of land utilization on the other. In general, we find that, as in Table 3 and Online Appendix Table A.8, it is the share of Europeans rather than the share of foreign-owned farms that has explanatory power for these outcomes. In Panel A, where we study the use of total farm land, we find that, besides a higher share of Europeans being associated with more farm land devoted to overall cultivation, it is also specifically related to more land being devoted to coffee, cash crops, and food crop production. Though not dispositive, this result is consistent with an interpretation in which more intense cultivation is driven in part by production of additional crops as part of an interrow cropping strategy. Panel B focuses on land use out of cultivated land, showing that a higher share of Europeans was associated with higher share of land devoted to coffee and less to food crops.

[51] This plausibility is enhanced by the fact that, as we have shown above, the bulk of the effects of immigration was the product of Italian immigrants, for whom the incentives described below were most salient.

in agriculture (e.g., Abramitzky et al., 2023). That we find no such effect in Brazil, and instead find an increase in the adoption of a complementary input—tools—is notable.

8. Implications for development and structural change

As outlined in Section 3, there are a variety of possible implications of agricultural development for the rest of the Brazilian economy and its transition to industrialization and sustained economic growth beyond the primary sector. The effects of immigration-driven agricultural development on these outcomes, in combination with immigration's direct effect, may have led immigration to impact Brazil's structural transformation. In this section, we ask whether immigration's effect on agricultural development was accompanied by effects on other sectors—either direct or as consequences of the effect on agriculture—that would limit Brazil's structural transformation. To be clear, we are not aiming to determine the effect of agricultural development driven by immigration on structural transformation; instead, our goal is to measure the net effect of immigration.

An important caveat is that we can only answer this question in the relatively short run, since we observe our outcomes shortly after the end of mass migration. In principle, development and structural transformation may have continued along the same lines or changed direction in unpredictable ways in the decades after our outcomes are observed, for instance due to a lag in the manifestation of the effects of migration through channels like human capital accumulation. At the same time, our shorter-term analysis has the advantage of not being influenced by subsequent events, such as the introduction of import substitution policies and the many regime changes experienced by Brazil from the 1930s until the 1980s, which had their own, sometimes substantial, economic consequences (Ferraz et al., 2024).

In Table 7, we implement our IV strategy for a variety of outcomes that capture differences in human capital formation (Panel A), economic structure (Panel B), economic specialization (Panel C), political economy (Panel D), and sex ratios (Panel E). For each set of variables, we present two specifications—one including the whole sample and another in which we exclude large population centers (municipalities in the top quartile of Brazil's population distribution). On the whole, we find no evidence that migration negatively affected other parts of the economy, or that it slowed down structural transformation. In fact there are indications that the reverse may have been true.

In Online Appendix Table A.11 we try to distinguish to some extent between the direct effects of immigration on structural transformation and the indirect effect, through agricultural development. In particular, we include the outcome of our main analysis (farm values) as a control in the structural transformation regressions of Table 7. The pattern that clearly emerges is that the effects of immigration on structural change and broader economic development are not mediated by the positive effects of immigration on the agricultural sector, as evidenced by the almost universally insignificant coefficients for farm values. At the same time, the estimated effects of immigration on broader development remain qualitatively unchanged even after including farm values.[52] That is, immigration had parallel effects on agricultural development and structural transformation, but the effect on structural transformation did not depend on the effect on agriculture.

8.1. Human capital

We capture human capital by analyzing the literacy rates of the whole population, of natives, of females (both native and foreign born), and of children between the ages of 7 and 14 (both native and foreign born). These variables are intended to capture spillovers from migrants to the rest of the population. We find in panel A of Table 7 that European immigration had a strong positive effect on all of our literacy measures: a one-standard deviation increase in the share of migrants in the population led to a 0.7–0.8-standard deviation increase in each of the measures of literacy. The positive link between immigration and human capital found in previous work is thus confirmed. This is not surprising: immigrants had greater human capital to begin with and they were often successful in lobbying for the creation of schools (de Carvalho Filho and Colistete, 2010). These schools were presumably one of the channels through which the provision of education improved at the local level, leading to higher literacy for the overall population.

8.2. Economic structure

Next, we focus on employment shares by sector, and specifically on the share of the labor force employed in agriculture, in industry, and in services. These are standard ways to measure structural transformation, as they capture the reallocation of labor from traditional to modern economic activities (e.g., Caselli and Coleman, 2001; Heldring et al., 2021). We operationalize these measures by using the share of the labor force employed in agriculture, industry, and services (excluding domestic service). These measures give an indication of the degree to which immigration aided or slowed down structural change.

We find in Panel B of Table 7 that municipalities with a higher share of immigrants in population had a lower share of workers employed in agriculture, even when excluding large urban centers. That is, our results on the whole point to a valuable agricultural sector that occupied a lower share of the labor force. This supports the view of agricultural development as a driver of overall economic development. We also find suggestive evidence that a greater immigrant share led to a higher share of workers employed in industry and services: the coefficients for industrial and services employment are positive, though not statistically significant. In any case, we find no evidence that immigration *reduced* industrial or service employment, which would indicate a slowing of structural change.

8.3. Economic specialization

To delve further in the effect of immigration on structural change and to account for skewness in the distribution of the share of employment in agriculture, industry and services, we classify certain areas as hubs of agriculture, industry, or services, defined as being in the upper quartile of the distribution of such employment across municipalities. We find that a higher share of Europeans generated a statistically significantly lower probability of being an agricultural hub in both the full and the non-urban samples (Panel C). Conversely, the probability of being an industry hub increases with the European share of the population, though the coefficient is not statistically significant when urban areas are excluded from the sample. For service hubs, instead, we find positive and statistically significant results, both in the full sample and in the non-urban sample. This set of results is consistent with immigration accelerating structural transformation in Brazil.

8.4. Political economy

We also study the effect of immigration on two variables capturing the presence and influence of local landed elites (Panel D). The first is the share of population made up of rentiers—people relying on returns to wealth as their main source of income. The second is the share of the population employed in public administration, capturing the tendency for local oligarchs to exert their influence by placing loyalists in public administration (Graham, 1990). Indeed, many parts of Brazil were characterized by a large-plantation-based agricultural sector and connected oligarchic system known as *coronelismo*, in which

[52] Because our identification strategy does not provide instruments for both immigration and agricultural development, we are not able to rule out the possibility that immigration may have had an even stronger impact on structural transformation if it had not also driven agricultural development. This is an interesting area of exploration for future work.

Public Disclosure Authorized

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Policy Research Working Paper 11444

Evaluating Spatial Underperformance

A Spatial Computable General Equilibrium Model Applied to Ho Chi Minh City

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September 2026

Policy Research Working Paper 11444

Abstract

Thu Duc, a rapidly urbanizing area of Ho Chi Minh City, is well located and heavily invested in policy ambition, yet its output and density remain far below what its position in the metropolitan economy would support. Much of it is also flood exposed. This paper asks how much of that shortfall flood risk explains, and what coordinated investment could recover. A Spatial Computable General Equilibrium model of the Greater Ho Chi Minh City Metropolitan Area, covering 116 cells across Ho Chi Minh City and neighboring provinces, evaluates interventions to 2040. The model endogenizes firm location, household residence and commuting, land use intensity, and agglomeration externalities, recovering cell-level productivity and amenity parameters by structural calibration against observed employment and residential distributions. Its methodological contribution is to integrate spatially explicit flood exposure as a determinant of those fundamentals, so flood risk is evaluated within the same general equilibrium framework as transport and land use, and local gains can be separated into net new metropolitan output and activity reallocated from elsewhere. Thu Duc's calibrated productivity is roughly one-third of the metropolitan average, implying that frontier output is more than five times the business-as-usual level. Flood mitigation raises sectoral total factor productivity by 12 to 21 percent and residential amenity by 10 to 13 percent, and it broadens the sectoral employment base more than the other interventions examined. A cumulative package of innovation zones, flood mitigation, and transport reaches about 31 percent of that frontier, raising output roughly two-thirds above business as usual, of which some 56 percent is net new output for the metropolis.

This paper is a product of the Urban, Disaster Risk Management, Resilience and Land Global Department. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at http://www.worldbank.org/prwp. The authors may be contacted at srubinyi@worldbank.org.

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Produced by the Research Support Team

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Keywords: Spatial CGE; urban economics; Viet Nam; Ho Chi Minh City; flood risk; agglomeration; transport infrastructure

JEL Codes: O18, R11, R13, R14, Q54

Funding: This research working paper was funded by the Swiss Confederation through the State Secretariat for Economic Affairs (SECO).

1. Introduction

Cities are the principal sites of productivity growth and income convergence in modern economies, yet the productivity realized in any given city is often well short of what its size, location, and endowments would support. Part of this gap reflects underperformance at the level of specific areas within the city: places whose output, density, and amenities fall systematically below the metropolitan frontier because overlapping frictions suppress the returns to spatial concentration. Diagnosing the sources of such spatial underperformance, ranking the interventions that could respond to them, and quantifying the scale of the impact on the metropolitan economy are central questions for urban and development economics. They are also the questions that conventional project appraisal is least equipped to answer. Evaluating a transport investment, a flood management scheme, or a land-use reform in isolation produces estimates of direct benefits within each domain. Insofar, it does not quantify how these interventions compound, how they reallocate activity across space, or how the local gains decompose into new output and activity drawn from elsewhere within the metropolis.

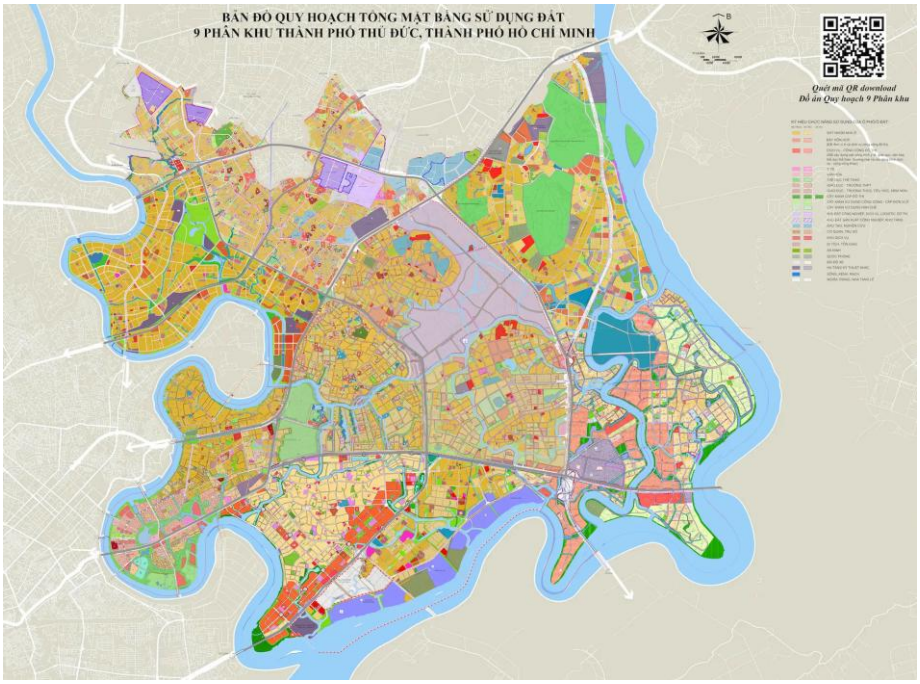
Figure 1. Thu Duc in the Greater Ho Chi Minh City as the study area

Thu Duc, an area in the east of Ho Chi Minh City (HCMC), Viet Nam, is a setting in which these questions can be studied with unusual precision.[1] The former Thu Duc City's masterplan designates the area as a

[1] Thu Duc operated as a city-within-a-city from 2021 until Viet Nam's July 2025 administrative restructuring, which abolished the district tier and enlarged HCMC through the merger of HCMC with the former provinces of Binh Duong and Ba Ria-Vung Tau. In this reform, the former Thu Duc City's 34 wards were consolidated into 12 new

future hub of higher education, research, high-tech manufacturing, and finance, with targets of 2.2 million to 2.64 million residents by 2040 (Government of Viet Nam, 2025) and initially envisaged to contribute 30 percent of HCMC's economic output (HCMC Party Committee, 2020), equivalent to approximately US$22 billion in present terms. The area contains genuine assets on which such an ambition can be built, including the Saigon Hi-Tech Park, Viet Nam National University HCMC, and established industrial and logistics functions that anchor it into the production networks of the wider metropolitan region (Figure 2). Yet it is substantially underdeveloped relative to these foundations. Employment density, population density, and local GVA all fall well below what proximity to the central business district and the existing asset base would predict.

Figure 2. Masterplan map for the land use of 9 sub-zones in the former Thu Duc City



Source: HCMC Department of Planning and Architecture.

Three constraints account for much of the gap. Thu Duc faces elevated flood risk from urbanization, climate change, and land subsidence, which has left large tracts undeveloped. Transport connectivity to the rest of HCMC is limited by the Saigon and Dong Nai rivers and a thin network of crossings. The productivity and amenities are substantially below the metropolitan average, suggesting a lack of supporting infrastructure and services.[2] These constraints are complementary in the technical sense that the returns to relaxing one depend on the state of the others. Flood risk suppresses the land values on which densification depends. Weak connectivity narrows the labor market catchment over which firms can draw and the set of destinations to which residents can commute. Lower density, in turn,

wards in the eastern part of the enlarged HCMC. "Thu Duc" is used throughout this paper as a spatial term referring to this eastern area, corresponding to the former Thu Duc City, which remains the relevant scale for analysis and for the plans, datasets, and policy documents that continue to reference it. This should not be confused with the new Thu Duc Ward, which is one of the 12 post-reform wards.

[2] Based on the parameters recovered from modeling the spatial distribution of employment and population, which is discussed in this paper's results section.

suppresses the Marshallian externalities through which the agglomeration premium is realized. Evaluated one at a time, as in conventional project appraisal, these frictions appear as separable engineering, transport, or land-use problems. Evaluated jointly, they compound.

This paper develops a Spatial Computable General Equilibrium (SCGE) model designed to quantify the gains of simultaneously applying these interventions in Thu Duc, while also measuring the impact on HCMC and its surrounding provinces. The model covers 116 cells across the metropolitan area. It endogenizes firm location across three production sectors, household residential and workplace choice across two skill groups under a discrete-choice logit specification, endogenous floor space supply, commuting and trade costs calibrated from a travel-time matrix, and Marshallian agglomeration externalities. Cell-level productivity and amenity parameters are recovered through structural calibration against observed 2019 employment and residential distributions. The paper extends the SCGE model developed for Dhaka by Bird and Venables (2019) in one central respect: spatially explicit flood exposure is integrated as a determinant of calibrated productivity and amenity, estimated from cross-sectional variation across the 116 cells, and then propagated through the full equilibrium response to flood mitigation. This places flood risk on a comparable analytical footing with transport and land-use interventions, evaluated through the same general equilibrium apparatus rather than through a separate engineering cost-benefit exercise.

The analysis proceeds in two stages. A hypothetical benchmark first asks what level of economic activity Thu Duc could sustain if its productivity and amenity parameters converged to the HCMC average. This quantifies the frontier of what the area could support and establishes the gap that transformation would need to close. A progression of realistic policy scenarios then evaluates how much of that gap a cumulative package of innovation zones, flood risk mitigation, transport improvements, and stronger agglomeration economies can close. The results, reported in full in Section 6, point to flood risk mitigation as the largest single contributor among the tractable interventions when they are evaluated in the cumulative order used here, to a substantial share of the resulting local gains taking the form of net new metropolitan output rather than reallocation across space, and to a residual gap that infrastructure alone cannot close. Taken together, the analysis suggests that flood risk is not merely one friction among many but a first-order determinant of whether flood-exposed parts of a city can realize their locational potential at all.

The paper is structured as follows. Section 2 situates the analysis within three bodies of literature: flood risk as a determinant of the urban spatial equilibrium, agglomeration economics and the formation of new urban centers, and quantitative spatial models. Section 3 sets out the paper's contribution. Section 4 presents the model. Section 5 describes the data, calibration, and parameters. Section 6 presents results. Section 7 discusses caveats and limitations. Section 8 concludes.

2. Literature Review

This paper sits at the intersection of three bodies of literature: the economics of flood risk in cities and its implications for the urban spatial equilibrium; agglomeration economics and the question of how new urban centers form; and the literature on quantitative spatial models, which provides the methodological apparatus used here.

2.1 Flood risks and the urban spatial equilibrium

Flood risk has traditionally been treated in applied economics as a damage phenomenon: a probability distribution over destructive events whose expected cost can be estimated and, in principle, offset through investment in protective infrastructure. The analytical tool of choice for evaluating such

The second contribution is empirical. The paper provides, to our knowledge, the first general-equilibrium evaluation of a metropolitan-scale coordinated flood mitigation package of the scale actually under consideration by policy makers in a flood-exposed city. The analysis traces the effects of a specific mitigation package through firm location, household residential and workplace choice, land use intensity, commuting and trade flows, and agglomeration externalities, reporting magnitudes at the metropolitan scale and at the scale of the specific area where mitigation is applied. This provides a benchmark for comparison with conventional engineering cost-benefit appraisal of the same class of investment and quantifies the scale of returns that the latter approach systematically does not capture.

The third contribution speaks to the literature on place-based policy. Reduced-form evaluations of place-based interventions can credibly identify local effects but have greater difficulty distinguishing net metropolitan gains from spatial reallocation of activity (Neumark and Simpson, 2015). The structural model used here makes this decomposition directly. Under the paper's cumulative realistic scenario, approximately 56 percent of the local output gain in the targeted area represents net new output for the metropolitan economy, with the remainder reflecting reallocation from other parts of the metropolis. This proportion is informative about the degree to which place-based investment produces genuine aggregate gains rather than spatially redistributed ones, and the analytical tool that produces it is applicable in other settings where the same decomposition is needed for policy appraisal.

The fourth contribution is the transferability of the methodological approach. The extended SCGE model developed here requires cell-level employment, residential, and flood exposure data, a travel-time matrix, and standard input-output structure. Where these are available, the model can be recalibrated to evaluate coordinated infrastructure and climate adaptation packages in other cities facing the same class of constraint: spatial underperformance driven by interacting flood, connectivity, and productivity-amenity frictions. This positions the paper as a demonstration of a methodology applicable beyond the specific case it evaluates, not only as a case study of Ho Chi Minh City.

4. Model Description

4.1 Overview

The model consists of three core components. First, it represents the geography of the Greater Ho Chi Minh City Metropolitan Area, which is divided, for analytical purposes, into $N = 116$ distinct geographic cells linked through a transport network. Economic interactions beyond these cells, with the rest of Viet Nam and the rest of the world, are described as imports and exports.

Second, the model incorporates the production side of the economy, comprising three sectors: (1) tradable business services, (2) local services including retail, and (3) land-intensive tradable manufacturing. Firms in each sector choose where to operate, how much labor to employ, and how much output to produce. In addition, the construction sector works with landowners to determine the intensity of land use in each cell.

Third, the model includes the household and labor supply side, which is divided into two household types: low-skilled ($k = 1$) and high-skilled ($k = 2$), distinguished by education level. Households choose both their residential location and their place of work, generating commuting flows across the city.

The model solves for the set of prices, wages, land rents, output, and employment levels that jointly coordinate these elements into an economic equilibrium.

4.2 Production

Technology in each sector s and cell i is Cobb-Douglas with Hicks-neutral productivity A_i^s and using four inputs to produce output x_i^s. The inputs are floor space with price q_i, skilled and unskilled labor with wage rates $w_i^k, k = 1, 2$, and imported intermediates with price p^f (the same in all cells). The price of output, p_i^s, equals unit cost in each sector and cell:

$$p_i^s = (q_i)^{\beta^{0s}} \left(w_i^1\right)^{\beta^{1s}} \left(w_i^2\right)^{\beta^{2s}} \left(p^f\right)^{\beta^{fs}} / A_i^s, \quad s = 1, 2, 3, \quad i = 1 \ldots N, \tag{1}$$

with $\beta^{0s} + \beta^{1s} + \beta^{2s} + \beta^{fs} = 1$. Denoting the quantity of output produced in sector s cell i by x_i^s, the value of the derived demand in cell i for each of the inputs is

$$w_i^k L_i^k = \sum_s \beta^{ks} p_i^s x_i^s, \quad k = 1, 2, \tag{2}$$

$$q_i z_i = \sum_s \beta^{0s} p_i^s x_i^s, \tag{3}$$

$$p^f I_i^f = \sum_s \beta^{fs} p_i^s x_i^s, \tag{4}$$

where L_i^k, z_i, I_i^f are respectively quantities of labour, land and intermediates.

4.3 Household utility, location choice, and labor market equilibrium

The number of households of each type, $\bar{L}_k$, is assumed to be exogenous to the model. The utility of a household of skill level k residing in cell i and working in cell j is:

$$u_{ij}^k = w_j^k \tau_{ij} B_i^k \left\{ (q_i)^{\alpha^0} \left(P_i^1\right)^{\alpha^1} \left(P_i^2\right)^{\alpha^2} \left(P_i^3\right)^{\alpha^3} \left(p^f\right)^{\alpha^f} \right\}^{-1}, \quad k = 1, 2 \tag{5}$$

Working in cell j gives wage income w_j^k, of which fraction $\tau_{ij} \leq 1$ remains after the cost of commuting from cell of residence, i. B_i^k is the residential amenity of place i for a household of type k. The denominator of (5) is the cost of living in i, depending on the price of floor place, q_i, price indices of the different types of output, P_i^s, and the price of imports, p^f. The exponents are Cobb-Douglas expenditure shares, summing to unity.

The price indices are constructed using the 'Armington' assumption, under which each cell produces a different variety of each product type. The indices are CES aggregates of the prices p_i^s of sector s varieties produced in cells i and sold at price $T_{ij}^s p_i^s$ in cell j

$$P_j^s = \left(\sum_i \left(T_{ij}^s p_i^s \right)^{1-\sigma^s} \right)^{\frac{1}{1-\sigma^s}}, \quad s = 1, 2, 3. \tag{6}$$

In this expression σ^s is the elasticity of substitution between sector s varieties, and T_{ij}^s is the cost of shipping from place of production, i, to each place of consumption, j. Foreign goods have fixed relative prices, the same price in all domestic cells, so the price index takes the simple form $P^f = p^f$.

Each household chooses where to live and where to work. This follows a logit-type specification in which the proportion of the type k population that chooses live in cell i and work in cell j is λ_{ij}^k, given by

$$\lambda_{ij}^k = \frac{\left(u_{ij}^k\right)^\epsilon}{\Sigma_i \Sigma_j \left(u_{ij}^k\right)^\epsilon} \quad \epsilon > 1. \tag{7}$$

The parameter ϵ measure the responsiveness of choice to differences in utility; micro-foundations of this approach are given in Ahlfeldt et al. (2015). Population resident in i, R_i^k, and employed in j, L_j^k are then

$$R_i^k = \bar{L}^k \sum_j \lambda_{ij}^k, \qquad L_j^k = \bar{L}^k \sum_i \lambda_{ij}^k, \quad k = 1, 2, \tag{8}$$

(so $\sum_i \sum_j \lambda_{ij}^k = 1$ and $\sum_i R_i^k = \sum_j L_j^k = \bar{L}^k$).

The equality of labor supply and demand (of each type in each place) is therefore, from (2) and (7),

$$\sum_s \beta^{ks} p_i^s x_i^s / w_i^k = \bar{L}^k \sum_i \lambda_{ij}^k . \tag{9}$$

The wage income of type k residents in cell i (accounting for commuting to j) is $\sum_j \lambda_{ij}^k w_j^k$, and total wage income in cell i is Y_i,

$$Y_i \equiv \sum_k \bar{L}^k \sum_j \lambda_{ij}^k w_j^k, \quad k = 1, 2. \tag{10}$$

4.4 Floor space, construction, and land market equilibrium

Usable land in each cell is denoted $\bar{M}_i$ and gets built to density (e.g. height) h_i to yield floor space $H_i = \bar{M}_i h_i$. Density is chosen to maximise the value of floor space per unit land $q_i h_i$ minus the cost of construction, this assumed to be iso-elastic, taking the form κh_i^ν, increasing and convex in density, $\nu > 1$. The density that maximises this objective is $h_i = (q_i/\kappa\nu)^{1/(\nu-1)}$ so supply of floor space in cell i is:

$$H_i = \bar{M}_i (q_i/\kappa\nu)^{1/(\nu-1)} . \tag{11}$$

Revenue from floor space per unit land is part construction cost and part pure rent, r_i, i.e. $q_i h_i = \kappa h_i^\nu + r_i$. With iso-elastic construction cost function share $1/\nu$ goes to construction costs and the remainder, $(\nu - 1)/\nu$ to pure rent. Expenditure on construction for the whole of cell i is therefore, $q_i H_i/\nu$, and pure rents are $r_i H_i = q_i H_i (\nu - 1)/\nu$. We assume that pure rents are spent by absentee landlords outside the city. Construction costs are spent on goods produced in the city and imported, using labor only in so far as it is embodied in these goods. This spending is distributed across sectors and imports with shares γ^s, so construction spending on sector s in cell i, denoted C_i^s, is

$$C_i^s = \gamma^s q_i H_i/\nu, \quad s = 1, 2, 3, f, \tag{12}$$

with shares $\gamma^1, \gamma^2, \gamma^3, \gamma^f$, summing to unity.

Demand for floor space in cell i comes from firms (equation 3) and from households, whose income is Y_i. With Cobb-Douglas preferences share α^0 of income is spent on housing, so the equality of supply and demand gives market clearing price of floor space as q_i solving

$$q_i H_i = \alpha^0 Y_i + q_i z_i. \tag{13}$$

4.5 Goods markets and trade

Expenditure on goods and services comes from households, construction, and exports. Expenditure in cell i takes the form

$$E_i^s = \alpha^s Y_i + C_i^s + F_i^s, \qquad s = 0, 1, 2, 3, f, \qquad i = 1 \ldots N. \tag{14}$$

The first term is share α^s of household income, this following from preferences, equation (5). The second is expenditure on goods used in construction, equation (12). The third is the value of exports, this depending on price P_i^s and the exogenous level of foreign demand, $expt^s$, so taking the form

$$F_i^s = (P_i^s)^{1-\eta^s} expt^s, \quad s = 0, 1, 2, 3, \tag{15}$$

where η^s the price elasticity of external demand for product s.

These expenditures in each cell, E_i^s, are supplied by varieties produced in different cells, and this is done using the 'Armington' assumption, as represented by the price indices equations (6). With this characterization of preferences, demand for products of each sector in each place, x_i^s, are given by

$$x_i^s = (p_i^s)^{-\sigma^s} \sum_j \left(T_{ij}^s\right)^{1-\sigma^s} \left(P_j^s\right)^{\sigma^s - 1} E_j^s, \quad s = 1, 2, 3. \tag{16}$$

Goods market clearing holds as these quantities x_i^s are met by supply, generating the demand for inputs given in equations (2)- (4). Two points are noteworthy. First, x_i^s is demand for products shipped, and these exceed quantities delivered by factor T_{ij}^s, the iceberg trade cost assumption. Second, elasticities of substitution between varieties of a particular product type produced at different places in the city, σ^s, are likely to be high. Demands are then quite sensitive to price differences, but all places produce all product types, albeit possibly in very small quantities.

4.6 Productivity and agglomeration

An important extension is to endogenize the level of productivity in each sector/cell. To capture the idea that there are economies of scale from clustering we assume that productivity A_i^s is a weakly increasing function $f(.)$ of production in the same sector in nearby places, so

$$A_i^s = a_i^s f\left(\sum_j \theta_{ij}^s x_j^S\right) = a_i^s \left(\sum_j \theta_{ij}^s x_j^S\right)^{\chi} \tag{17}$$

where θ_{ij}^s is the spillover from cells j to i, with elements smaller the further away is j from i and a_i^s is a scaling factor. The second equation gives an isoelastic form for the function $f(.)$, with elasticity $\chi \geq 0$.

4.7 Equilibrium and solution

The equilibrium is found when all equations are solved simultaneously. Import supply is perfectly price-elastic and the numeraire is foreign goods, $p^f = 1$. Each cell is on its budget constraint, so balance of payments balance holds for each cell (and in aggregate), i.e.

$$F_i^1 + F_i^2 + F_i^3 = \alpha^f Y_i + \gamma^f C_i + \sum_s \beta^{fs} p_i^s x_i^s + \sum_i q_i H_i (\nu - 1)/\nu \tag{18}$$

The left side of this equation is the exports from each cell, and the right side is imports, these meeting demand from consumption out of wage income and construction expenditure. The final term is transfer of revenue from each cell as pure rents are paid to landlords outside the city.

6.1 Business-as-usual (BAU)

Assumptions

The BAU scenario projects the Greater HCMC Metropolitan Area to 2040 under 45 percent population growth (Hoornweg and Pope, 2014) and 6.5 percent annual GDP growth, consistent with Viet Nam's development trajectory and ambition to become a high-income country. Floor space grows in line with the 2010 to 2020 elasticity of floor space to population, recovered from the GHSL data.

Such anticipated growth means that even under the Business-As-Usual scenario, Thu Duc is expected to grow substantially. However, this growth is limited by the increasing flood risks in the area associated with climate change. While such changes are anticipated across the greater HCMC Metropolitan Area, Thu Duc is particularly impacted, meaning the area's anticipated growth rate is lower than some surrounding, better protected, areas.

These changes are modeled by using the change in flood risk from today through to 2040 under RCP8.5 to forecast a potential change in resultant productivities and amenities, all else equal. These changes affect areas across the wider city, not just Thu Duc. The share of land at flood risk (inundation above 30 centimeters under a 1-in-20-year event) rises by approximately 2.9 percentage points across the metropolitan area by 2040. As a result, this increase in flood risk causes productivity to be 3.9% lower than otherwise in business services, 4.7 percent in local services, and 6.3 percent in manufacturing.

Impact

The effects of climate change are not evenly distributed. Thu Duc is particularly exposed to increased flood risk. As a result, the productivity in Thu Duc is expected to be 14.6 percent lower than otherwise in business services, 18.6 percent in local services, and 21.8 percent in manufacturing. Amenity values show the same pattern: the average amenity in Thu Duc will be 11 percent lower than otherwise expected by 2040. The area still grows, and it grows at a substantial pace. But the scale of that growth is constrained by worsening flood risks.

The result is that during HCMC's period of rapid growth, Thu Duc is becoming gradually relatively less attractive relative to safer parts of the metropolitan area, and so a smaller share of the city's population growth is expected to occur there.

Under BAU, Population in Thu Duc would grow from 930,000 today through to 1.02 million, with employment rising from 430,000 to 530,000, and GVA reaching nearly USD14 billion. These represent substantial growth from today but fall well short of the 2.6 million masterplan target.

One third of Thu Duc's residents are educated beyond secondary level, nearly twice the metropolitan average. The economy remains dominated by manufacturing, which employs 45 percent of Thu Duc's workforce. Thu Duc accounts for approximately 5 percent of the metropolitan area's 10 million jobs. Daily commuter flows between Thu Duc and the rest of the metropolitan area reach approximately 850,000.

6.2 Hypothetical economic potential: Uplifting to HCMC average conditions

Assumptions

A hypothetical benchmark establishes a model-derived analytical upper bound for Thu Duc, conditional on the calibration. The productivity and amenity parameters underlying it are equilibrium residuals recovered through model inversion rather than direct measurements, and they reproduce the observed 2019 employment and residential distributions by construction. The benchmark should therefore be

read as a yardstick for measuring the gap, not as a measured statement of attainable potential. Baseline calibrated productivity and amenity parameters reveal a substantial gap between Thu Duc and the rest of HCMC. Across all three sectors, Thu Duc's productivity is on average approximately one-third that of the rest of HCMC, while residential amenity values are approximately two-thirds of the metropolitan average. These gaps reflect decades of underdevelopment, compounded by flood risk, weak connectivity, and limited public services. The scale of this divergence is difficult to overstate: reaching metropolitan-average fundamentals would require Thu Duc's productivity to roughly triple, a transformation far larger than any single infrastructure or policy package could deliver, and one that frames the frontier as an upper bound rather than an attainable near-term target.

The hypothetical scenario asks what Thu Duc could become by 2040 if its underlying productivity and amenity parameters converged to at least the levels observed elsewhere in HCMC. This would require productivity increases of approximately 240 percent in business services, 270 percent in local services, and 275 percent in manufacturing, alongside amenity increases of 65 to 85 percent. The scenario is not a policy prescription. It does not specify the investments required; it quantifies the frontier that targeted policies must aim to approach and identifies the magnitude of the underlying productivity and amenity gap relative to other determinants of underdevelopment.

Impact

Table 4. Hypothetical Potential – Uplifting to HCMC Average Conditions (2040)

Indicator	BAU	Hypothetical	Change	% vs BAU
GVA (USD Billions)	13.9	74.7	+60.7	435.5%
Total Wage Income (USD Billions)	9.3	53.0	+43.7	471.7%
Population (Millions)	1.02	3.40	+2.37	231.8%
Employment (Millions)	0.53	2.16	+1.63	309.0%
Employment – Business Services (M)	0.12	0.40	+0.28	244.5%
Employment – Local Services (M)	0.18	0.96	+0.78	433.6%
Employment – Manufacturing (M)	0.23	0.67	+0.44	190.7%
Employment – Innovation Zone (M)	0.00	0.12	+0.12	—
Total Floor Space (index)	100	187	+87	87.0%
Commuters per day (thousands)	850	2,891	+2,041	240.1%

Table 5. Hypothetical Potential – Greater Metropolitan Area Impacts (2040)

Indicator	BAU	Hypothetical	Change	% vs BAU
GVA (USD Billions)	445.6	469.4	+23.8	5.3%
Total Wage Income (USD Billions)	308.0	324.2	+16.2	5.3%
Average Real per capita income (Index = 100)	100.0	105.3	—	5.3%
Real per capita income – Low Skill	100.0	103.3	—	3.3%
Real per capita income – High Skill	100.0	108.8	—	8.8%

Source: Authors' SCGE model estimates. All values are 2040 projections in constant USD. The hypothetical scenario models exogenous uplift of Thu Duc productivity and amenity parameters to HCMC average levels and should be interpreted as a growth frontier, not a policy forecast.

The hypothetical scenario quantifies a substantial frontier. Thu Duc's GVA rises to US$74.7 billion, with employment reaching 2.16 million and population 3.40 million. The implied population density exceeds that envisaged in the former Thu Duc City masterplan. The scale is an upper bound, and real-world constraints could reduce it, but the magnitude is meaningful. Manufacturing, local services, and business services all expand substantially under the productivity uplift, supported by an 87 percent increase in floor space (without which GVA would be US$11 billion lower). Of the US$60.7 billion gain in Thu Duc GVA, approximately 39 percent represents net new output for the metropolitan economy; metropolitan GVA rises by US$23.8 billion (5.3 percent), with high-skilled residents gaining an average of 8.8 percent in real per capita income and low-skilled residents gaining 3.3 percent.

Even at the frontier, policy-induced improvements in Thu Duc generate substantial reallocation of activity across the metropolitan area. Under the hypothetical scenario, approximately 61 percent of Thu Duc's local gains reflect reallocation from other parts of the metropolis rather than net new output. This is consistent with a spatial equilibrium in which firms and households respond to the relative attractiveness of locations, and it has direct implications for the design of place-based policy.

It should be noted that the growth is delivered through greater development and increased density within this development; total floorspace in Thu Duc City could grow by 87%, but without this densification, Thu Duc City's output would be USD 11 billion lower than this frontier. Conditions need to facilitate this building growth to support the economic transformation.

6.3 Scenario A: Innovation Zones

Assumptions

Scenario A adds to BAU the construction and operation of innovation zones as specified in the masterplan: 8 percent of Thu Duc's land area, directly employing approximately 120,000 workers in knowledge-intensive activities. This figure is an exogenous input taken from the masterplan. The zones are assumed to generate a 1 percent productivity boost to manufacturing citywide from knowledge spillovers. The required land is costed at local market rent, and the 10,000 high-skilled employees per km^2 are assumed to work at the market wage. These costs are financed through a lump-sum tax on land rents.

Impact

Thu Duc's GVA rises by US$4.8 billion under Scenario A, a 34 percent increase over BAU, with a net increase of approximately 90,000 additional jobs in Thu Duc; although the zones house roughly 120,000 workers, net Thu Duc employment rises by less because some non-zone activity is displaced. Substantially all of this job growth is located within the innovation zones themselves. The zones displace alternative land uses, and non-zone employment in business services, local services, and manufacturing in Thu Duc declines slightly. Metropolitan GVA rises by US$4.2 billion, implying that approximately 88 percent of Thu Duc's local gain is net new output for the metropolitan economy rather than reallocation; the corresponding displacement from elsewhere in the metropolis is approximately US$600 million. Growth is concentrated in manufacturing, whose share of total Thu Duc jobs rises by 0.5 percentage points as zone-induced productivity supports higher sectoral output.

The welfare effects are skill-differentiated. The average high-skilled resident gains 3.2 percent in real per capita income metropolitan-wide, reflecting higher wages in the innovation zones and in manufacturing. Low-skilled residents experience a marginal welfare loss of 0.6 percent, as demand for lower skilled jobs in local services decreases, and land rents push people away from the city center. These metropolitan-average welfare figures are near-identical across Scenarios A, B, and C because Thu Duc accounts for

bind in practice is partly a matter of policy choice. The model also abstracts from the informal sector, which is economically significant in HCMC and particularly in less-developed parts of the metropolitan area. Informal firms and households may respond differently to policy change than their formal counterparts, and the distributional implications of the welfare results may differ if informal activity is explicitly modeled. This is a limitation shared with much of the quantitative urban economics literature in developing-country settings.

7.6 Assumption on static metropolitan population

A related limitation concerns the treatment of displacement across the metropolitan area. The model holds the size of the metropolitan population fixed at the exogenous growth path and therefore attributes all reallocation across cells to intra-metropolitan adjustment. In reality, stronger conditions in Thu Duc could attract in-migration from beyond the modeled area, reducing the apparent displacement effect. The net-new versus reallocation decomposition reported in Section 6 should therefore be interpreted as a lower bound on net new output for the broader economy.

7.7 Macroeconomic assumptions

All projections are conditional on 6.5 percent annual GDP growth and a specific external environment. Deviations, including lower growth, changes in Viet Nam's export competitiveness, or shifts in global manufacturing supply chains, would alter the baseline levels and the absolute scale of the results. The model generates conditional projections rather than unconditional forecasts, and the results should be interpreted accordingly. The spatial pattern of relative growth across cells, on which the paper's main findings depend, is not, however, sensitive to this assumption.

The 2040 horizon is itself a modeling choice. 2040 is seen as sufficient time for the city to adjust to any proposed investments, and a new equilibrium to be reached. Earlier horizons would show less of the BAU deterioration in flood risk and less scope for infrastructure-driven transformation. Longer horizons would show both greater flood risk under BAU and larger potential gains from mitigation and magnify the importance of the dynamic limitations noted in Section 7.3.

8. Conclusions

This paper has applied a Spatial Computable General Equilibrium model of the Greater Ho Chi Minh City Metropolitan Area to evaluate the economic consequences of policy interventions in Thu Duc, an area whose current output and density fall substantially below what its location, endowments, and position within the metropolitan economy would support. The analysis proceeded in two stages. A hypothetical benchmark quantified the full productivity-amenity gap, showing that Thu Duc's GVA under convergence to HCMC-average fundamentals would be approximately US$74.7 billion in 2040, compared to US$13.9 billion under the business-as-usual trajectory. A progression of realistic policy scenarios then evaluated how far a cumulative package of innovation zones, flood risk mitigation, transport improvements, and stronger agglomeration economies could bring Thu Duc toward that frontier. The cumulative package brings Thu Duc to approximately 31 percent of the frontier level, which corresponds to closing approximately 15 percent of the gap between the business-as-usual path and the frontier (US$9.1 billion of US$60.7 billion). Flood risk mitigation produces the most meaningful broadening of Thu Duc's economic base among the evaluated interventions, raising sectoral productivity by 12 to 21 percent and residential amenities by 10 to 13 percent and expanding the scope of developable land available for productive use. This comparison is drawn from the breadth of the sectoral response rather than from the size of the marginal increments alone, which depend on the order in which the interventions are

stacked. The remaining distance to the frontier reflects the broader productivity and amenity shortfall relative to HCMC-average conditions, which the modeled interventions only partially address.

The paper's methodological contribution is the integration of spatially explicit flood exposure as a determinant of calibrated productivity and amenity within a Spatial CGE framework. This places flood risk on an analytical footing comparable to transport and land-use interventions, evaluated through the same general equilibrium apparatus rather than through a separate engineering cost-benefit exercise. The resulting estimates of the economic response to coordinated flood mitigation substantially exceed those captured by damage-avoided cost-benefit analysis of the same investments, because the general equilibrium response includes the induced adjustment of firm location, household sorting, floor space supply, and agglomeration externalities. The decomposition of local gains into net new metropolitan output and reallocation from elsewhere in the metropolis, which conventional appraisal cannot provide, shows that approximately 56 percent of Thu Duc's gains under the cumulative realistic package represent net new output for the metropolitan economy. The framework developed here is transferable to other cities where spatial underperformance is driven by interacting flood, connectivity, and productivity-amenity frictions.

Several extensions follow naturally from the limitations noted in Section 7. A dynamic specification of the model would quantify the time profile of returns and the path dependence of sequenced investments. Solving the model under alternative orderings of the same interventions would, in addition, permit an order-invariant attribution of the cumulative gain, for example by averaging each intervention's marginal contribution across orderings. Endogenous congestion would refine the gains attributed to high-activity scenarios, particularly under the hypothetical frontier. Panel variation in flood exposure, together with instrumental variables based on elevation and historical hydrology, would sharpen the identification of the flood-to-fundamentals relationship and reduce the reliance on cross-sectional inference. Application of the framework to other flood-exposed urban economies, beginning with delta megacities in Southeast Asia and extending to coastal cities more broadly, would test the generality of the findings reported here and contribute to a more systematic evidence base on the general equilibrium returns to coordinated climate adaptation in cities.

Journal of Development Economics 122 (2016) 63–75

Contents lists available at ScienceDirect

Journal of Development Economics

journal homepage: www.elsevier.com/locate/devec

Can a poverty-reducing and progressive tax and transfer system hurt the poor?

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ARTICLE INFO

Article history:
Received 8 April 2015
Received in revised form 25 December 2015
Accepted 1 April 2016
Available online 13 April 2016

Keywords:
Poverty
Horizontal equity
Progressivity
Fiscal impoverishment

ABSTRACT

To analyze anti-poverty policies in tandem with the taxes used to pay for them, comparisons of poverty before and after taxes and transfers are often used. We show that these comparisons, as well as measures of horizontal equity and progressivity, can fail to capture an important aspect: that a substantial proportion of the poor are made poorer (or non-poor made poor) by the tax and transfer system. We illustrate with data from seventeen developing countries: in fifteen, the fiscal system is poverty-reducing and progressive, but in ten of these at least one-quarter of the poor pay more in taxes than they receive in transfers. We call this fiscal impoverishment, and axiomatically derive a measure of its extent. An analogous measure of fiscal gains of the poor is also derived, and we show that changes in the poverty gap can be decomposed into our axiomatic measures of fiscal impoverishment and gains.

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1. Introduction

Anti-poverty policies are often evaluated in isolation from the taxes used to pay for them.[1] If, however, taxes cancel out the benefits of transfers for many poor households, so that some poor pay more in taxes than they receive in transfers, the objective of these policies might be compromised. This is especially important when poverty traps exist at the individual level (e.g., Ghatak, 2015, Ravallion, 2015): a tax and transfer system in which many poor pay more in taxes than they receive in transfers risks pushing the transiently poor into chronic poverty by shifting their after tax and transfer incomes below their individual-specific poverty trap thresholds.

Recently, the connection between anti-poverty policies and the taxes used to pay for them has come into the spotlight in the debates over the United Nations' Post-2015 Sustainable Development Goals. In recognition of the resources necessary to achieve these ambitious development goals, and partly as a consequence of austerity in advanced countries (and thus lower anticipated flows of international aid to developing countries), much of the discussion has focused on how developing countries should collect the revenue necessary to achieve the goals.[2] Influential organizations such as the International Monetary Fund and World Bank emphasize the importance of efficient taxes with minimal exemptions (International Monetary Fund, 2013, World Bank, 2013). When concerns are raised about these taxes—such as a no-exemption value added tax—falling disproportionately on the poor, many argue that higher tax burdens on the poor are acceptable if they are accompanied by sufficiently large targeted transfers: "spending instruments are available that are better targeted to the pursuit of equity concerns" (Keen and Lockwood, 2010, p.141). Similarly, Engel et al. (1999, p. 186) assert that "it is quite obvious that the disadvantages of a proportional tax are moderated by adequate targeting" of transfers, since "what the poor individual pays in taxes is returned to her." These taxes "might conceivably be the best way to finance pro-poor expenditures, with the net effect being to relieve poverty" (Ebrill et al., 2001, p. 105).

How can we be sure that what the poor individual pays in taxes is returned to her? Even if the net effect of taxes and transfers is

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[1] We focus on anti-poverty policies that are redistributive in nature, one of the three categories of anti-poverty policies described in Ghatak (2015).

[2] See, for example, the focus on domestic resource mobilization in United Nations (2015).

http://dx.doi.org/10.1016/j.jdeveco.2016.04.001
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to relieve poverty, are some poor made worse off? When taxes and transfers are analyzed in tandem to determine how they affect the poor, it is common to compare poverty before taxes and transfers ("pre-fisc") to poverty after taxes and transfers ("post-fisc"). As we show in this paper, however, a fiscal system can be unambiguously poverty-reducing for a range of poverty lines and any poverty measure, yet still make a substantial proportion of the poor worse off. This phenomenon does not only occur with regressive taxes: we show that taxes and transfers can be globally progressive, unambiguously equalizing, and unambiguously poverty-reducing and *still* make many poor worse off. In other words, conventional tools used to measure how the poor are affected by the tax and transfer system are inadequate to measure whether some of the poor pay more in taxes than they receive in transfers, a phenomenon we call fiscal impoverishment (FI).

We also show that in practice, there are a number of countries with poverty-reducing and progressive tax and transfer systems that nevertheless make a substantial proportion of the poor poorer (or non-poor poor), illustrating with data from seventeen developing countries.[3] In fifteen of these countries, post-fisc poverty is unambiguously lower than pre-fisc poverty (measured with any poverty line up to $1.25 per person per day in low and lower-middle income countries and $2.50 per day in upper-middle income countries)[4] and the tax and transfer system is globally progressive and unambiguously equalizing, i.e., we would conclude that the tax and transfer system unambiguously benefits the poor using conventional measures, potentially overlooking impoverishment. In all of these countries, some degree of FI occurs, and in ten of them we find that at least one-quarter of the poor pay more in taxes than they receive in transfers.

In light of the debate about financing anti-poverty policies and the Sustainable Development Goals, it is necessary to fill this gap in the measurement arsenal and develop a measure of this phenomenon that adheres to certain properties. We axiomatically derive a measure of FI, as well as an analogous measure for fiscal gains of the poor (FGP), which captures the extent to which some poor receive more in transfers than they pay in taxes.[5] We then show how a commonly used measure of poverty that overlooks the extent of FI, the poverty gap, can be decomposed into FI and FGP components using our axiomatic measures, again illustrating with data from seventeen developing countries. Because the extent of FI and FGP depends on the particular poverty line used, we also propose dominance criteria that can be used to determine whether one fiscal system (such as the one that would occur after a proposed reform) causes unambiguously less FI or more FGP than another (such as the current system) over a range of poverty lines. We analyze FI and FGP over a range of poverty lines in Brazil, which is a pertinent example due to the coexistence of high tax burdens on the poor (Baer and Galvão, 2008, Goñi et al., 2011) and lauded poverty-reducing cash transfer programs: a large-scale conditional cash transfer program that reaches over one-fourth of all Brazilian households and a non-contributory pension program for the elderly poor that reaches one-third of all elderly (Levy and Schady, 2013, Table 1).

Section 2 uses hypothetical and empirical examples to show that common tools to assess how the tax and transfer system affects the poor can fail to capture FI. Section 3 axiomatically derives a measure that does capture FI; it then proposes a partial FI ordering that can be used to compare the level of FI induced by two fiscal systems for any poverty line. Section 4 derives an analogous measure and partial ordering for FGP and shows that the poverty gap can be decomposed into our axiomatic measures of FI and FGP. Section 5 uses data from seventeen developing countries to illustrate the axiomatic measures and poverty gap decomposition. Section 6 concludes, and the formal axioms and proofs are collected in the Appendix.

2. The problems with conventional measures

Through a number of examples, we illustrate and explain the problems with conventional measures of poverty, horizontal equity, and progressivity. Of course, these measures are still quite important for assessing a tax and transfer system; we merely aim to show that they do not capture everything we are interested in. First, in Section 2.1 we show the problem with poverty measures when they are used to compare poverty before and after taxes and transfers. Although comparisons of pre-fisc and post-fisc poverty are common in empirical studies (e.g., DeFina and Thanawala, 2004, Hoynes et al., 2006), poverty measures can overlook fiscal impoverishment because they obey the anonymity axiom (which is usually taken as an innocuous and desirable axiom): the tax and transfer system can reduce poverty while simultaneously making a substantial portion of the poor poorer, or making some non-poor poor. The anonymity axiom is not the only culprit for the shortcomings of existing measures, however: in Section 2.2 we show that measures designed to incorporate information about individuals' pre-fisc positions, such as measures of horizontal equity and progressivity, can also fail to capture FI.[6] To show that these shortcomings of conventional measures are not confined to contrived hypothetical examples, but rather occur frequently in practice, in Section 2.3 we present examples from seventeen developing countries: in ten, the tax and transfer system is poverty-reducing and progressive, but hurts a substantial portion of the poor by pushing them deeper into poverty.

2.1. Poverty measures

Suppose the change in poverty caused by the fiscal system will be evaluated over a range of poverty lines, including lines greater than 6 and less than or equal to 10. Suppose there are three individuals in society with pre-fisc incomes of 5, 8, and 20, and (retaining the order of the individuals) post-fisc incomes 9, 6, and 18. For any poverty line in the range we are considering, and for any poverty measure in a broad class of measures, poverty has either not changed or decreased. This is because the poorest individual in the pre-fisc

[3] Our illustration uses results provided to us by the authors of country studies conducted as part of the Commitment to Equity (CEQ) Institute, located at Tulane University (www.commitmenttoequity.org). The countries included are Armenia (Younger and Khachatryan, forthcoming), Bolivia (Paz Arauco et al., 2014), Brazil (authors' calculations), Chile (Martínez-Aguilar and Ortiz-Juarez, 2015), the Dominican Republic (Aristy-Escuder et al., forthcoming), Ecuador (Llerena Pinto et al., 2015), El Salvador (Beneke et al., 2015), Ethiopia (Hill et al., forthcoming), Ghana (Younger et al., 2015), Guatemala (Cabrera et al., 2015), Indonesia (Afkar et al., forthcoming), Mexico (Aranda and Scott, 2015), Peru (Jaramillo et al., 2015), Russia (Lopez-Calva et al., forthcoming), South Africa (Inchauste et al., forthcoming), Sri Lanka (Arunatilake et al., forthcoming), and Tunisia (Shimeles et al., forthcoming). For an overview of the impact of taxes and social spending on inequality and poverty in many of these countries, see Lustig (2015).

[4] The $1.25 per person per day poverty line (in 2005 US dollars adjusted for purchasing power parity) is approximately equal to the median poverty line of the fifteen poorest countries for which poverty line data are available, and the $2.50 line to the median of the world's low and middle income countries excluding the fifteen poorest (Chen and Ravallion, 2010).

[5] Our axioms are adapted from the axiomatic poverty and mobility measurement literatures (see Foster, 2006 and Zheng, 1997 for surveys of axiomatic poverty measurement and Fields, 2001 for a survey of axiomatic mobility measurement). Our resulting measure can be viewed as a censored directional version of the mobility measure derived by Fields and Ok (1996).

[6] Other measures that are sometimes used, such as the percent of income gained or lost by each pre-fisc income decile, overlook FI for a distinct reason: they average over individuals, so for example the poorest decile could gain income on average while a substantial number of poor within the first decile lose income. We do not include these measures in this paper since the reason they overlook FI is obvious.

income distribution has an income of 5 and the second-poorest 8, while in the post-fisc distribution, the poorest has an income of 6 and the second-poorest 9. Poverty comparisons do not take into account that the poorest individual in the post-fisc distribution, with an income of 6, is not the poorest individual in the pre-fisc distribution who has an income of 5, but instead had an income of 8 in the pre-fisc distirbution and paid 2 more in taxes than she received in transfers. Depending on the exact poverty line chosen within the range we are considering, this individual was either pre-fisc poor and lost income to the fiscal system, or pre-fisc non-poor and pushed into poverty by the fiscal system.

It is clear, then, that poverty measures are inadequate to measure whether some of the poor pay more in taxes than they receive in transfers. Stochastic dominance tests, which are used to determine whether poverty is unambiguously lower in one income distribution than another for any poverty line and a broad class of poverty measures (Atkinson, 1987, Foster and Shorrocks, 1988), are also inadequate. This is because poverty measures and stochastic dominance tests are anonymous with respect to pre-fisc income: they compare the pre- and post-fisc income distributions without paying attention to the specific pre-fisc to post-fisc trajectory of particular individuals' incomes. The anonymity axiom, normally considered an innocuous and desirable property, becomes problematic when we are concerned with how the fiscal system affects the poor: in the words of Amiel and Cowell (1994, p. 448–9), "anonymity itself may be questionable as a welfare criterion when the social-welfare function is to take into account something more than the end-state distribution of incomes." Anonymity implies that poverty measures fail to take into account individuals' initial positions, and thus whether some are being made poorer by the tax and transfer system.[7]

To illustrate visually, Fig. 1 shows a stylistic representation of the pre- and post-fisc incomes of a population ordered by pre-fisc income. The increasing curve represents pre-fisc income, the wavy curve post-fisc income, and the dashed line the poverty line; because some individuals receive more in transfers than they pay in taxes, while others pay more in taxes than they receive in transfers, the post-fisc income curve is sometimes above and sometimes below the pre-fisc income curve. Although post-fisc poverty is lower than pre-fisc poverty because the losses of some poor are more than compensated by the gains of other poor, there is FI. The extent of FI is shown by the dark-shaded areas, while the light-shaded areas represent the extent of FGP (using the measures we axiomatically derive in Sections 3 and 4).

2.2. Horizontal equity and progressivity

Anonymity is not the only reason conventional measures overlook fiscal impoverishment: non-anonymous measures such as horizontal equity and progressivity, which are designed to incorporate information about an individual's pre-fisc position, can fail to capture FI because they are not concerned with whether her net tax burden (taxes paid minus transfers received) is positive or negative. Denote income before taxes and transfers by $y_i^0 \in \mathbb{R}_+$ and income after taxes and transfers by $y_i^1 \in \mathbb{R}_+$ for each $i \in S$, where S is the set of individuals in society. Consider a range of potential poverty lines $\mathcal{Z} \subset \mathbb{R}_+$. Each individual's income before or after taxes and transfers is arranged in the vector y^0 or y^1, both ordered in ascending order of pre-fisc income y_i^0—even if reranking occurs, the order of the y^1 vector reflects the pre-fisc income ranking.

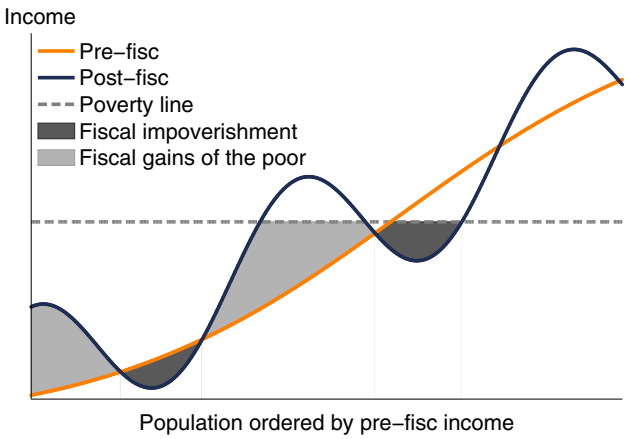

Fig. 1. Stylistic illustration of fiscal impoverishment and gains to the poor.

Horizontal equity can be defined in two ways: the reranking definition, which requires that no pair of individuals switch ranks, and the classical definition, which requires that pre-fisc equals are treated equally by the tax and transfer system. Under either definition, the existence or absence of horizontal equity among the poor does not tell us whether FI has occurred. Even if some are impoverished by the tax and transfer system, the ranking among the poor may not change (so there is horizontal equity by the reranking definition) and pre-fisc equals may be impoverished to the same degree (so there is classical horizontal equity): e.g., $\mathcal{Z} = (6, 10]$, $y^0 = (1, 1, 7, 7, 13)$, $y^1 = (3, 3, 6, 6, 11)$. Nor does horizontal inequity among the poor necessarily imply FI, because there could be reranking among the poor or unequal treatment among pre-fisc equals when the tax and transfer system lifts incomes of some of the poor without decreasing incomes of any poor: e.g., $\mathcal{Z} = (6, 10]$, $y^0 = (5, 5, 6, 20)$, $y^1 = (5, 7, 6, 18)$.

A tax and transfer system is everywhere progressive when net taxes (i.e., taxes minus benefits), relative to pre-fisc income, increase with income (Duclos, 1997, Lambert, 1988). The tax and transfer system can be progressive (and unambiguously equalizing) but cause fiscal impoverishment: e.g., $\mathcal{Z} = (6, 10]$, $y^0 = (1, 3, 7, 13)$, $y^1 = (3, 4, 6, 11)$; net taxes relative to pre-fisc income increase with income, but the third individual whose income falls from 7 to 6 is fiscally impoverished. Thus, progressivity is not a sufficient condition to ensure that FI does not occur. Nor is progressivity a necessary condition for the absence of FI: e.g., $\mathcal{Z} = (6, 10]$, $y^0 = (1, 3, 7, 14)$, $y^1 = (1, 5, 8, 11)$, which involves no FI but is not everywhere progressive because net taxes first decrease with income when moving from the poorest to the second-poorest, then increase with income thereafter.

Table 1 summarizes the examples presented in Sections 2.1 and 2.2 to show that conventional tools—specifically, poverty measures (and stochastic dominance tests) and measures of or tests for horizontal equity and progressivity—can overlook FI.

[7] Amiel and Cowell (1994) also point out that the respect for income dominance axiom is only equivalent to the monotonicity axiom when anonymity is imposed. In the example from the previous paragraph, the post-fisc income distribution first order stochastically dominates the pre-fisc distribution on the domain from 0 to the maximum poverty line, so it would be evaluated as superior by any measure satisfying poverty focus and respect for income dominance (or, equivalently, poverty focus and both monotonicity and anonymity). It would not necessarily be evaluated as superior by a measure satisfying poverty focus and monotonicity but not anonymity, however. Other concerns with the anonymity axiom have also been pointed out: for example, it can clash with the Pigou–Dalton transfer axiom when there are households of different types (Ebert, 1997) and with the subgroup sensitivity axiom, an extension of the Pigou–Dalton transfer axiom to subgroups (Subramanian, 2006).

Table 1
Summary of the problems with conventional measures.

Measure	Issue	Example with $\mathcal{Z} = (6, 10]$
Poverty (and stochastic dominance)	↓ poverty $\not\Rightarrow$ no FI (anonymity)	$y^0 = (5, 8, 20), y^1 = (9, 6, 18)$
Horizontal equity	Horizontally equitable $\not\Rightarrow$ no FI	$y^0 = (1, 1, 7, 7, 13), y^1 = (3, 3, 6, 6, 11)$
	No FI $\not\Rightarrow$ horizontally equitable	$y^0 = (5, 5, 6, 20), y^1 = (5, 7, 6, 18)$
Progressivity	Progressive $\not\Rightarrow$ no FI	$y^0 = (1, 3, 7, 13), y^1 = (3, 4, 6, 11)$
	No FI $\not\Rightarrow$ progressive	$y^0 = (1, 3, 7, 14), y^1 = (1, 5, 8, 11)$

2.3. Real-world examples

The problems with conventional measures are not limited to contrived hypothetical examples. In a number of countries, we observe an unambiguous reduction in poverty and a globally progressive tax and transfer system, while a significant proportion of the poor are fiscally impoverished. Using the income concepts from Higgins et al. (2015), we compare market income (before taxes and transfers) to post-fiscal income (after direct and indirect taxes, direct cash and food transfers, and indirect subsidies) in seventeen developing countries. We use post-fiscal income as the after taxes and transfers income concept even though taxes are used to fund more than just direct cash and food transfers and indirect subsidies from the government (e.g., they are used to fund public goods and services, many of which also reach the poor) because this is the income concept relevant for measuring poverty: it is "disposable money and near-money income" that should be compared to the poverty line when the latter is based on "a poverty budget for food, clothing, shelter, and similar items" (Citro and Michael, 1995, p. 212, 237). For low and lower-middle income countries, we use a poverty line of $1.25 per person per day; for upper middle income countries, $2.50 per day. Table 2 column 1 shows the pre-fisc (market income) poverty headcount and column 2 shows the change in poverty from the pre-fisc to the post-fisc income distribution; countries in which poverty increased due to the fiscal system are excluded.[8]

Moving to the progressivity of the tax and transfer system and change in inequality in each country, column 3 shows the pre-fisc Gini coefficient and column 4 shows the Reynolds and Smolensky (1977) index, which is a summary indicator corresponding to tests of global progressivity; the Reynolds–Smolensky equals the pre-fisc Gini minus the concentration coefficient of post-fisc income with respect to pre-fisc income, and thus globally progressive systems have a positive Reynolds–Smolensky index. Column 5 shows the change in inequality, with negative numbers indicating that inequality fell as a result of the tax and transfer system.[9]

Since we do not derive an axiomatic measure of FI until Section 3, here we use two intuitively appealing measures likely to have policy traction. Column 6 shows the percent *of the population* that are fiscally impoverished and column 7 the percent *of the post-fisc poor* that are fiscally impoverished. Although all of the countries in Table 2 experienced a reduction in poverty and inequality due to the tax and transfer system, the amount of FI varies greatly between countries. In ten countries—Armenia, Bolivia, Brazil, El Salvador, Guatemala, Indonesia, Mexico, Russia, Sri Lanka, and Tunisia—between one-quarter and two-thirds of the post-fisc poor lost income to the fiscal system.[10] In other countries, this figure is much lower, at 13.3% of the post-fisc poor in South Africa (but, due to the high proportion of the total population that is poor, still 5.9% of the total population) and 3.2% of the post-fisc poor in Ecuador.

Even when poverty increases from pre-fisc to post-fisc income and hence we know that FI has occurred (as in Ghana and Ethiopia), it is impossible to tell its extent without explicit measures like the ones we propose in Section 3. A stark example of this comes from Ethiopia, where looking at poverty and progressivity numbers alone greatly masks the extent of FI: the headcount ratio at $1.25 per day increases from 31.9% to 33.2% of the population, while the squared poverty gap and Gini coefficient fall as a result of taxes and transfers (World Bank, 2015). Nevertheless, applying our measures to the same data, Hill et al. (forthcoming) find that 28.5% of Ethiopians and over 80% of the post-fisc poor experience FI.

Even if we add the value of public spending on education and health (imputed at their government cost to families who report a child attending public school or who report using public health facilities), fiscal impoverishment is still high in several countries: in Armenia, Ethiopia, Indonesia, Russia, and Tunisia, between 25 and 50% of those who are fiscally impoverished before adding in benefits from public spending on health and education are still fiscally impoverished when these benefits are included as transfers.

3. Measures of fiscal impoverishment

To assess anti-poverty policies in tandem with the taxes used to finance them, it is important to have measures of the extent of fiscal impoverishment. In the last section, we provided a glimpse of FI in several developing countries using two simple, straightforward, and intuitive measures that—given these features—can be useful for policy discussions. These two measures also have drawbacks, however. To illustrate their limitations, we begin by providing more detail about the two measures. For a particular poverty line $z \in \mathcal{Z}$, there is *fiscal impoverishment* if $y_i^1 < y_i^0$ and $y_i^1 < z$ for some individual $i \in S$. In other words, the individual could be poor before taxes and transfers and made poorer by the fiscal system, or non-poor before taxes and transfers but poor after. Both straightforward measures count the number of individuals who meet this condition (and are thus fiscally impoverished) in the numerator. The proportion *of the population* who are fiscally impoverished (column 6 of Table 2) divides this numerator by the number of individuals in society, while the proportion *of the post-fisc poor* who are fiscally impoverished (column 7) divides it by the number who are post-fisc poor (with $y_i^1 < z$).

[8] Although the table only shows poverty for a particular poverty line and poverty measure, it is also true that the post-fisc distribution first order stochastically dominates the pre-fisc distribution from 0 to the poverty line used for each country, meaning that poverty unambiguously fell for all poverty lines up to $1.25 or $2.50 and all poverty measures in a broad class.

[9] We test global progressivity by dominance of the concentration curve of post-fisc with respect to pre-fisc income over the pre-fisc Lorenz curve, and test unambiguously equalizing by comparing the post-fisc and pre-fisc Lorenz curves.

[10] If we instead scale down taxes so that they equal the transfers included in our analysis, which we avoid in the main analysis for the reasons mentioned above in defense of post-fiscal income as the after taxes and transfers income concept, FI is lower: for example, in Brazil 10.8% of the post-fisc poor are fiscally impoverished using this method.

Table 2
Poverty, inequality, and fiscal impoverishment in developing countries.

Country (survey year)	(1) Pre-fisc poverty headcount (%)	(2) Change in poverty headcount (p.p.)	(3) Pre-fisc inequality (Gini)	(4) Reynolds–Smolensky (post-fisc w.r.t. pre-fisc)	(5) Change in inequality (ΔGini)	(6) Fiscally impoverished as % of population	(7) Fiscally impoverished as % of post-fisc poor
Panel A: Upper-middle income countries, using a poverty line of $2.50 per day							
Brazil (2008–2009)	16.8	−0.8	57.5	4.6	−3.5	5.6	34.9
Chile (2013)	2.8	−1.4	49.4	3.2	−3.0	0.3	19.2
Ecuador (2011–2012)	10.8	−3.8	47.8	3.5	−3.3	0.2	3.2
Mexico (2012)	13.3	−1.2	54.4	3.8	−2.5	4.0	32.7
Peru (2011)	13.8	−0.2	45.9	0.9	−0.8	3.2	23.8
Russia (2010)	4.3	−1.3	39.7	3.9	−2.6	1.1	34.4
South Africa (2010–2011)	49.3	−5.2	77.1	8.3	−7.7	5.9	13.3
Tunisia (2010)	7.8	−0.1	44.7	8.0	−6.9	3.0	38.5
Panel B: Lower-middle income countries, using a poverty line of $1.25 per day							
Armenia (2011)	21.4	−8.4	47.4	12.9	−9.2	6.2	52.3
Bolivia (2009)	10.9	−0.5	50.3	0.6	−0.3	6.6	63.2
Dominican Republic (2007)	6.8	−0.9	50.2	2.2	−2.2	1.0	16.3
El Salvador (2011)	4.3	−0.7	44.0	2.2	−2.1	1.0	27.0
Guatemala (2010)	12.0	−0.8	49.0	1.4	−1.2	7.0	62.2
Indonesia (2012)	12.0	−1.5	39.8	1.1	−0.8	4.1	39.2
Sri Lanka (2009–2010)	5.0	−0.7	37.1	1.3	−1.1	1.6	36.4

Sources: For Brazil, authors' calculations. For other countries, provided to us by the authors of the studies cited in Footnote 3.
Notes: p.p.=percentage points. w.r.t.=with respect to. Ethiopia and Ghana are not included in the table because poverty with a $1.25 per day poverty line increased from pre-fisc to post-fisc income (and hence they do not illustrate shortcomings of conventional measures). Country classifications are from the World Bank for the year of the survey.

In the context of poverty measurement, Sen (1976, p. 219) proposes a monotonicity axiom requiring that, all else equal, "a reduction in income of a person below the poverty line must increase the poverty measure." We propose a similar axiom for FI measures requiring that a larger decrease in post-fisc income for an impoverished person, all else equal, must increase the FI measure. Monotonicity is violated by the straight-forward measures, which do not increase when an impoverished person becomes more impoverished because she counts as one impoverished individual in the measure's numerator regardless of how much income she loses to the fiscal system.[11]

3.1. Axioms

We propose eight properties desirable for a robust measure of FI; we describe these properties here and formally define them in the Appendix. Throughout, we assume that income is measured in real terms and has been converted to a common currency such as US dollars adjusted for purchasing power parity, thereby simplifying away concerns about inflation or currency conversions if comparing FI over time or across countries.

Our *FI monotonicity* axiom described above implies not only that the FI measure must be strictly increasing in the extent to which an impoverished individual is impoverished (ceteris paribus), but also that the measure must be strictly increasing in the number of individuals that are impoverished, holding fixed the amount of FI experienced by others. The *focus* axiom, analogous to Sen's (1981) focus axiom for poverty measurement, says that different income changes to the non-impoverished—provided that they remain non-impoverished—leave the FI measure unchanged. Given the focus axiom, it is natural to impose a *normalization* that if no one is impoverished, the FI measure equals zero. Note that this normalization axiom is not instrumental to our result: if we did not impose it, our result would be that our axioms uniquely determine a measure of FI up to a linear (rather than proportional) transformation.[12]

Similar to Chakravarty's (1983) *continuity* axiom for poverty measures, we require the FI measure to be continuous in pre-fisc income, post-fisc income, and the poverty line (since we may want to assess FI for a range of possible poverty lines). This is stronger than Foster and Shorrocks's (1991) restricted continuity axiom which only requires the measure to be continuous in incomes *below* the poverty line and left-continuous *at* the poverty line, thus allowing the measure to jump discontinuously at the poverty line; see Zheng (1997) and Permanyer (2014) for arguments in favor of using the stronger continuity axiom in the contexts of unidimensional and multidimensional poverty measures.

Because "the names of income recipients do not matter" (Zheng, 1997, p. 131), we impose a *permutability* axiom requiring that if we take each individual's pre- and post-fisc income pair and (keeping each pre- and post-fisc income pair as a bundle) shuffle these around the population, FI is unchanged. We use the term "permutability" rather than symmetry or anonymity because—although both have been used in the same way we use permutability above (e.g., Cowell, 1985, Fields and Fei, 1978, Plotnick, 1982)—symmetry and anonymity have also taken on different definitions. Symmetry can instead mean, for two income distributions X and Y and a distance measure d, that $d(X,Y) = d(Y,X)$; the two income distributions are treated symmetrically: losses are not distinguishable from gains (Ebert, 1984, Fields and Ok, 1999). Anonymity can instead mean that the measure compares the cumulative distribution of pre-fisc income, F_0, to that of post-fisc income, F_1,

[11] Another simple tool to examine FI is the $q \times q$ transition matrix P, whose typical element p_{kl} represents the probability of being in post-fisc income group $l \in \{1,\ldots,q\}$ for an individual in pre-fisc income group $k \in \{1,\ldots,q\}$. Measures based on P also fail to satisfy FI monotonicity and have the large drawback of not capturing FI among the poorest pre-fisc group ($k = 1$).

[12] It is also possible to normalize by the measure's upper bound so that it always lies on the interval $[0,1]$ by specifying an axiom that if everyone loses all of their income to the fiscal system (the maximum possible FI), the measure of FI equals 1. We prefer to avoid normalizing in this way so that the class of axiomatic FI measures is more general.

without regard to where a particular individual at position j in F_0 ended in F_1 (e.g., Bourguignon, 2011a,b). In other words, an anonymous measure would compare the pre-fisc income of the jth poorest individual in F_0 to the post-fisc income of the jth poorest individual in F_1, even though "they are not necessarily the same individuals" because of reranking (Bourguignon, 2011a, p. 607).

Next, we must decide whether our measure of FI should be absolute or relative (recalling that we assume income to be in real terms of a constant currency, so arguments about inflation or currency exchange should not affect the decision). Suppose each poor individual's pre-fisc income increases by \$1, taxes and transfers are held fixed, and the price of one essential good in the basic goods basket, normalized to have one unit in the basket, also increases by \$1 per unit.[13] Each poor individual remains the same distance below the poverty line; that distance represents the amount of additional income she needs to afford adequate nutrition and other basic necessities. For those who experience FI, it is the absolute increase in the distance between that individual's income and the poverty line that matters in terms of the quantity of basic goods she can buy. Hence, we assume that if all pre- and post-fisc incomes increase by \$1 and the poverty line also increases by \$1, FI should remain unchanged. We thus impose *translation invariance*.

Given our above argument for absolute measures, we also impose *linear homogeneity*: if all incomes and the poverty line are multiplied by the same factor, the measure of FI changes by that factor. Instead, specifying homogeneity of degree zero (scale invariance) would be incompatible with translation invariance for the reasons explored in Zheng (1994). Since we assume that income is expressed in real terms and a common currency, our measure is nevertheless insensitive to inflation or currency changes. The translation invariance and linear homogeneity axioms have been used together in axiomatic derivations of measures of inequality (Kolm, 1976), poverty (Blackorby and Donaldson, 1980), economic distance (Chakravarty and Dutta, 1987, Ebert, 1984), and mobility (Fields and Ok, 1996, Mitra and Ok, 1998).[14]

Our final axiom is based on a concept introduced to the poverty literature by Foster et al. (1984, p. 761), who argue that "at the very least, one would expect that a decrease in the poverty level of one subgroup ceteris paribus should lead to less poverty for the population as a whole." Similarly, it would be desirable for a measure of FI if a decrease in the measured FI for one subgroup of the population and no change in the measured FI for all other subgroups results in a decrease in the measured FI of the entire population. Hence, we impose a *subgroup consistency* axiom analogous to the one used for poverty measurement by Foster and Shorrocks (1991). In his survey of axiomatic poverty measurement, Zheng (1997, p. 137) notes that subgroup consistency "has gained wide recognition in the literature."

3.2. An axiomatic measure of fiscal impoverishment

Proposition 1. *A measure satisfying FI monotonicity, focus, normalization, continuity, permutability, translation invariance, linear homogeneity, and subgroup consistency is uniquely determined up to a proportional transformation, and given by*

$$f\left(y^0,\, y^1;\, z\right) = \kappa \sum_{i \in S} \left(\min\left\{y_i^0,\, z\right\} - \min\left\{y_i^1,\, z\right\}\right). \tag{1}$$

The summand for individual i behaves as follows. For an individual who was poor before taxes and transfers and is impoverished ($y_i^1 < y_i^0 < z$), it is equal to her fall in income, $y_i^0 - y_i^1$. For an individual who was non-poor before taxes and transfers and is impoverished ($y_i^1 < z \le y_i^0$), it equals her post-fisc poverty gap, or the amount that would need to be transferred to her to move her back to the poverty line (equivalently, to prevent her from becoming impoverished), $z - y_i^1$. For a non-impoverished pre-fisc non-poor individual ($y_i^0 \ge z$ and $y_i^1 \ge z$) it equals $z - z = 0$. For a non-impoverished pre-fisc poor individual ($y_i^0 < z$ and $y_i^1 \ge y_i^0$) it equals $y_i^0 - y_i^0 = 0$. Hence, f sums the total amount of FI, multiplied by a factor of proportionality. This constant can be chosen based on the preferences of the practitioner: for example, $\kappa = 1$ gives total FI (the dark-shaded area in Fig. 1), while $\kappa = |S|^{-1}$ gives per capita FI.[15]

3.3. Fiscal impoverishment dominance criteria

Having identified the existence of FI in a country, a useful implementation of our FI measure would be to compare the degree of FI in two situations, e.g. by comparing the current fiscal system to a proposed reform. The choice of poverty line might, however, influence our conclusion about which situation entails higher FI. We thus present a partial FI ordering that can be used to determine if FI is unambiguously lower in one situation than another for any poverty line and any measure that satisfies FI monotonicity, focus, normalization, continuity, permutability, translation invariance, linear homogeneity, and subgroup consistency. Since we have already shown that a FI measure satisfies these axioms if and only if it takes the form in Eq. (1), a simple way to test for FI dominance for any measure satisfying those axioms and any poverty line in the domain of poverty lines $\mathcal{Z}$ is to simply compare the curves $f(y^0, y^1; z)$ and $f(x^0, x^1; z)$ across $\mathcal{Z}$. Interestingly, if the minimum poverty line being considered is 0 (so $\mathcal{Z} = [0, z^+]$, where z^+ is the maximum poverty line), there is an alternative (equivalent) way to test whether FI is unambiguously lower in one situation than another that uses a dominance test already developed in the mobility literature: Foster and Rothbaum's (2014) second order downward mobility dominance.

Proposition 2. *The following are equivalent.*

a) *FI is unambiguously lower in (y^0, y^1) than (x^0, x^1) for any poverty line in $[0, z^+]$ and any measure satisfying FI monotonicity, focus, normalization, continuity, permutability, translation invariance, linear homogeneity, and subgroup consistency.*
b) *$f(y^0, y^1; z) < f(x^0, x^1; z)$ for all $z \in [0, z^+]$.*
c) *(y^0, y^1) second order downward mobility dominates (x^0, x^1) on $[0, z^+]$.*

4. Fiscal gains of the poor

Most likely, we will be interested in more than just the extent to which some poor are not compensated for their tax burden with transfers: we will also want to know about the gains of other poor families, and the way in which a comparison of poverty before and after taxes and transfers can be decomposed into the losses and gains of different poor households. In this section, we formally define fiscal gains of the poor, briefly present the axioms for a measure of FGP

[13] To avoid inflation in this thought experiment, assume that there is an offsetting fall in the price of a good *not* in the basic goods basket and not consumed by the poor.
[14] By requiring translation invariance and linear homogeneity, we are deriving a measure of *absolute* FI; from there, the measure can nevertheless be modified to obtain other types of desired measures such as a scale invariant measure. This is similar to the approach taken by Fields and Ok (1996), who axiomatically derive a measure of absolute mobility from which other desired measures such as mobility proportional to income can be obtained.
[15] We do not impose a population invariance axiom; this axiom is commonly imposed but is criticized by Hassoun and Subramanian (2012). A subset of measures of form (1) are population invariant: choosing $\kappa = |S|^{-1}$ gives a measure that satisfies population invariance, while $\kappa = 1$ gives a measure that does not.

analogous to those in Section 3.1 for a measure of FI, and present an axiomatic measure and partial ordering of FGP. We then show that a commonly used measure of poverty, the poverty gap, can be decomposed into our axiomatic measures of FI and FGP.

4.1. An axiomatic measure of fiscal gains of the poor

There are *fiscal gains of the poor* if $y_i^0 < y_i^1$ and $y_i^0 < z$ for some individual $i \in S$. The individual may or may not receive enough in net transfers to be post-fisc non-poor (i.e., it is possible that $z \leq y_i^1$ or $y_i^1 < z$). Consider a pre-fisc poor individual who receives more in transfers than she pays in taxes. If she is given even more transfer income, while the pre- and post-fisc incomes of all others experiencing FGP do not change, FGP should not decrease; if she would have remained in poverty post-fisc without the additional transfer income, FGP should increase with the additional transfer. We impose these conditions in the *FGP monotonicity* axiom; we also impose FGP analogues of the other axioms from Section 3.1.

Proposition 3. *A measure satisfying FGP monotonicity, focus, normalization, continuity, permutability, translation invariance, linear homogeneity, and subgroup consistency is uniquely determined up to a proportional transformation, and given by*

$$g\left(y^0,\, y^1;\, z\right) = \kappa \sum\nolimits_{i \in S} \left(\min\left\{y_i^1,\, z\right\} - \min\left\{y_i^0,\, y_i^1,\, z\right\}\right). \tag{2}$$

An individual who is pre-fisc poor and gains income from the tax and transfer system, but remains post-fisc poor ($y_i^0 < y_i^1 < z$), contributes the amount of her income gain, $y_i^1 - y_i^0$, to the measure of FGP. A pre-fisc poor individual that gains income and as a result has post-fisc income above the poverty line ($y_i^0 < z \leq y_i^1$) contributes the amount of net transfers that pulled her pre-fisc income to the poverty line, $z - y_i^0$. Someone who is pre-fisc poor and does not gain income ($y_i^1 \leq y_i^0 < z$) contributes $y_i^1 - y_i^1 = 0$. Someone who is pre-fisc non-poor ($z < y_i^0$) also contributes 0 (for her, the summand equals $z - z$ if she remains non-poor or $y_i^1 - y_i^1$ if she loses income and becomes poor). For $\kappa = 1$, g equals the light-shaded area in Fig. 1.

As with fiscal impoverishment orderings, a fiscal gain partial ordering can be used to make unambiguous FGP comparisons for any poverty line and any measure satisfying our axioms. The ordering compares $g(y^0, y^1; z)$ to $g(x^0, x^1; z)$ for all $z \in \mathcal{Z}$, and for $\mathcal{Z} = [0,\, z^+]$ coincides with Foster and Rothbaum's (2014) second order upward mobility dominance (the proof proceeds similarly to the proof of Proposition 2 for FI).

4.2. Decomposition of the difference between pre-fisc and post-fisc poverty

The most common measures of poverty used in both policy circles and scholarly papers (e.g., Chen and Ravallion, 2010, Ravallion, 2012) are the poverty headcount ratio, which enumerates the proportion of the population that is poor, and the poverty gap, which takes into account how far the poor fall below the poverty line. The latter might be expressed in absolute terms, summing the gap between each poor person's income and the poverty line, in which case it can be thought of as the total amount that would need to be given to the poor to eliminate poverty (if targeting were perfect). Or it can be normalized, dividing the absolute poverty gap by the poverty line and population size, for example, to create a scale- and population-invariant measure. We use a general definition of the poverty gap that encompasses its absolute and normalized forms:

$$p(y;\, z) = \nu(S,\, z) \sum_{i \in S} \left(z - y_i\right) \mathbb{I}\left(y_i < z\right), \tag{3}$$

where $\nu(S,z)$ is a normalization factor. Two special cases are the *absolute poverty gap*, where $\nu(S,z) = 1$, and the *poverty gap ratio*, where $\nu(S,z) = (z|S|)^{-1}$. For simplicity and because a comparison of pre- and post-fisc poverty usually occurs for a fixed population and given poverty line, we assume that S and z are fixed in what follows.

Proposition 4. *A change in the poverty gap before and after taxes and transfers is equal to the difference between the axiomatic measures of FI and FGP from* Eqs. (1) *and* (2)*, multiplied by a constant.*

Given the assumption that the population and poverty line are fixed, $\nu(S,z)$ is a constant that we denote $\bar{\nu}$. The poverty gap in Eq. (3) can be rewritten as $p(y;z) = \bar{\nu}\sum_{i \in S}(z - y_i)\,\mathbb{I}(y_i < z) = \bar{\nu}\sum_{i \in S}(z - \min\{y_i,\, z\})$, so we have $p\left(y^1;\, z\right) - p\left(y^0;\, z\right) = \bar{\nu}\sum_{i \in S}\left(z - \min\left\{y_i^1,\, z\right\}\right) - \bar{\nu}\sum_{i \in S}\left(z - \min\left\{y_i^0,\, z\right\}\right)$, or

$$\begin{aligned} p\left(y^1;\, z\right) - p\left(y^0;\, z\right) &= \bar{\nu}\Big[\sum\nolimits_{i \in S}\left(\min\left\{y_i^0,\, z\right\} - \min\left\{y_i^0,\, y_i^1,\, z\right\}\right) \\ &\quad - \sum\nolimits_{i \in S}\left(\min\left\{y_i^1,\, z\right\} - \min\left\{y_i^0,\, y_i^1,\, z\right\}\right)\Big] \\ &= \frac{\bar{\nu}}{\kappa}\left[f\left(y^0,\, y^1;\, z\right) - g\left(y^0,\, y^1;\, z\right)\right]. \end{aligned}$$

Comparisons of pre- and post-fisc poverty are often used to assess whether the tax and transfer system helps or hurts the poor. This decomposition can be used to dig deeper into that net effect and observe the extent to which a net reduction in poverty masks the offsetting gains of some poor and impoverishment of others at the hands of the (possibly progressive) tax and transfer system.

5. Illustration

5.1. Results for seventeen developing countries

We saw in Section 2 that in fifteen of seventeen developing countries for which we have data, the tax and transfer system is poverty-reducing and progressive but, in many cases, fiscally impoverishes a significant proportion of the poor. In Table 3, we present FI and FGP results for these countries using the axiomatic measures derived in Sections 3 and 4. Column 1 gives total FI (i.e., the axiomatic measure from Eq. (1) with $\kappa = 1$) and column 2 total FGP, both expressed in millions of 2005 US dollars per year using purchasing power parity adjusted exchange rates. Because the axiomatic measure with $\kappa = 1$ is population variant, FI and FGP tend to be higher in more populous countries; these absolute amounts of FI and FGP can be useful, for example, in comparisons to the size of a country's main cash transfer program, as we show for Brazil below. To ease interpretation and comparison across countries, column 3 shows FI expressed as a percent of FGP, while columns 4 and 5 show FI and FGP per capita (where per capita refers to dividing by the entire population), normalized by the poverty line; each of these is population invariant.

There is large heterogeneity in the extent to which some poor are hurt by the tax and transfer system relative to the extent to which other poor gain, despite that the same range of policies, including direct taxes, direct cash and near-cash transfers, indirect consumption taxes, and indirect subsidies were considered in each country study. Among the upper-middle income countries, FI as a percent of FGP (using a poverty line of $2.50 per day) ranges from less than 1% in Ecuador to 40% in Tunisia. In low and lower-middle income countries, FI as a percent of FGP (using a poverty line of $1.25 per day) is even higher in some countries, reaching 55% in Guatemala and

Table 3
Fiscal impoverishment and gains of the poor in developing countries.

Country (survey year)	(1) Total FI ($ millions per year)	(2) Total FGP ($ millions per year)	(3) FI as % of FGP	(4) Per capita FI as % of z	(5) Per capita FGP as % of z	(6) Change in poverty gap ratio (p.p.)
Panel A: Upper-middle income countries, using a poverty line of $2.50 per day						
Brazil (2008–2009)	676.0	3503.6	19.3	0.39	2.02	−1.63
Chile (2013)	2.0	93.3	2.1	0.01	0.59	−0.58
Ecuador (2011–2012)	1.1	277.8	0.4	0.01	2.00	−1.99
Mexico (2012)	227.7	1446.5	15.7	0.21	1.35	−1.14
Peru (2011)	53.7	177.0	30.3	0.20	0.65	−0.45
Russia (2010)	84.9	1561.4	5.4	0.07	1.24	−1.17
South Africa (2010–2011)	186.6	5964.0	3.1	0.41	12.96	−12.56
Tunisia (2010)	20.8	52.0	40.0	0.23	0.59	−0.35
Panel B: Low and lower-middle income countries, using a poverty line of $1.25 per day						
Armenia (2011)	6.3	117.9	5.3	0.44	8.17	−7.74
Bolivia (2009)	25.9	32.2	80.6	0.55	0.68	−0.13
Dominican Republic (2007)	4.4	105.1	4.2	0.02	0.53	−0.51
El Salvador (2011)	1.2	11.1	11.1	0.04	0.39	−0.35
Ethiopia (2010–2011)	408.9	392.8	104.1	1.18	1.13	0.05
Ghana (2013)	25.9	9.9	262.1	0.22	0.08	0.13
Guatemala (2010)	20.7	37.8	54.9	0.33	0.61	−0.27
Indonesia (2012)	150.2	531.5	28.3	0.13	0.47	−0.34
Sri Lanka (2009–2010)	4.4	25.5	17.1	0.05	0.27	−0.23

Sources: For Brazil, authors' calculations. For other countries, provided to us by the authors of the studies cited in Footnote 3.
Notes: p.p. = percentage points. z denotes the poverty line. "$ millions" denotes millions of 2005 US dollars, at purchasing power parity adjusted exchange rates. Country classifications are from the World Bank for the year of the survey.

81% in Bolivia; in Ethiopia and Ghana—the two countries in which post-fisc poverty is higher than pre-fisc poverty—FI exceeds FGP.

Column 6 shows the change in the poverty gap ratio from pre-fisc to post-fisc income, which by Proposition 4 can be decomposed into FI per capita minus FGP per capita, both normalized by the poverty line like the poverty gap ratio. This decomposition reveals some interesting traits of each country's tax and transfer system. For example, Ecuador achieves the same FGP per capita as Brazil but with nearly no FI, compared to substantial FI in Brazil; as a result, the poverty gap is reduced by more in Ecuador. The difference in FI might be attributable to the multiple consumption taxes levied at the state and federal levels in Brazil: these are high and often cascading, and consumption tax exemptions for basic goods are almost non-existent (Corbacho et al., 2013), compared to a system that exempts food, basic necessities, and medicine in Ecuador (Llerena Pinto et al., 2015). Interestingly, most of those experiencing FI are *not* excluded from the safety net; they *do* receive government transfers or subsidies: 65% of the impoverished in Brazil receive cash transfers from Bolsa Família, for example.

It is also noteworthy that Peru, one of the countries in which *less than* a quarter of the post-fisc poor experience FI, nevertheless redistributes low amounts to the poor, and thus has a low reduction in the poverty gap; this is consistent with Jaramillo's (2014, p. 391) finding that Peru's low poverty reduction induced by fiscal policy is "associated with low social spending rather than with inefficient spending." Among three lower-middle income countries that each reduce the poverty gap ratio by about 0.3 percentage points (El Salvador, Guatemala, and Indonesia), Guatemala has high FI but also higher FGP, while El Salvador has lower FGP but very low FI, and Indonesia falls in the middle. We do not attempt to answer whether a lower-FI, lower-FGP or higher-FI, higher-FGP system is preferable from a welfare perspective, but note that this decomposition enables a substantially richer analysis than the typical comparison of poverty before and after taxes and transfers.

5.2. Results for a range of poverty lines in Brazil

So far, the FI and FGP results we have presented use a fixed poverty line ($1.25 in low and lower-middle income countries and $2.50 in upper-middle income countries). We now extend the analysis to a range of poverty lines, focusing the illustration on data from Brazil, using the Pesquisa de Orçamentos Familiares (Family Expenditure Survey) 2008–2009. The precise direct and indirect taxes, direct cash and food transfers, and indirect subsidies included in our analysis are described in detail in Higgins and Pereira (2014).

As we stated in Section 2.3, the tax and transfer system in Brazil is unambiguously poverty-reducing for any poverty line up to $2.50 per person per day, globally progressive, and unambiguously equalizing.[16] This is shown in Fig. 2, where cumulative distribution functions reveal that the post-fisc distribution first order stochastically dominates the pre-fisc distribution on the domain $[0, 2.5]$, which implies an unambiguous reduction in poverty for any poverty line in this domain and any measure in a broad class (Atkinson, 1987, Foster and Shorrocks, 1988);[17] the post-fisc concentration curve with respect to pre-fisc income dominates the pre-fisc Lorenz curve, which implies global progressivity (in the income redistribution sense; see Duclos, 2008); and the post-fisc Lorenz curve dominates the pre-fisc Lorenz curve, which implies that the fiscal system is unambiguously equalizing (Atkinson, 1970). If, however, we extend the maximum poverty line to, say, $4 per person per day—a poverty line frequently used by the World Bank when studying middle-income Latin American countries (e.g., Ferreira et al., 2013)—poverty is no longer unambiguously lowered by the fiscal system: for poverty lines above about $3 per day, the poverty headcount is higher after taxes and transfers than before. We would thus know

[16] Nevertheless, the tax and transfer system reduces poverty by less than its potential under the type of optimal redistribution considered by Fellman et al. (1999), which follows a lexicographic maximin principle. Replacing the actual tax system with optimal taxes of this type (which, in total, equal the size of actual taxes), and replacing the actual distribution of Bolsa Família benefits with the optimal one (redistributing all transfers this way would completely eliminate poverty, so we only optimally redistribute Bolsa Família for illustration), the lowest income in the population would be $1.92 per day, the post-fisc poverty gap ratio would be 2.6% of the poverty line rather than 5.5%, and the post-fisc Gini would be 45.3 rather than 53.9.

[17] We verify that this first order dominance is statistically significant at the 5% level using the asymptotic sampling distribution derived by Davidson and Duclos (2000) with a null hypothesis of non-dominance; the result is also robust to the type of data contamination considered in Cowell and Victoria-Feser (2002).

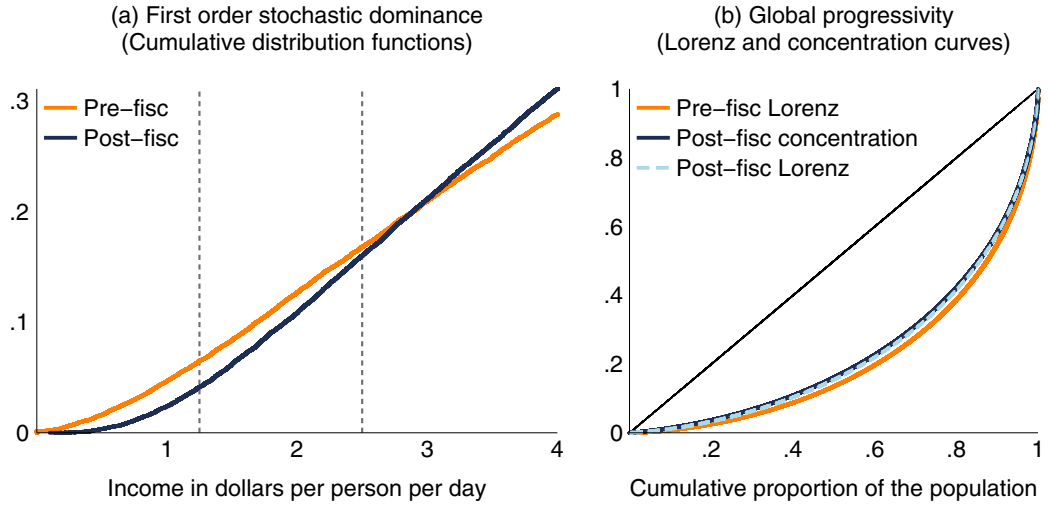

Fig. 2. Conventional tools to assess the tax and transfer system in Brazil. Note: Dashed vertical lines included at common "international" poverty lines of $1.25 and $2.50 per person per day.

that FI occurred using conventional measures and a poverty line above $3 per day, but would still be unaware of its extent without FI measures.[18]

Using the $2.50 line, we know that 5.6% of Brazil's population and over one-third of its post-fisc poor experience FI (Table 2); these impoverished individuals pay a total of $676 million more in taxes than they receive in transfers annually (Table 3), which is equivalent to 10% of the 2009 budget of Bolsa Família, Brazil's flagship anti-poverty program that reaches over one-fourth of the country's population. While substantial in size, this FI is dwarfed by FGP from Brazil's transfer programs, which totals over $3.5 billion. The absolute poverty gap, or the minimum amount that would need to be transferred to the poor to eliminate poverty if transfers were perfectly targeted, falls from $12.4 billion before taxes and transfers to $9.6 billion after. The change in the absolute poverty gap, $2.8 billion, looks impressive, but masks differential trends in two groups of the poor: those who gain (a total of $3.5 billion) and those who lose (a total of $676 million), as revealed by the decomposition of the change in the poverty gap derived in Section 4.

Fig. 3 shows how this decomposition and our axiomatic measures of total FI and FGP in Brazil vary with the poverty line. For low poverty lines, FI is essentially non-existent: at $1.25 per day, for example, total FI is $28 million per year, or 0.4% of the 2009 budget of Bolsa Família (Fig. 3a). This is not surprising in light of the unconditional component of the government cash transfer program Bolsa Família, available to households with income below 70 reais per person per month ($1.22 per day), regardless of whether the household has children or elderly members, and without conditions. At higher poverty lines, FI begins to increase more rapidly, and at a poverty line of $2.88 the rate of increase of FI exceeds the rate of increase of FGP: this can be seen by comparing the slopes of the solid curves in Fig. 3a, or by looking at the point where the difference between the two curves (plotted as the dashed curve in Fig. 3a) is at its maximum. By Proposition 4, this is also the point at which the absolute poverty gap reduction acheived by the fiscal system reaches its maximum, as seen by the dashed curve in Fig. 3b.

At this poverty line of $2.88 per day, where maximum poverty reduction is achieved, the difference between the pre-fisc and post-fisc poverty gaps is $2.9 billion. The eligibility cut-off for the conditional component of Bolsa Família, available to families with children who comply with certain education and health requirements, is $2.45 per person per day. Just above this line, a number of families still receive benefits due to program leakages, variable and mismeasured income, or components of income we are measuring that are not taken into account in the estimation of eligible income; not far above the line, however, families become much less likely to receive the program and we see a simultaneous deceleration of fiscal gains and acceleration of impoverishment.

6. Conclusions

Anti-poverty policies are increasingly being discussed in the same breath as the taxes used to pay for them. One example is the focus on mobilizing domestic resources to finance the policies necessary to achieve the United Nations' Post-2015 Sustainable Development Goals. To analyze transfers, subsidies, and taxes together, poverty comparisons and progressivity measures are often used. These measures, however, can lead us to conclude that the tax and transfer system unambiguously benefits the poor, when in fact a substantial number of poor are not compensated with transfers for their tax burdens. Indeed, we observe this in a number of developing countries: out of seventeen developing countries for which we have data, fifteen have tax and transfer systems that unambiguously reduce poverty and are globally progressive, but in ten of these at least one-quarter of the poor pay more in taxes than they receive in transfers and subsidies. In Brazil, for example, over one-third of the post-fisc poor experience fiscal impoverishment, paying a total of $676 million more in taxes than they receive in transfers and subsidies.

Given this shortcoming of conventional criteria and the debate about anti-poverty policies and the taxes used to pay for them, we propose a set of axioms that should be met by a measure of FI, and show that these uniquely determine the measure up to a proportional transformation. We also propose a partial ordering to determine when one fiscal system, such as that under a proposed reform, induces unambiguously less FI than another, such as the current system, over a range of possible poverty lines. To obtain a complete picture of the fiscal system's effect on the poor, we propose an analogous measure of fiscal gains of the poor, and show that the difference between the pre-fisc and post-fisc poverty gaps can be decomposed into our axiomatic measures of FI and FGP.

Our results can be extended to comparisons between two points in time or before and after a policy reform, rather than pre- and

[18] It is easy to show that if the post-fisc distribution does not first order stochastically dominate the pre-fisc distribution on the domain from 0 to the maximum poverty line, then FI has occurred.

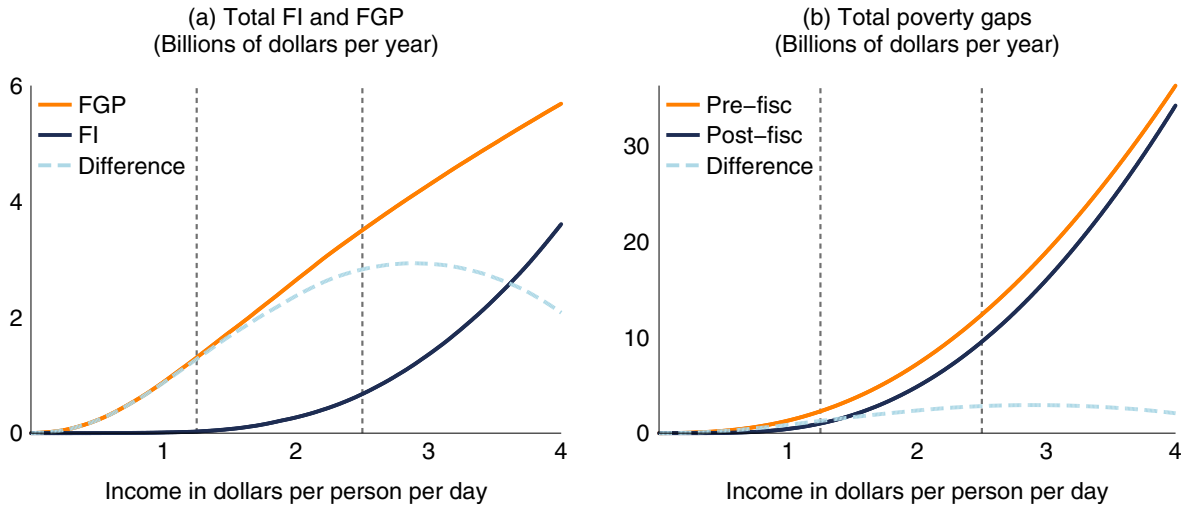

Fig. 3. FI, FGP, and poverty gaps in Brazil for various poverty lines. Note: Dashed vertical lines included at common "international" poverty lines of $1.25 and $2.50 per person per day.

post-fisc. In comparison to the tools used to assess whether the tax and transfer system hurts the poor, tools from the literatures on pro-poor growth and policy reforms (tax and subsidy reforms, trade liberalization, etc.) suffer from similar limitations. For pro-poor growth,[19] poverty measures and stochastic dominance tests are often used to assess whether poverty is unambiguously reduced over time; it directly follows from the first row of Table 1 that these will not necessarily capture that some of the poor become poorer over time. Hence, growth can appear unambiguously pro-poor even if a significant proportion of the poor are immiserized. Growth incidence curves (Ravallion and Chen, 2003) and related pro-poor partial orderings (Duclos, 2009) can fail to capture impoverishment for the same reason that stochastic dominance tests do: they are anonymous with respect to initial income. Although their non-anonymous counterparts (Bourguignon, 2011a, Grimm, 2007, Van Kerm, 2009) resolve this issue in theory, in practice—to become graphically tractable—they average within percentiles, and hence impoverishment can still be overlooked if within some percentiles, some poor are "hurting behind the averages" (Ravallion, 2001, p. 1811).

For consumption tax and subsidy reform, Besley and Kanbur (1988) derive poverty-reducing conditions for reallocating food subsidies; these results are extended to commodity taxes and a broader class of poverty measures by Makdissi and Wodon (2002) and Duclos et al. (2008). Again, by the first row of Table 1, unambiguous poverty reduction does not guarantee that a substantial portion of the poor are not hurt by the reform. Studies that evaluate indirect tax reform with measures that take pre-fisc positions into account but average within groups, such as the percent gain or loss caused by the reform for each income or expenditure decile (Mirrlees et al., 2011, Chapter 9), can again overlook FI that occurs within each group.

In the literature on trade liberalization, Harrison et al. (2003 p. 97) note that "even the most attractive reforms will typically result in some households losing," and recent efforts to measure welfare impact at the household level have been made following Porto (2006). Nevertheless, because results are presented at some aggregate level (e.g., by state or percentile), impoverishment due to trade reform could still be overlooked. For example, Nicita's (2009, p. 26) finding that "on average all income groups benefited from [Mexico's] trade liberalization, but to a varying extent" does not tell us the extent to which some households within each group were made worse off by the reform.

In each of these cases, our axiomatically derived FI measure could be used to quantify the impoverishment of those becoming poorer over time or the extent to which losers are hurt by policy reforms. Our decomposition could be used to examine the extent to which a decrease in poverty over time or due to a reform balances out the gains and losses of different households. Doing so, we will cease to overlook cases where growth, policy reform, or the tax and transfer system is poverty-reducing and progressive, yet hurts a substantial proportion of the poor.

[19] Here, we are using the poverty-reducing or weak absolute definition of pro-poor (in the respective taxonomies of Kakwani and Son (2008) and Klasen (2008)), by which "growth is pro-poor if the poverty measure of interest falls" (Kraay, 2006, p. 198). We could instead adopt a relative definition of pro-poor growth (Kakwani and Pernia, 2000); growth-adjusted stochastic dominance tests can be used to determine when growth is unambiguously relatively pro-poor (Duclos, 2009), and it can be shown that this type of dominance can also occur despite a significant portion of the poor becoming poorer.

Acknowledgments

For detailed comments on earlier versions of the paper, we are grateful to Francesco Andreoli, Alan Barreca, Jean-Yves Duclos, John Edwards, Charles Kenny, Peter Lambert, Darryl McLeod, Mauricio Reis, Kathleen Short, Rafael Salas, Jay Shimshack, Harry Tsang, Paolo Verme, two anonymous referees, and Maitreesh Ghatak, the Editor. We are also grateful to Karim Araar, Abhijit Banerjee, François Bourguignon, Satya Chakravarty, Nachiketa Chattopadhyay, Chico Ferreira, Gary Fields, James Foster, Ravi Kanbur, Doug Nelson, Jukka Pirttilä, John Roemer, Jon Rothbaum, and Shlomo Yitzhaki for their useful feedback. For applying our measures to microdata in a number of developing countries and sharing the results with us for this paper, we are grateful to the many country study authors we cite in the paper. We thank Claudiney Pereira for collaboration on the analysis of Brazilian data and Ruoxi Li, Sandra Martínez-Aguilar, Adam Ratzlaff, Mel Reitcheck, and William Smith for research assistance. Work on this project was partially completed when S. Higgins was visiting Haas School of Business at UC Berkeley and the Center for Economic Studies at El Colegio de México with funding from the Fulbright–García Robles Public Policy Initiative. This paper is part of a larger project that received funding from the Bill & Melinda Gates Foundation (grant numbers OPP1097490 and OPP1335502). These institutions and funders are gratefully acknowledged.

Appendix A

A.1. FI axioms

Consider pre- and post-fisc incomes $y_i^0, y_i^1 \in \mathbb{R}_+$ for each $i \in S$; denote the vectors of pre- and post-fisc income for these individuals by y^0 and y^1, both ordered by pre-fisc income y_i^0. Now consider income vectors for the same individuals under different pre- and post-fisc scenarios, denoted by x^0 and x^1, both ordered by pre-fisc income x_i^0. The sets of impoverished individuals in scenarios (y^0, y^1) and (x^0, x^1) are denoted $I_y \equiv \{i \in S \mid y_i^1 < y_i^0 \text{ and } y_i^1 < z\}$ and $I_x \equiv \{i \in S \mid x_i^1 < x_i^0 \text{ and } x_i^1 < z\}$. A measure of FI is a function $f : \bigcup_{n=1}^{\infty} \mathbb{R}_+^n \times \bigcup_{n=1}^{\infty} \mathbb{R}_+^n \times \mathbb{R}_+ \to \mathbb{R}$, which takes as arguments the pre- and post-fisc income vectors and the poverty line.

Axiom 1 (*FI Monotonicity*). If $y_i^0 = x_i^0$ for all $i \in S$ and there exists $j \in I_y \cup I_x$ such that $y_j^1 > x_j^1$, while $y_k^1 = x_k^1$ for all $k \in I_y \cup I_x \setminus \{j\}$, then $f(y^0, y^1; z) < f(x^0, x^1; z)$.

Axiom 2 (*Focus*). If $y_i^0 = x_i^0$ and $y_i^1 = x_i^1$ for all $i \in I_y \cup I_x$, then $f(y^0, y^1; z) = f(x^0, x^1; z)$.

Axiom 3 (*Normalization*). $I_y = \emptyset \Rightarrow f(y^0, y^1; z) = 0$.

Axiom 4 (*Continuity*). f is jointly continuous in y_i^0, y_i^1, and z.

Axiom 5 (*Permutability*). $f(y^0, y^1; z) = f(y_\sigma^0, y_\sigma^1; z)$ for any permutation function $\sigma : S \to S$, where $y_\sigma^0 \equiv \left(y_{\sigma(1)}^0, \ldots, y_{\sigma(|S|)}^0\right)$ and $y_\sigma^1 \equiv \left(y_{\sigma(1)}^1, \ldots, y_{\sigma(|S|)}^1\right)$.

Axiom 6 (*Translation Invariance*). $f(y^0 + \alpha 1_{|S|}, y^1 + \alpha 1_{|S|}; z + \alpha) = f(y^0, y^1; z)$ for all $\alpha \in \mathbb{R}$, where $1_{|S|}$ denotes a vector of ones with length $|S|$.

Axiom 7 (*Linear Homogeneity*). $f(\lambda y^0, \lambda y^1; \lambda z) = \lambda f(y^0, y^1; z)$ for all $\lambda \in \mathbb{R}_{++}$.

Axiom 8 (*Subgroup Consistency*). Partition S into m subsets $S_1, \ldots, S_m$, and denote the vectors of pre- and post-fisc incomes for individuals belonging to subset S_a, $a \in \{1, \ldots, m\}$, by y_a^0 and y_a^1 or x_a^0 and x_a^1. If $f(y_a^0, y_a^1; z) < f(x_a^0, x_a^1; z)$ for some $a \in \{1, \ldots, m\}$ and $f(y_b^0, y_b^1; z) = f(x_b^0, x_b^1; z)$ for all $b \in \{1, \ldots, m\} \setminus \{a\}$, then $f(y^0, y^1; z) < f(x^0, x^1; z)$.

A.2. FGP axioms

Let the sets of pre-fisc poor individuals experiencing fiscal gains under two scenarios be denoted $G_y \equiv \{i \in S \mid y_i^0 < y_i^1 \text{ and } y_i^0 < z\}$ and $G_x \equiv \{i \in S \mid x_i^0 < x_i^1 \text{ and } x_i^0 < z\}$. A measure of FGP is a function $g : \bigcup_{n=1}^{\infty} \mathbb{R}_+^n \times \bigcup_{n=1}^{\infty} \mathbb{R}_+^n \times \mathbb{R}_+ \to \mathbb{R}$, which takes as arguments the pre- and post-fisc income vectors and the poverty line.

Axiom 1′ (*FGP Monotonicity*). If $y_i^0 = x_i^0$ for all $i \in S$ and there exists $j \in G_y \cup G_x$ such that $y_j^1 < x_j^1$, while $y_k^1 = x_k^1$ for all $k \in G_y \cup G_x \setminus \{j\}$, then $g(y^0, y^1; z) \le g(x^0, x^1; z)$, with strict inequality if $y_j^1 < z$.

The remaining axioms for FI are desirable for a measure of FGP as well, and carry over directly to FGP after replacing f with g, I_y with G_y, and I_x with G_x.

A.3. Proofs

Proof of Proposition 1. We begin with a lemma analogous to one of the propositions in Foster and Shorrocks (1991). To simplify notation, $y_a \equiv (y_a^0, y_a^1)$ for a subset S_a of a partition of S into m subgroups $a = 1, \ldots, m$; similarly, $x_a \equiv (x_a^0, x_a^1)$. We also define vectors $y_{-a}^t \equiv (y_b^t)_{b \neq a \in \{1,\ldots,m\}}$, $t \in \{0, 1\}$ as the vector of pre- or post-fisc incomes of all $i \notin S_a$ (similarly for x_{-a}^t) and $y_{-a} \equiv (y_{-a}^0, y_{-a}^1)$, $x_{-a} \equiv (x_{-a}^0, x_{-a}^1)$. □

Lemma 1. $f(y_a, y_{-a}; z) \ge f(x_a, y_{-a}; z) \Rightarrow f(y_a, x_{-a}; z) \ge f(x_a, x_{-a}; z)$.

Proof. By subgroup consistency, $f(y_a, y_{-a}; z) \ge f(x_a, y_{-a}; z) \Rightarrow f(y_a; z) \ge f(x_a; z)$. (Suppose not. Then $f(y_a; z) < f(x_a; z)$, which by subgroup consistency implies $f(y_a, y_{-a}; z) < f(x_a, y_{-a}; z)$, a contradiction.) $f(y_a; z) \ge f(x_a; z)$ implies either $f(y_a; z) > f(x_a; z)$ or $f(y_a; z) = f(x_a; z)$. In the former case, it immediately follows by subgroup consistency that $f(y_a, x_{-a}; z) \ge f(x_a, x_{-a}; z)$. In the latter case, the implication is shown by contradiction. Suppose that $f(y_a, x_{-a}; z) < f(x_a, x_{-a}; z)$. Then by subgroup consistency we have (since $f(y_a; z) = f(x_a; z)$) $f(y_a, x_{-a}, x_a; z) < f(x_a, x_{-a}, y_a; z)$, which contradicts permutability. □

This lemma shows that a subgroup-consistent and permutable measure of FI is separable by group, using a definition of separability analogous to that used for preferences in the utility literature. Because the lemma can be reiterated within any particular subgroup to further separate individuals in that subgroup, we have that each set of individuals is separable (which is analogous to the "each set of sectors is separable" requirement in Gorman (1968, p. 368)). Hence, from Debreu (1960, Theorem 3), there exists a continuous FI function determined up to an increasing linear transformation of the form

$$f\left(y^0, y^1; z\right) = \alpha + \beta \sum\nolimits_{i \in S} \phi_i\left(y_i^0, y_i^1, z\right)$$

where ϕ_i is a real-valued function for each $i \in S$. The additional requirement for Debreu's (1960) proof that more than two of the $|S|$ elements of S are essential is satisfied as long as $|S| \ge 3$ and f is non-constant on $[0, z]$, which in turn is implied by monotonicity as long as at least one individual is impoverished.[20]

Permutability implies that $\phi_i = \phi_j$ for all $i, j \in S$, so we have $f(y^0, y^1; z) = \alpha + \beta \sum \phi(y_i^0, y_i^1, z)$ where ϕ is a real-valued function. By the focus and normalization axioms:

$$\phi\left(y_i^0, y_i^1, z\right) = \begin{cases} \tilde{\phi}\left(y_i^0, y_i^1, z\right) & \text{if } y_i^1 < y_i^0 \text{ and } y_i^1 < z \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

By the continuity of f, ϕ and $\tilde{\phi}$ must also be continuous. Consider an individual with $y_i^0 > z$ and $y_i^1 = z$. Since y_i^1 is not less than z, i is not impoverished, so by Eq. (4), $\phi(y_i^0, y_i^1, z) = 0$. Now consider an alternative situation where $\tilde{y}_i^1 = z - \epsilon$ for a sufficiently small $\epsilon > 0$. In this scenario, $\tilde{\phi}$ cannot be a direct function of y_i^0 or ϕ would be discontinuous at z; instead, $\tilde{\phi}$ must be a direct function of just y_i^1 and z so that an infinitesimal decrease in y_i^1 below z results in an infinitesimal increase in ϕ. By a similar argument, for an individual with $y_i^0 < z$, $y_i^1 = y_i^0$, and $\tilde{y}_i^1 = y_i^0 - \epsilon$, $\tilde{\phi}$ cannot be a direct function of z and instead must directly depend only on y_i^1 and y_i^0 so that an infinitesimal decrease in y_i^1 below $y_i^0 < z$ results in an infinitesimal increase in ϕ.

[20] The assumptions of at least three individuals in society and at least one impoverished individual are innocuous for any real-world application.

Given this, we can rewrite $\tilde{\phi}\left(y_i^0, y_i^1, z\right) = \tilde{\phi}\left(\min\left\{y_i^0, z\right\}, y_i^1\right)$. Since $\tilde{\phi}$ is only defined for those who are impoverished (i.e., those for whom $\min\left\{y_i^0, y_i^1, z\right\} = y_i^1$), we have

$$\tilde{\phi}\left(y_i^0, y_i^1, z\right) = \tilde{\phi}\left(\min\left\{y_i^0, z\right\}, \min\left\{y_i^0, y_i^1, z\right\}\right) \tag{5}$$

$$= \tilde{\phi}\left(\min\left\{y_i^0, z\right\} - \min\left\{y_i^0, y_i^1, z\right\}, 0\right) \tag{6}$$

$$= \left(\min\left\{y_i^0, z\right\} - \min\left\{y_i^0, y_i^1, z\right\}\right)\tilde{\phi}(1,0) \tag{7}$$

where Eq. (6) follows from translation invariance and Eq. (7) from linear homogeneity. Noting that $\tilde{\phi}(1,0)$ is a constant (that is positive by monotonicity) and denoting it γ, we have

$$\phi\left(y_i^0, y_i^1, z\right) = \begin{cases} \left(\min\left\{y_i^0, z\right\} - \min\left\{y_i^0, y_i^1, z\right\}\right)\gamma & \text{if } i \in I_y \\ 0 & \text{otherwise.} \end{cases}$$

For $i \notin I_y$ we can also write $\phi\left(y_i^0, y_i^1, z\right) = \left(\min\left\{y_i^0, z\right\} - \min\left\{y_i^0, y_i^1, z\right\}\right)\gamma$ since the non-impoverished are either non-poor before taxes and transfers and non-poor after ($\Rightarrow \min\left\{y_i^0, z\right\} = \min\left\{y_i^0, y_i^1, z\right\} = z$) or poor before taxes and transfers but do not lose income to the fiscal system ($\Rightarrow \min\left\{y_i^0, z\right\} = \min\left\{y_i^0, y_i^1, z\right\} = y_i^0$). Therefore $f\left(y^0, y^1; z\right) = \alpha + \beta\gamma\sum_{i\in S}\left(\min\left\{y_i^0, z\right\} - \min\left\{y_i^0, y_i^1, z\right\}\right)$. By normalization, $\alpha = 0$, which completes the proof. □

Proof of Proposition 2. (a)⟺(b) follows immediately from Proposition 1. For (b)⟺(c), we begin by defining Foster and Rothbaum's (2014) second order downward mobility dominance.

Definition 1. (y^0, y^1) *second order downward mobility dominates* (x^0, x^1) on $[0, z^+]$ if

$$\int_0^z m\left(y^0, y^1; c\right) dc < \int_0^z m\left(x^0, x^1; c\right) dc \quad \forall\, z \in [0, z^+],$$

where $m\left(y^0, y^1; z\right) = |S|^{-1}\sum_{i\in S}\mathbb{I}\left(y_i^1 < z < y_i^0\right)$ is Foster and Rothbaum's (2014) *downward mobility curve*, measuring the proportion of the population that begins with income above each poverty line and ends with income below the line.

A sufficient condition for (b) being equivalent to (c) is $f\left(y^0, y^1; z\right) \propto \int_0^z m\left(y^0, y^1; c\right) dc$. For a given poverty line $z = \hat{z}$, partition the set S into four subsets: $S_1 = \left\{i \in S \mid y_i^1 < y_i^0 < \hat{z}\right\}$, $S_2 = \left\{i \in S \mid y_i^1 < \hat{z} \le y_i^0\right\}$, $S_3 = \left\{i \in S \mid y_i^0 \ge \hat{z}, y_i^1 \ge \hat{z}\right\}$, $S_4 = \left\{i \in S \mid y_i^0 < \hat{z}, y_i^0 \le y_i^1\right\}$. For any subset $S_a \subset S$, denote $f_a(\cdot; z) \equiv \kappa\sum_{i\in S_a}\left(\min\left\{y_i^0, z\right\} - \min\left\{y_i^0, y_i^1, z\right\}\right)$ and $m_a(\cdot; z) \equiv |S|^{-1}\sum_{i\in S_a}\mathbb{I}\left(y_i^1 < z < y_i^0\right)$.

Each $i \in S_1$ experiences downward mobility on the interval $[0, \hat{z}]$ for all $z \in \left(y_i^1, y_i^0\right) \Rightarrow$ individual $i \in S_1$ increases $m_1(\cdot; z)$ by $|S|^{-1}$ for $z \in \left(y_i^1, y_i^0\right)$ and by zero for $z \le y_i^1$ and $z \ge y_i^0 \Rightarrow$ individual $i \in S_1$ increases $\int_0^{\hat{z}} m_1(\cdot; c)dc$ by $|S|^{-1}\left(y_i^0 - y_i^1\right)$. Summing over all $i \in S_1$, $\int_0^{\hat{z}} m_1(\cdot; c)dc = \sum_{i\in S_1}|S|^{-1}\left(y_i^0 - y_i^1\right)$.

$y_i^1 < y_i^0 < \hat{z}\ \ \forall\, i \in S_1 \Rightarrow f_1(\cdot; \hat{z}) = \kappa\sum_{i\in S_1}\left(y_i^0 - y_i^1\right) \Rightarrow$

$$f_1(\cdot; \hat{z}) = \kappa|S|\int_0^{\hat{z}} m_1(\cdot; c)dc. \tag{8}$$

Each $i \in S_2$ experiences downward mobility on the interval $[0, \hat{z}]$ for all $z \in \left(y_i^1, \hat{z}\right]$, which increases $m_2(\cdot; z)$ by $|S|^{-1}$ for $z \in \left(y_i^1, \hat{z}\right]$ and by zero for all other $z \Rightarrow$ individual $i \in S_2$ increases $\int_0^{\hat{z}} m_2(\cdot; c)dc$ by $|S|^{-1}\left(\hat{z} - y_i^1\right)$. Summing over all $i \in S_2$, $\int_0^{\hat{z}} m_2(\cdot; c)dc = \sum_{i\in S_2}|S|^{-1}\left(\hat{z} - y_i^1\right)$.

$y_i^1 < \hat{z} \le y_i^0\ \ \forall\, i \in S_2 \Rightarrow f_2\left(\hat{z}, \cdot\right) = \kappa\sum_{i\in S_2}\left(\hat{z} - y_i^1\right) \Rightarrow$

$$f_2\left(\cdot; \hat{z}\right) = \kappa|S|\int_0^{\hat{z}} m_2(\cdot; c)dc. \tag{9}$$

Each $i \in S_3$ does not experience downward mobility on the interval $[0, \hat{z}]$; summing over all $i \in S_3$ and integrating over our domain, we have $\int_0^{\hat{z}} m_3(\cdot; c)dc = 0$. $y_i^0 \ge \hat{z}$ and $y_i^1 \ge \hat{z}\ \ \forall\, i \in S_3 \Rightarrow$

$$f_3(\cdot; \hat{z}) = \kappa\sum_{i\in S_3}(\hat{z} - \hat{z}) = 0 = \kappa|S|\int_0^{\hat{z}} m_3(\cdot; c)dc. \tag{10}$$

Similarly $\int_0^{\hat{z}} m_4(\cdot; c)dc = 0$ because each $i \in S_4$ does not experience downward mobility on $[0, \hat{z}]$. $y_i^0 < \hat{z}$ and $y_i^0 \le y_i^1\ \ \forall\, i \in S_4 \Rightarrow$

$$f_4(\cdot; \hat{z}) = \kappa\sum_{i\in S_4}\left(y_i^0 - y_i^0\right) = 0 = \kappa|S|\int_0^{\hat{z}} m_4(\cdot; c)dc. \tag{11}$$

Given the definitions of $f_a(\cdot; z)$ and $m_a(\cdot; z)$ and that $S = S_1 \cup S_2 \cup S_3 \cup S_4$ and $S_1 \cap S_2 \cap S_3 \cap S_4 = \emptyset$, we have $f\left(y^0, y^1; z\right) = \sum_{a=1}^4 f_a(\cdot; z)$ and $m\left(y^0, y^1; z\right) = \sum_{a=1}^4 m_a(\cdot; z)$. Hence, by Eqs. (8)–(11), $f\left(y^0, y^1; \hat{z}\right) = \kappa|S|\int_0^{\hat{z}} m(\cdot; c)dc$. This holds for all $\hat{z} \in [0, z^+]$ since the choice of $\hat{z}$ was arbitrary, which completes the proof. □

Proof of Proposition 3. Analogous to the proof of Proposition 1 for FI. □

Proof of Proposition 4. Given in text. □

Appendix B. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.jdeveco.2016.04.001.

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